Semiparametric Continuous Difference-in-Differences with Causal Forests: The Case of Charter School Effect Heterogeneity*

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Abstract

A large and growing literature has advanced the understanding and use of difference-in-differences (DiD) and semi-parametric machine learning methods. A much smaller literature has combined DD with causal forests (CF) for causal effect heterogeneity analysis. We build on this DiD-CF literature by extending it to cases with continuous treatment. We then apply the method to charter schools, which serve 7 percent of all U.S. school children. While prior research suggested charter school effectiveness was driven by student demographics, we find that this mostly reflects the something correlated with those demographics: low baseline achievement levels. Also, higher school funding is the only policy variable that seems to improve charter effectiveness. Two dozen other factors are unrelated to charter effects.

Keywords: Charter schools; causal forests; effect heterogeneity; student achievement

JEL Classification: H75; I21; I28

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1 Introduction

The results of social science research often vary considerably across studies and samples and understanding this effect heterogeneity is important for optimal policy design. In education research, in particular, we often see variation in treatment effects across units (e.g., some teachers and schools are much better than others) but less attention is paid to explaining this variation.

Charter schools are a case in point. Research consistently shows that charter schools have positive effects on the achievement of students who attend them (e.g., Sass, 2006; Booker et al., 2007; Hoxby and Rockoff, 2004; Abdulkadiroğlu et al., 2017; Angrist et al., 2016) as well as perhaps high school graduation (Chen and Harris, 2023).[1] However, these charter school participant effects also vary across states (CREDO, 2023) and across cities (Angrist et al., 2016; Harris and Larsen, 2023). The overall distribution of school-level charter effects is also wider than traditional public schools, i.e., a large share of charter schools perform worse than the lowest-performing traditional public schools—and, conversely, the best charter schools are better than the highest-performing traditional public schools (Baude et al., 2020).

To better understand the sources of this variation, we apply and extend prior methods for understanding effect heterogeneity. More than a thousand school districts, across 46 states, have had at least one charter school open in the last 20 years. If charter effects depend, for example, on what types of students they serve, then it might be possible to target charter schools to particular subgroups. Indeed, researchers have commonly found that the participant effects of charter schools on test scores are more positive in urban areas and for disadvantaged students (Cohodes and Parham, 2021). But what we attribute to student demographics could reflect differences in settings and policy that happen to be correlated with student disadvantage. Given the strong correlation between student race/income and student achievement levels, for example, it may be that the apparent role of the former reflects a role for the latter. Also, if state policy differences – regarding funding, charter accountability, regulations, and more – explain variation in treatment effects then this would have implications for charter policy design. While subgroup heterogeneity could be useful regardless of the reasons behind it,[2] it is still useful to identify those reasons and understand the underlying mechanisms.

We begin by following Chen and Harris (2023) and apply various standard difference-in-differences (DiD) methods using national data on most charter schools that have ever been opened

[1] See Cohodes and Roy (2025) for a review of lottery studies that include both test scores and high school graduation. They report four lottery studies that have examined effects on high school graduation. Three of these have negative point estimates (Angrist et al., 2016; Cohodes and Feigenbaum, 2023; Angrist et al. (2023)) and one shows positive effects (Dobbie and Fryer, 2015). However, this is a selected sub-sample of over-subscribed schools, whereas Chen and Harris (2023) find positive market-level effects when looking at the entire country.

[2] Targeting based on student subgroup may improve policy effects even if the underlying cause is some unobservable that is correlated with subgroup status.

in the United States. We also share a focus on "market-level" or net effects. The dependent variable is the weighted average of outcome across traditional public and charter schools, so that the DiD estimates captures participant effects and various competitive and match effects. Using this approach, Chen and Harris (2023) find that a 10-percentage point increase in charter market share increases student test scores by 0.004 standard deviations and increases high school graduation by 1-2 percentage points. As these are market-level effects, the magnitudes reflect a weighted average of effects on both charter participants and the vast majority of students who are non-participants.

Given the positive, but small, average treatment effects, combined with the growth and relevance of the charter school sector, it is important to understand whether this average is masking a wide distribution of effects. To capture this distribution, we build on other recent and novel advances in non-parametric methods for estimating heterogeneous treatment effects. First, we find evidence of significant treatment effect heterogeneity at the school district level for both achievement and graduation rates. For test score effects, the 5th-95th percentile range is -0.15 to +0.15 s.d.. The latter coefficient means that an increase in the charter market share by 10 percentage points, which is common in practice, increased student test scores by +0.015 s.d..

Prior research had found that charter schools were more effective for disadvantaged students, e.g., racial ethnic minorities, low-students, and those in urban areas (Cohodes CITE). In contrast, we find a smaller role for student advantage. Instead, the strongest moderator in our analysis is students' baseline achievement levels, which are strongly correlated with student disadvantage.

Only one policy variable seems important in our analysis: school spending. We included five different spending measures and three of those are in the top-11 on the Variable Importance Factor (VIF). This aligns with other education research showing that this factor is important in traditional public schools (Jackson, YEAR). None of the other policy measures, such as the strength of state-specified accountability or the presence of an independent charter authorizer are ever in the top-10 for any student outcome or method. We also considered measures of market conditions such as district size, the availability of young, educated teachers, market size, and urbanicity. These factors do not seem to play a role.

While we mainly focus on test scores, because of data issues and to keep the analysis concise, we also find that the results often differ for high school graduation. Others, too, have noted the discrepancies in charter results across outcomes (Jackson, 2018). But if we view our outcomes as equally important, this makes it more difficult to target charter schools to improve outcomes. The exceptions, again, are baseline outcome levels and charter funding, which are important for all the outcomes.

Another important policy topic is whether there is some optimal charter market share. These

shares range from 0 percent in most districts, to a national average of 7 percent, and to an extreme of essentially 100 percent in New Orleans and elsewhere. The number of high-charter-share districts is very small, however, which makes this analysis somewhat speculative and imprecise. Nevertheless, we see some indications of diminishing returns with the effects crossing zero in the 50-60 percent market share range.

Our findings for charter schools are based on advances in both causal inference and effect heterogeneity analysis. Each subgroup in the data can have a different counterfactual and selection process. Moderator variables may be subject to measurement error, making treatment effects appear more similar across subgroups than they really are. When many potential dimensions of effect heterogeneity arise, two additional problems arise: (a) hypothesis testing is subject to multiple comparisons and fishing; and (b) subgroups are more likely to be in sparse cells where a plausible, and sufficiently sized, comparison group is difficult to come by. Conventional econometric methods also typically impose functional form assumptions that are convenient and minimize dimensionality but are often tenuous. Causal effect heterogeneity therefore inherits all the challenges of causal inference and then some.

To address these challenges, we extended the generalized random forest (GRF) method (Athey et al., 2019), an extension of the machine learning causal forest (CF) (Wager and Athey, 2018), and we will henceforth refer to these interchangeably. Briefly, CF takes a training sample of the data and splits observations according to the moderator variable that maximizes the sum of squared differences in outcomes between the treated and untreated units. This process repeats and identifies important dimensions of effect heterogeneity, based in part on how many splits are driven by a given moderator. To avoid overfitting, the results are obtained by splitting the data in the same way in the remaining, estimation sample. This general CF method has many advantages over conventional interaction terms; it is non-parametric, not susceptible to multiple comparisons, less susceptible to fishing, and can handle high-dimensional data (large k relative to n).

The CF method, however, was designed for randomized control trials and premised on the idea that the differences between treated and untreated units can be interpreted as "effects." A few studies have recently started marrying CF with difference-in-differences (DiD). Gavrilova et al. (2025) provide what they call the 'Difference-in-Differences Causal Forest (DiDCF)' while Kattenberg et al. (2023) provide the 'Causal Forest Fixed Effects (CFFE).' Briefly, these methods demean outcomes, in a way similar to two-way fixed effects, prior to the data splitting. To our knowledge, prior studies have not applied these methods to continuous treatment and stacked DiD settings, as we have do in our application to charter schools.

In both the DiD and DiD-CF methods, we identified the correlates market-level effect heterogeneity by using nationwide data from the National Longitudinal School Database (NLSD),

which includes nearly all districts in the U.S. from school years 1995 to 2018. To capture the market-level effect, the dependent variable is a weighted average of outcomes of all traditional public schools (TPS) and charter schools located within each local market, i.e., the geographic school district. Prior research on the average market-level treatment effects suggests small positive effects on both achievement and high school graduation nationally (Chen and Harris, 2023). The authors' analyses pass various endogeneity tests (e.g.,parallel trends, and changes in observable demographics), providing empirical support for the idea that we are capturing heterogeneity in causal effects.

Since the DiD-CF methods are new, we test their practical import by comparing the above results to those from a DiD using the usual linear interactions. While the signs on many of the variables change, both methods point in the same general direction—that demographics explain more than policy and that the few moderator variables associated with positive test score effects are not present, or even reverse signs, for graduation effects. This suggests that while the DiD-CF has many advantages and imposes fewer assumptions, it does, at least in some cases yield similar conclusions to standard econometric techniques.

The paper proceeds as follows. Section 2 describes the data, with particular attention to the market-level structure of the analysis, and reviews the research motivating the moderator variables. Section 3 introduces the methods and identification strategies for the DiD and DiD-CF approaches. Section 4 presents the results, and Section 5 concludes.

2 Data: Outcomes and Moderator Variables

2.1 Outcome Data and Structure

Most treatment effect studies, especially in the charter school arena, focus on participant effects, i.e., the effect of attending a charter school compared with counterfactual (usually nearby) TPS. This is less useful in the charter school arena, however, where the rationale for the policy relies not only on participant effects but a range of other market mechanisms whereby charter schools may affect non-participants, too. The entry of a charter school in a given location has been shown to have: *near competitive effects* on the closest public schools (Hoxby, 2003; Bettinger, 2005; Bifulco and Ladd, 2006; Sass, 2006; Zimmer and Buddin, 2009; Imberman, 2011; Linick, 2014; Ridley and Terrier, 2025; Griffith, 2019; Slungaard Mumma, 2022), *far competitive effects* on other traditional public schools located within geographic school district (Hess, 2002; Holley et al., 2013; Figlio et al., 2023),[3] *closure effects* on low-performing traditional public schools (Engberg et al.,

[3]There is very little evidence on far competitive effects. These citations are mainly case studies of individual districts. Also, the Figlio et al. citation refers to district responses to private schools.

2012; Bross et al., 2023; Carlson and Lavertu, 2016), and *match effects* (Abdulkadiroğlu et al., 2017; Campos and Kearns, 2024). Given the evidence on all of these mechanisms, it is important to capture all of them – what we call the "market-level" effect – to understand charter school policy.

Our focus on market-level effects also shapes our data structure. The unit of analysis is the geographic district, which generally follows the boundaries of the local administrative school districts. This is for two reasons: first, distance affects the choice sets of families and therefore the geographic scope of competitive processes; second, most schools are traditional public schools (TPSs) that are governed and led by elected school boards whose authority resides only within the district's geographic boundaries.[4]

The shape files and other data for these geographic districts come from the National Longitudinal School Database (NLSD) and contain a near-census of all TPSs, charter schools, and private schools in the U.S. from 1991 to the most recent academic year.[5] The NLSD merges school and district-level data from Common Core of Data (CCD), Stanford Education Data Archive (SEDA), Census Small Area Income and Poverty Estimates (SAIPE), and other sources. We are most interested in the student outcomes, test scores and graduation rates, but also include a vector of district-level covariates in some models. The street addresses of charter schools also allow us to place the location of each charter school within its geographic school district. These districts define the scope of the market, i.e., which schools are assumed to compete with one another.[6]

We focus mainly on test score outcomes, i.e., student math and English Language Arts (ELA) test scores in grades 3-8 from 2009-2018 from the Stanford Education Data Archive (SEDA). The SEDA provides nationally comparable, publicly available test score data for U.S. public school districts (Ho, 2020).[7] We normalize test score levels to a mean of zero and standard deviation of unity within the grade, year, and subject.

[4]We recognize that the size of these districts varies regionally, e.g., the geographic districts in the U.S. South are much larger geographically than those in the North and some other states.

[5]All the school years mentioned in this paper are spring school years unless specifically stated otherwise.

[6]The SEDA team uses the 2019 Elementary and Unified School District Boundaries https://nces.ed.gov/programs/edge/Geographic/DistrictBoundaries) to define the Geographic School District for test score sample (except for special education or virtual schools). We use the same boundaries to define the Geographic School District for the graduation sample.

[7]To make estimates are comparable across states, grades, and years. The SEDA research team took the following steps: (1) estimate the location of each state's proficiency "thresholds" in the distribution of scores; (2) place the proficiency thresholds on the same scale using the National Assessment of Educational Progress (NAEP), a test taken by a representative sample of students in each state; (3) estimate the mean test scores in each school, district, county, metropolitan statistical area, commuting zone, and state from the raw data and the threshold estimates, and (4) create estimates of average scale scores and achievement growth measures. See details in SEDA website https://edopportunity.org/methods/. While this method is of course based on assumptions (e.g., that the NAEP sample is truly state representative and the distribution of state scores is the same as the NAEP distribution despite the differing content of these various tests), this could only result in bias if test error in charter-entering districts differs from never-charter districts (within grades and states and conditional on other included covariates) and changes over time in conjunction with charter school entry. We could not come up with potential examples where this might be the case.

The AFGR is defined by the U.S. Department of Education and is calculated from aggregate student enrollment data to estimate the size of an incoming freshman class and aggregate counts of the number of diplomas awarded four years later.[8] For example, the AFGR for a school year in 2006 is the total number of diploma recipients in 2006 divided by the average enrollment of the 8th-grade class in 2002, the 9th-grade class in 2003, and the 10th-grade class in 2004. The graduation rate sample includes schools covering grades 9-12 annually from 1995 to 2010.[9]

Treatment status is defined as the charter enrollment share, or the percentage of students attending publicly funded schools who are enrolled in charter schools. These market share measures are created separately by grade level under the theory that charter schools only affect the geographic district outcomes in the specific grades being served (Jinnai, 2014; Slungaard Mumma, 2022).

For both outcomes, data are aggregated to the geographic district level by averaging TPS outcomes with those of charter schools located within their boundaries[10] (weighted by sector enrollment share). This is crucial to the analysis as it allows us to capture charter effects operating through all the various market mechanisms. In contrast, prior studies of charter schools have focused on the outcomes of individual schools in the examination of school participant effects or the competitive effects of individual charter schools on nearby TPS. Here, we are interested in the total effect of all the mechanisms, which means we need to account for the outcomes of all traditional and charter schools. Aggregating outcomes to the geographic district level also allows us to sidestep the main problem with prior studies, i.e., selection bias from students moving across schools (within districts).

The test score (graduation) data are available for 11,399 (10,658) districts, including 1,597 (1,073) districts that have at least one charter school. The final samples for the test scores (graduation) analyses include 97 (98) percent of the nation's publicly funded schools and 99 (97) percent of the nation's charter schools.[11] Table 1 reports weighted summary statistics for the graduation-rate and test-score samples, including the outcome variables, district demographics, financial variables, and charter-school policy measures.

We are identifying total charter effects leveraging the timing of charter entry. One key factor affecting this is the timing of the passage of charter laws that allow them to enter. The first law

[8]The AFGR was the standard federal metric in the years before student-level data systems were prevalent. It was not possible to see where students moved to when they exited district in these earlier years that are important to our analysis.

[9]We noticed some clear errors in the AFGR data, e.g., wild swings from year to year. In these cases, we imputed the errant observations using linear interpolation. (When the errant observation was in the first or last year of the panel, we imputed using the adjacent year.)

[10]SEDA provides test score data at the geographic district level. We converted the graduation rate data to the geographic district level ourselves. Please refer to Appendix B for the description that how the sample was created.

[11]Since geographic districts are the unit of analysis, an observation is only missing if all the schools in the unit-by-school-type are missing. There are some missing schools within non-missing districts.

allowing the establishment of public charter schools was passed in Minnesota in 1991. As of fall 2022, charter school legislation had been passed in 45 states and the District of Columbia.[12] As a result of the growing number of states allowing charter schools, the percentage of publicly funded schools that are charter schools increased from 0 to 7.3 percent, and their percentage of enrollment increased from 0 to 6.2 percent.[13]

To highlight the variation in charter market share we are exploiting, Figure **XXXX** lists the trajectory of charter school growth from the point of the first charter entrant. We report separate lines based on the maximum charter (enrollment) market share that ever appears in our data. As expected, growth rates are faster in geographic districts that have higher eventual market shares and diminish over time.[14]

2.2 Moderator Variables

Pre-specification is important in effect heterogeneity analysis because there are usually large numbers of potential dimensions that can be studied and the odds of finding some statistically significant patterns by chance are high, i.e., the multiple comparisons problem (Athey and Imbens, 2019). It is also easy to "fish" for some statistically significant pattern. Selecting variables based on theory and prior research make this less likely as it directs analysis to dimensions where we have good reasons to expect meaningful patterns. In this section, we summarize theory and research that led us to focus on the moderators used later. This section also highlights some of the potential complexities in the interpretation of results regarding these moderators, setting the stage for the later discussion of results. We organize the moderator variables into four categories: student demographics, market characteristics, school district financing, and charter-related policies.[15]

2.2.1 Student Subgroups

Race/ethnicity and parent income. Education is meant to be an equalizer across student background, so such measures are easily the most widely studied moderators. The fact that it is commonly studied also partially explains why this is a commonly identified dimension of effect heterogeneity. for charter schools and other education policies. We also note that student background is closely correlated with education outcome levels. It may be easier to raise student outcomes when they are already low (i.e., the "low hanging fruit" argument). One objective of

[12] The states in which public charter school legislation had not been passed by that time were Montana, Nebraska, North Dakota, South Dakota, and Vermont.

[13] Appendix Figure **A1** presents the trends in charter school share and charter enrollment share from spring 1991 to spring 2018.

[14] This non-linear pattern partially reflects the opening of the first charter school followed by enrollment growth in those schools, not just opening more schools.

[15] These variables were pre-specified in proposal written for this study and is available upon request.

this analysis is to try and disentangle these potential interpretations.[16]

Charter schools tend to locate in areas that are more diverse on income and race, perhaps because of heterogeneity of preferences (Glomm et al., 2005) or a desire for same-race peers. While it is not obvious that this would affect test scores, it could affect high school graduation. Being closer to the preferences of students and parents might improve the utility of staying in high school relative to the alternatives like work.

These demographic variables are captured in essentially all K-12 education data sets. The income measures are usually in the form of the percent of students eligible for free or reduced-price lunches. The underlying source for these variables in our analysis is the federal Common Core of Data (CCD).

2.2.2 Market Setting Moderators

Baseline school quality. As noted earlier, the common finding that charter schools are more effective for racial/ethnic minorities and low-income students might reflect that their traditional public schools are low-performing, making improvement easier through charter schools. To test this, we use the first available baseline test scores and graduation rates (see discussion above on data sources).

Teacher supply. The importance of teachers is now widely recognized (Chetty et al., 2014). This is unsurprising given that education is a labor-intensive service and the vast majority of resources go to funding and supporting classroom teachers.

In the charter sector, the strongest evidence of effectiveness also comes from large cities that are attractive to young college graduates (e.g., Boston, Denver, and New Orleans). Also, charter school tend to rely on young, highly educated teachers, often from elite colleges and universities and/or teachers from outside the local area. Therefore, the variation in charter effectiveness might partially reflect the ability of charter operators to attract and retain productive teachers who fit their educational and organizational models. We measure teacher supply in two ways: by whether there is a major university located within the geographic district and the share of the local population between the ages of 25-34.

Enrollment growth. We noted above how competitive responses may vary depending on funding formulas and authorizer type. District enrollment growth is another possibility. TPSs have more difficulty expanding when enrollment growth is high. While they are required to serve all students within their boundaries, they are less flexible, e.g., their buildings have to meet fairly specific standards and they are disinclined to open in shopping malls the way that some charter schools do. Anecdotally, California school districts did not generally oppose the entry of charter

[16]We could also describe this as a "diminishing returns" explanation but we use that term later to refer to the non-linearities charter market share effects.

schools in the 1990s and 2000s when enrollments were growing rapidly, but they have more recently as charter shares have grown and total, cross-sector enrollments began declining. This might also imply that districts with more enrollment growth experience a smaller competitive response. In addition, authorizers might be less stringent in accepting applications from charter operators, out of desperation to provide additional capacity. Both factors suggest that we should expect smaller effects of charter entry in districts where enrollment growth is high (though the very fact of enrollment growth also leads to questions about changes in demographics that could bias our estimates).

However, our approach also has more direct implications for policy. We will learn whether various state policies are associated with improved charter market effects and whether new policies might be able to better target charter schools to those contexts where they do the most good. Our analysis also evaluates the performance of charter schools in particular states and of particular charter school authorizers. In the latter case, this is only possible with independent authorizers (in the states that have them) as the analysis requires isolating the effects of charter authorization and charter school improvement efforts from other district behaviors.

2.2.3 Charter Sector Characteristics

Charter school types. The above discussion pointed to two key differences in charter schools: whether they are managed by CMOs (versus standalone schools) and which type of charter authorizer they operate under. Based on prior evidence, we would expect that the entry of charter schools would be more positive when they are managed by CMOs.

Diminishing marginal returns. We have written about the general direction of charter school effects, which might imply a linear relationship between charter market share and school quality. First, if authorizers are picking from menu of potential charter entrants, and they have some accurate signals of future quality then the initial charter schools may be of higher quality than subsequent entrants (Bross et al., 2023).[17] Second, and similarly, the TPS competitive response seems more likely to arise with initial entrants as this is when competition first becomes an issue and is most salient. When the first charter school enters, this is likely to draw newspaper headlines and discussion about the emergence of charter schools.

2.2.4 State Policy Variables

Effect heterogeneity also depends on variation in the treatment itself. In the case of charter schools, this might depend on the state policies that fund, restrict, and guide the work of charter

[17] A potential counterargument here is that, as the charter sector gains traction, more supporting structures may emerge to incubate better charter schools. This occurred in New Orleans for example, where a non-profit began incubating charter schools once it became clear that the city was going to be opening more such schools.

operators.

Following Harris and McKenzie (2024), we also relied on assessments of state policies by charter advocacy organizations. Specifically, we gathered state ratings of charter school policies published in 2014 by the National Alliance for Public Charter School (NAPCS). These rankings are useful for understanding regulation because these organizations are well-informed about charter laws and are likely to focus their rankings on policies that are genuinely important, at least to their charter constituents.[18]

NAPCS evaluates state charter laws based on charter school laws that promote the creation of "high-quality charter schools while holding underperforming schools and authorizers accountable."[19] The five most heavily weighted components used for our study include state ratings by performance-based contract required, clear processes for renewal/nonrenewal, automatic exemption of charter schools from laws applied to traditional public schools (TPS), no caps on the growth of public charter schools, and equitable operational funding. The performance-based contract requirement component provides higher ratings to state laws that have requirements for contracts that provide academic performance expectations, operational performance expectations, and school and authorizer rights and duties. The process for renewal component examines whether there are "clear processes" for renewal, nonrenewal, and revocation decisions, including school closure and dissolution procedures. The no charter caps category determines whether there are caps on growth for public charter schools in a state. Equitable operational funding compares general operational, transportation, and other categorical funding of charter schools to TPS.

State policy is obviously important here and that is no more true than with statewide charter authorizer policies. In most situations, school districts have little incentive to authorize charter schools because this creates competition with the district's existing TPS. It is therefore no surprise that the presence of an independent charter authorizer (i.e., an authorizer other than the school district itself) increases the number of charter schools that open (Harris and McKenzie, 2023). There is also some, more correlational evidence that independent authorizers tend to select charter applications that turn out to have higher value-added (Bross et al., 2023; Harris and McKenzie, 2023). The presence of an independent authorizer might also change the way that districts authorize. For example, if a district believes that the state will authorize a given charter school, then the district might be more likely to recruit and/or authorize that same school itself, so that it can maintain control over it. This is an example of a far competitive response.

[18]We selected the highest weighted components from each sub-index that are likely to be relevant for school performance, openings, and closures.

[19]https://www.publiccharters.org/publications/model-law-supporting-high-quality-charter-public-schools

2.2.5 School Funding

Another policy category, school funding, deserves a separate discussion. Funding is one of the most widely studied elements in education policy and seems to affect student outcomes (Jackson, YEAR). However, the role of funding with market-level charter effects is theoretically complex.

If we were studying only participant effects, then we would hypothesize that higher per-pupil charter funding (absolute or relative to nearby traditional public schools (TPS)) would improve student outcomes. Higher charter funding would also increase the competitive effect because better-funded charter schools would pose more of a threat and possibly involve taking more money from traditional public schools.

But the competitive effect also depends on the resources available to TPS. This makes the hypothesized direction of charter school spending ambiguous. Higher charter spending might mean that TPS has more to lose and but also they have fewer resources to work with in responding to that competition.

Which effect dominates depends on the school funding formula. TPS revenue is usually partially buffered from enrollment losses to charter schools either because: (a) some TPS funding comes from local property taxation, irrespective of enrollment; or (b) state funding includes hold-harmless provisions. In the extreme, where TPS funding is completely independent of enrollment, increased charter funding would have no competitive effect. The market-level effect would still be positive because higher charter spending increases the participant effect.

The above case would be costly because TPS revenue would remain constant even as charter funding grows with enrollment. This has an under-recognized implication: when TPS revenue is buffered from enrollment losses, an increased charter enrollment share also increases market-level per-pupil spending. Thus, buffering TPS means increases spending/costs and reducing competitive effects. For our analysis, the implication is that we also need to consider market-level spending per pupil.

Finally, we note that the above funding measures are subject to considerable measurement error. The above measures require accurate measurement of funding at the school level (in the case of charter schools) and accurate accounting of how funding "passes through" school districts to charter schools, neither of which is the case in U.S. education data. This attenuates the estimated moderator role. Taken together, the above points suggest that we should still expect charter effects to be larger when charter spending is higher, but perhaps less than one might expect.

3 Identification Strategies

We use two complementary methods to study effect heterogeneity in charter school markets. Both methods incorporate the insights of Callaway et al. (2025) to account for continuous treatment dosage. The first is a linear parametric estimator akin to a generalized difference-in-differences (which we will refer to as the CS-DiD), which permits linear interaction heterogeneity within a fixed-effects regression. The second is the Difference-in-Differences Causal Forest (DiDCF) of Gavrilova et al. (2025), extended here to settings with staggered treatment adoption and continuous treatment dosage.

3.1 Notation and Definitions

In this section, we define our treatment effect parameters of interest and discuss how they are identified under the CS-DiD and DiDCF approaches, respectively. We observe school districts (or district-grades), indexed with $i \in \{1, ..., N\}$, over time periods $t \in \{1, ..., T\}$. Our outcomes of interest, $Y_{i,t} \in \mathbb{R}$, include average high school freshman graduation rate (AFGR), standardized elementary math test scores (Math), and standardized elementary English/language arts scores (ELA). Each district has a set of time-invariant observable characteristics/moderators $X_i \in \mathcal{X}$ and are observed with some continuous charter intensity, $W_{i,t} \in [0, 1]$. Some of the characteristics in X_i may be measured confounders of the relationship between $W_{i,t}$ and $Y_{i,t}$, and some may not confound but may nonetheless be related to heterogeneity.

In our setting, charter adoption is staggered over time. Each district belongs to a cohort of districts which are first treated with a positive charter intensity in the same calendar year, or to the group of districts who are never treated. We denote these timing cohorts by $G_i \in \mathcal{G}$, where $G = \infty$ for never-treated districts by convention. Also by convention, we exclude always-treated districts so that $\mathcal{G} \subseteq \{2, ..., T, \infty\}$, and $\bar{\mathcal{G}} = \mathcal{G} \backslash \{\infty\}$ is the set of timing cohorts among ever-treated districts. The dose that district i experiences in period t is therefore $W_{i,t} = W_i \cdot \mathbb{1}\{t \geq G_i\}$, where G_i is the first period in which the district is treated.

Denote $Y_{i,t}(g, w)$ as the potential outcome for district i in time period t if that district is first treated in cohort g under dose w. We write $Y_{i,t}(0) = Y_{i,t}(\infty, 0)$ as the untreated potential outcome. In each period, the observed outcome for district i is $Y_{i,t} = Y_{i,t}(G_i, W_i)$. We define the base period for cohort g as the period before first treatment, which we denote $b_g = g - 1$.

Our first causal parameter of interest is a conditional-on-covariates version of what Callaway et al. (2025) refer to as the "level treatment effect." This is defined as the difference in potential outcomes for district i in timing cohort g if that district experienced dose w vs. if that same district had experienced zero dose in period t: $Y_{i,t}(w, g, x) - Y_{i,t}(0, g, x)$. This is a straightforward extension of the binary treatment effect to a continuous "treatment effect function" or "dose-

response function." The second causal parameter of interest is a conditional-on-covariates version of what Callaway et al. (2025) call the "causal response." A unit's (conditional) causal response is the derivative of the potential outcome function with respect to dose: $\frac{\partial Y_{i,t}(w,g,x)}{\partial w}$. That is, what is the marginal impact on district i from timing group g and covariates x if that district increases its charter intensity from w to w'?

We aim to conduct inference on averages of these two causal parameters. The first is the conditional average treatment effect on the treated ($CATT$) in levels:

$$CATT(g,t,w,x \mid g,w,x) \equiv \mathbb{E}[Y_{i,t}(g,w) - Y_{i,t}(0) \mid G_i = g,\ W_i = w,\ X_i = x] \tag{1}$$

This is the average effect of dose w relative to zero dose in period t, among cohort g districts with covariates x that actually received dose w. This parameter serves as a building block for more easily interpretable aggregates which we will define later in this section. The second building block is the conditional average causal response to treatment. It is the derivative of the first with respect to a counterfactual dose margin, evaluated at the dose the district received:

$$\begin{aligned} CACRT(g,t,w,x \mid g,w,x) &\equiv \frac{\partial\, CATT(g,t,l,x \mid g,w,x)}{\partial l}\bigg|_{l=w} \\ &= \frac{\partial\, \mathbb{E}[Y_{i,t}(g,l) \mid G_i = g,\ W_i = w,\ X_i = x]}{\partial l}\bigg|_{l=w} \end{aligned} \tag{2}$$

The $CACRT$ is interpreted as the average effect of a marginal increase in the dose from w among units that actually experienced dose w (for a district in period t, among cohort g, with covariates x). The second equality holds because the untreated potential outcome $Y_{i,t}(0)$ does not depend on the size of the dose change l. Equations 1 and 2 are, notably, just conditional-on-covariates versions of the staggered-adoption building blocks of Callaway et al. (2025). We also consider versions of these building block parameters that are averages over the entire dose distribution:

$$CATT(g,t,w,x) \equiv \mathbb{E}[Y_{i,t}(g,w) - Y_{i,t}(0) \mid G_i = g,\ W_i > 0,\ X_i = x] \tag{3}$$

$$CACRT(g,t,w,x) \equiv \frac{\partial\, CATT(g,t,w,x)}{\partial w} = \frac{\partial\, \mathbb{E}[Y_{i,t}(g,w) \mid G_i = g,\ W_i > 0,\ X_i = x]}{\partial w} \tag{4}$$

Equations 3 and 4 average over all treated districts in cohort g with covariates x, rather than just those districts that received dose w.

The above causal functions are high dimensional objects which are difficult to interpret and report (though we will estimate them directly). Therefore, we also consider aggregations of Equations 1 – 4 into event study-like parameters. Denote $e = t - b_g$ as the number of periods since initial treatment for units in group g. We first aggregate over these event-times and within dose

groups:

$$\widetilde{CATT}^{dose,es}(w, x \mid w, e, x) = \sum_{g \in \bar{\mathcal{G}}} \sum_{t=2}^{T} \phi^{dose,es}(g, t, w, x \mid e)\ CATT(g, t, w, x) \tag{5}$$

$$\widetilde{CACRT}^{dose,es}(w, x \mid w, e, x) = \sum_{g \in \bar{\mathcal{G}}} \sum_{t=2}^{T} \phi^{dose,es}(g, t, w, x \mid e)\ CACRT(g, t, w, x) \tag{6}$$

where the weight on cohort g is the share of districts with dose w that belong to that cohort:

$$\begin{aligned}\phi^{dose,es}(g, t, w, x \mid e) = \mathbb{1}\{g + e \in [2, T]\}\ \mathbb{1}\{g + e = t\} \\ \times \mathbb{P}(G_i = g \mid W_i = w,\ G_i + e \in [2, T],\ G_i \leq T)\end{aligned}$$

Next, we aggregate again over dose to report conditional event-study parameters which are interpreted as the average effect of treatment for districts with covariates x that have been exposed to treatment for e periods:

$$CATT^{es}(e, x) = \mathbb{E}\left[\widetilde{CATT}^{dose,es}(W_i, x \mid W_i, e, x) \,\middle|\, G_i + e \in [2, T],\ G_i \leq T\right] \tag{7}$$

$$CACRT^{es}(e, x) = \mathbb{E}\left[\widetilde{CACRT}^{dose,es}(W_i, x \mid W_i, e, x) \,\middle|\, G_i + e \in [2, T],\ G_i \leq T\right] \tag{8}$$

3.1.1 Assumptions

Going forward, we adopt the following assumptions:

Assumption 1 (Random Sampling)**.** The observed data is drawn from an i.i.d. distribution and consists of $\{Y_{i,1}, ..., Y_{i,T}, W_i, G_i, X_i\}_{i=1}^{n}$.

Assumption 2 (Overlap)**.** The conditional density of treatment is bounded away from zero uniformly: there exists $\varepsilon > 0$ such that $f(W_i \mid X_i = x) \geq \varepsilon$ for all $x \in \text{supp}(X_i)$ and all W_i in the interior of $[0, 1]$. This rules out regions of the covariate space in which all districts receive the same charter intensity, ensuring the causal forest has non-degenerate treatment variation at every x.

Assumption 3 (No Anticipation)**.** For all $g \in \mathcal{G}$ and $t = 1, ..., T$ with $t \leq b_g$, $Y_{i,t}(g, w) = Y_{i,t}(0)$

Assumption 4 (Dose Support and Smoothness)**.** $\mathbb{P}(W_i = 0) > 0$. In addition, there is an open interval $\mathcal{W}_+^c = (w_L, w_U) \subseteq \mathcal{W}_+$ such that, for all $g \in \bar{\mathcal{G}}$, $t = 2, ..., T$, and $x \in \mathcal{X}$, the conditional density of W_i given $G_i = g$ and $X_i = x$ is positive on $\mathcal{W}_+^c$, and $\mathbb{E}[Y_{i,t} - Y_{i,b_g} \mid G_i = g, W_i = w, X_i = x]$ is continuously differentiable in w on $\mathcal{W}_+^c$.

We also adopt one of two conditional parallel trends assumptions depending on which causal function we are attempting to estimate:

Assumption 5 (Conditional Parallel Trends)**.** For all $g \in \bar{\mathcal{G}}$, $t = 2, ..., T$, $w \in \mathcal{W}_+$, and $x \in \mathcal{X}$:

$$\begin{aligned} &\mathbb{E}[Y_{i,t}(0) - Y_{i,b_g}(0) \mid G_i = g,\ W_i = w,\ X_i = x] \\ &\quad = \mathbb{E}[Y_{i,t}(0) - Y_{i,b_g}(0) \mid G_i = \infty,\ W_i = 0,\ X_i = x] \end{aligned} \tag{9}$$

In other words, the expected counterfactual change in outcomes between period t and b_g for districts in cohort g who experienced dose w with covariates x, had they not been treated, is equal to the observed change in outcomes among never-treated districts, conditional on X. This implies that all districts have the same counterfactual trends under $w = 0$ which are equal to those of the never-treated cohort.

Assumption 6 (Conditional Strong Parallel Trends)**.** Both of the following are true. (a) For all $g \in \bar{\mathcal{G}}$, $t = 2, ..., T$, $w, l \in \mathcal{W}_+$, and $x \in \mathcal{X}$:

$$\begin{aligned} &\mathbb{E}[Y_{i,t}(g, w) - Y_{i,b_g}(g, w) \mid G_i = g,\ W_i = l,\ X_i = x] \\ &\quad = \mathbb{E}[Y_{i,t}(g, w) - Y_{i,b_g}(g, w) \mid G_i = g,\ W_i = w,\ X_i = x] \end{aligned}$$

(b) Assumption 5 holds.

Assumption 6 states that districts in the same cohort and covariate cell share a common path under any given dose. Assumption 6 is typically a stronger parallel trends assumption than Assumption 5, and requires that every dose group (within x) follow the same causal function with respect to dose, which implies that districts with the same covariate values have the same treatment effects at all dose levels. Assumptions 5 and 6 are relaxations of Assumptions PT-MP and SPT-MP in Callaway et al. (2025), respectively, such that they must only hold within groups of districts with the same covariate values. It is worth noting that only the subset of X_i variables which are confounders are required in order to satisfy Assumptions 5 and 6. The non-confounding variables in X_i are not required to obtain consistent estimates of causal parameters, but we nonetheless include them in order to further explore heterogeneity and because the set of confounders is unknown.

3.1.2 Identification Under Parallel Trends vs. Strong Parallel Trends

Our first result is that under Assumptions 1 – 5, the $CATT$ is identified via a difference-in-differences estimator which compares the change in outcomes for districts in dose group w, timing group g, period t, and covariates x to districts with covariates x who were never treated:

Lemma 1 (Identification of the $CATT$)**.** Under Assumptions 1 – 5, for all $g \in \bar{\mathcal{G}}$, $t \geq g$, $w \in \mathcal{W}_+$, and $x \in \mathcal{X}$:

$$CATT(g,t,w,x \mid g,w,x) = \mathbb{E}[Y_{i,t} - Y_{i,b_g} \mid G_i = g,\ W_i = w,\ X_i = x] - \mathbb{E}[Y_{i,t} - Y_{i,b_g} \mid G_i = \infty,\ W_i = 0,\ X_i = x]$$

For proof, see Appendix D.

If Assumption 6 also holds, then the $CACRT$ is identified as the derivative of the $CATT$ function evaluated at w

Lemma 2 (Identification of the $CACRT$)**.** Under Assumptions 1 – 6, for all $g \in \bar{\mathcal{G}}$, $t \geq g$, $w \in \mathcal{W}_+^c$, and $x \in \mathcal{X}$:

$$CACRT(g,t,w,x \mid g,w,x) = \frac{\partial\, \mathbb{E}[Y_{i,t} - Y_{i,b_g} \mid G_i = g,\ W_i = l,\ X_i = x]}{\partial l}\bigg|_{l=w} \tag{10}$$

For proof, see Appendix D. Equation 10 carries no never-treated term. The never-treated comparison in Equation 1 does not vary with l, so it vanishes when we differentiate. This is the conditional analogue of Theorem C.1 of Callaway et al. (2025).

3.2 Generalized Difference-in-Differences (CS-DiD) Identification and Estimation

Callaway et al. (2025) propose estimators which identify the unconditional $ATT(g,t,w)$ and $ACRT(g,t,w)$ under unconditional versions of our Assumptions 1 – 6 (Assumption 6 only required for $ACRT(\cdot)$). Their estimators are constructed to ensure the regression weights implicit in the selected model match the true dose distribution. We extend their estimators to identify the $CATT(g,t,w,x)$ and $CACRT(g,t,w,x)$ under Assumptions 1 – 6 which condition on covariates (and are therefore relaxations of their unconditional parallel trends and strong parallel trends assumptions). First, recall from Lemma 1 that for some positive dose w, between period t and b_g for a district in group g we have: $CATT(g,t,w,x) = E[\Delta Y_{i,t}|G = g, W_i = w, X_i = x] - E[\Delta Y_{i,t}|G_i = \infty, W_i = 0, X_i = x]$. With an estimate of the baseline conditional mean function $\hat{m}_{g,t}(x) = E[\Delta Y_{i,t}|G_i = \infty, W_i = 0, X_i = x]$, the general form of their estimators can be expressed as the following:

$$\widetilde{\Delta} Y_{i,g,t} = CATT(G,T,W,X) + \varepsilon_{i,g,t}$$

Where $\widetilde{\Delta} Y_i = \Delta Y_i - \hat{m}_{g,t}(x)$ and $CATT(G,T,W,X)$ can be any researcher-selected functional

form of the $CATT$. We will propose both parametric and semi-parametric functional forms in our case. In our parametric model, we specify a linear function akin to a generalized difference-in-differences (DiD), where the following regression is estimated on treated districts separately within cohort g and time t:

$$\widetilde{\Delta} Y_{i,g,t} = \alpha_0 + \alpha_1 X_i + \tau_0 W_i + \tau_1 (W_i X_i) + \varepsilon_{i,g,t} \tag{11}$$

Where $\hat{\alpha}(x) = \hat{\alpha}_0 + \hat{\alpha}_1 x$ and $\hat{\tau}(x) = \hat{\tau}_0 + \hat{\tau}_1 x$. This implies that $\widehat{CATT}(g,t,w,x) = \hat{\alpha}(x) + \hat{\tau}(x) w$ and, under Assumption 6, $\widehat{CACRT}(g,t,w,x) = \hat{\tau}(x)$. Our estimate of the baseline conditional mean in this model comes from an OLS regression of $\hat{m}_{g,t}(x) = \hat{\gamma}_0 + \hat{\gamma}_1 x = \hat{\mathbb{E}}[\Delta Y_{i,g,t} | G_i = \infty, W_i = 0, X_i = x]$, estimated from never treated units.

To see that this model closely matches a traditional two-way fixed effects DiD, consider a 2-period regression between period t and b_g estimated on all units in cohort g and never-treated units $\mathcal{S}_g = \{i : G_i = g\} \cup \{i : G_i = \infty\}$:

$$Y_{i,g,t} = \mu_i + \lambda_t + Post_t \Big[\Big(\gamma_0 + \gamma_1 X_i \Big) + \mathbb{1}\{G_i = g\}(\alpha_0 + \alpha_1 X_i) + W_i(\tau_0 + \tau_1 X_i) \Big] + \varepsilon_{i,g,t}$$

Where μ_i and λ_t are unit and time fixed effects, respectively, and $Post_t$ is an indicator for a unit being in a post-treatment period. In period b_g, $Post_t = 0$, so taking the first difference between periods t and b_g yields:

$$\begin{aligned}
\Delta Y_{i,g,t} &= \underbrace{(\lambda_t - \lambda_{b_g}) + \gamma_0}_{\text{constant}} + \gamma_1 X_i + \mathbb{1}\{G_i = g\}(\alpha_0 + \alpha_1 X_i) + W_i(\tau_0 + \tau_1 X_i) + \Delta \varepsilon_{i,g,t} \\
&= \underbrace{\gamma_0 + \gamma_1 X_i}_{\hat{m}_{g,t}(x)} + \mathbb{1}\{G_i = g\}(\alpha_0 + \alpha_1 X_i) + W_i(\tau_0 + \tau_1 X_i) + \Delta \varepsilon_{i,g,t}
\end{aligned}$$

Rearranging yields Equation 11.

One potential challenge with so many different dimensions of effect heterogeneity (see Section 2) is that the factors in X_i may be correlated with each other. For example, state policies may be correlated with household demographics. Also, states with more accountability might also have distinctive funding formulas. We are able to overcome this challenge because the richness of our data allows us to include essentially all of these factors at the same time to isolate the role of each.

The error term $\varepsilon_{i,g,t}$ is clustered at the district level in the main specification. The estimates are weighted by the district size, so that the estimates are nationally representative; specifically, we weight by graduation rate denominator in the high school sample and by grade-level enrollment for math and ELA. In our baseline analyses, we estimate the CS-DiD model for both the

current year charter enrollment share and with a one-year lag to allow delayed effects.

Our method accounts for time-invariant unobserved differences across districts. Identification of the CATT requires that, conditional on observed covariates, districts receiving a given level of charter-school exposure would, in the absence of treatment, have experienced the same average change in outcomes as never-treated districts. In other words, the counterfactual outcome trend under zero charter exposure must be common across treated and never-treated districts within covariate cells. This assumption would be violated if unobserved time-varying factors jointly affected charter-school expansion and student outcomes. We note that, even if charter market share changed based on time-invariant unobserved district characteristics that affected student outcomes, this would not introduce bias if the timing of charter entry were conditionally random, which is possible given all the steps involved in opening a charter school (e.g., putting in an application, getting it approved, gaining access to a building, recruiting students, and hiring staff).

Still, it is worth considering specific scenarios under which this might be violated. First, charter schools are more likely to locate in districts with low contemporaneous student outcomes (Glomm et al., 2005), signaling that we also might expect charter schools to also locate where *expected* future TPS performance is lower, though this is unobservable in our analysis.[20] This would lead to a downward bias in our estimates because TPS outcomes comprise the majority of the weighted average outcomes.

Alternatively, local governments might change non-charter education policies at the same time they introduce charter schools, as part of a package of reforms. Since school districts are often charter authorizers, it may be that a change in school board politics leads both to an increase in charter schools and, for example, new reading and math curricula in TPS. In this case, we might falsely attribute outcome changes to charter schools that were actually caused by other policy changes. We discuss various tests for these and related types of violations in the next section, after describing our main results.

3.3 Causal Forest Identification and Estimation

The CS-DiD identifies the $CATT/CACRT$ within a linear parametric model. However, there exists a large literature which considers partially linear models of conditional average treatment

[20]Relatedly, charter schools might open where districts experience idiosyncratic shocks, in which case future outcomes regress toward the mean. If charter schools prefer to locate near low-performing schools, as noted above, then this would yield an upward bias (i.e., charter schools enter because of the negative shock in existing schools, but then those outcomes bounce back in the next period). We do not see this scenario as very likely because it takes several years to create an organization that can assemble a charter application, submit the application and gain approval, hire personnel, purchase necessary capital, and recruit students. Also, with such a long-term investment, charter organizations are likely to consider multiple years of information in making their entry decisions, which reduces the chance of regression to the mean.

effects (e.g. Robinson (1988)), which can be written generally as:

$$Y_i = \alpha(X_i) + \tau(X_i)W_i + \varepsilon_i \tag{12}$$

Where the coefficient on W_i, $\tau(X_i)$, is an unknown function of X_i. Under a conditional independence condition (i.e. $\mathbb{E}[W_i\varepsilon_i|X_i] = 0$), $\tau(X)$ identifies the conditional average treatment effect ($CATE$)[21]. This translates into a generalization of the CS-DiD framework under our Assumptions 1 – 6, where $\tau(X)$ now identifies the $ACRT$[22]:

$$Y_{i,g,t} = \mu_i + \lambda_t + Post_t\mathbb{1}\{W_{i,t} > 0\}\underbrace{\left(\alpha(X_i) + \tau(X_i)W_{i,t}\right)}_{CATT} + \varepsilon_{i,g,t} \tag{13}$$

It is worth noting that Equation 13 is not a structural model of potential outcomes, as interpreting it as such would require X_i to contain all moderators that contribute to heterogeneity (Frölich, 2007), some of which may be unobservable in practice. Nonetheless, extrapolating the result of Abadie (2003) allows us to conduct inference on the $CATT/CACRT$ without additional assumptions beyond Assumptions 1 – 6.

In order to conduct inference on the $CATT/CACRT$, our task is to non-parametrically estimate $\left\{\alpha(X_i), \tau(X_i)\right\}$ in Equation 13. We begin by briefly reviewing $CATE$ estimation using causal forests in order to build intuition for the extensions we employ, and refer the reader to other studies which provide more detailed expositions of $CATE$ estimation using casual forests (e.g. Knaus et al., 2021; Caron et al., 2022; Strittmatter, 2023). Next, we discuss the method we employ to operationalize the causal forest in DiD scenarios: the DiDCF of Gavrilova et al. (2025).

3.3.1 Causal Forest Overview

To discuss the basics of causal forest estimation, consider the case of a completely randomized experiment with a binary treatment. In this case, we seek to estimate the $CATE$ for a set of pre-specified covariates, which corresponds to $\tau(X_i)$ in Equation 12. A *causal forest* (Wager and Athey, 2018) is built from many *causal trees* (Athey and Imbens, 2016). Causal trees group observations by recursively splitting the data into finer and finer subgroups. With each split, the causal tree aims to partition units such that the $ATEs$ within each partition are as different

[21] Abadie (2003) takes this further and shows that even if W_i is endogenous, if one obtains a valid instrument then $\tau(X)$ identifies the conditional *local* average treatment effect ($CLATE$) within the same partially linear model without any further assumptions beyond those of a standard IV estimator.

[22] Traditionally, parallel trend assumptions like Assumption 5 only assume that counterfactual trends of *treated* units are parallel to the observed trends of control units. PTA's do not assume that the counterfactual trends of control units had they been treated are parallel to the observed outcomes of units that were actually treated. This is why we do not attempt to identify the CATE: because doing so would require a stronger PTA that stipulates the observed outcomes of treated districts are representative of the counterfactual outcomes of control districts.

as possible. Splits occur at "nodes," and the resulting partitions of the data which exist after the initial split are called "children nodes." With continuously distributed covariates, splitting can in principle continue until each unit is contained in its own partition. To prevent this level of overfitting, hyperparameters are used which define stopping rules such that a tree knows when to cease splitting along a branch. These may include (1) a minimum number of treatment/control units in a child node, (2) a minimum share of a parent node that must be contained in a child node, (3) imbalance penalty parameters, etc. The final partition along a branch is called a "leaf" or "terminal node," $L(x)$, and results in a small set of units defined by a similar vector of covariate values, x. Within any given leaf, the $CATE$ is then estimated as the difference in outcomes between treated vs. untreated units in that leaf. Causal forests then provide an estimate of each unit's individualized $CATE$ by averaging estimates across causal trees.[23] The purpose of growing many trees is to improve accuracy, as singular trees do not produce smooth estimates of $CATE$ functions (Wager and Athey, 2018).

Athey et al. (2019) extend the causal forest approach such that it can be applied to more general settings of causal effect estimation such as observational studies with non-randomized treatment, which they refer to as *generalized random forests* (GRF). They do this by re-framing the causal forest algorithm such that it no longer views forests as an average of causal trees, but rather as an adaptive nearest-neighbor estimator of any causal parameter that can be identified with a local moment condition. In the case of Equation 12, $\tau(X_i)$ is identified via the (conditional) local moment condition corresponding to conditional independence of W_i and Y_i:

$$\begin{aligned} & \mathbb{E}[W_i \varepsilon_i | X_i] = 0 \quad \forall X_i \in \mathcal{X} \\ \implies & \mathbb{E}[W_i(Y_i - \alpha(X_i) - W_i \tau(X_i)) | X_i] = 0 \quad \forall X_i \in \mathcal{X} \end{aligned} \tag{14}$$

Based on Equation 14, the GRF estimates of $(\hat{\alpha}(X_i), \hat{\tau}(X_i))$ are given by:

$$\big(\hat{\alpha}(X_i), \hat{\tau}(X_i)\big) \in \arg\min_{\alpha,\tau} \left\{ \Big|\Big| \sum_{i=1}^{n} \omega_i(x)[W_i(Y_i - \alpha(X_i) - W_i\tau(X_i))] \Big|\Big|_2 \right\} \tag{15}$$

Where $\omega_i(x)$ is the GRF-defined similarity weight which captures the relevance of the i-th observation for estimating $\hat{\alpha}(X_i)$ and $\hat{\tau}(X_i)$; which is how often observation i appears in the same terminal leaf as x across all causal trees. It is straightforward to see that Equation 15 is simply a GRF-weighted OLS estimator. The task is therefore to derive the similarity weights such that the resulting $CATE$ estimates from Equation 15 are consistent and asymptotically nor-

[23] All units will fall into some leaf within each tree, and thus will receive a treatment effect estimate from each tree.

mal. We will outline the GRF algorithmic approach to growing causal trees that accomplishes this.

There are two important concepts to note before explaining the GRF-training algorithm. First, GRF performs Neyman orthogonalization (i.e. local recentering) (Robinson, 1988; Chernozhukov et al., 2018; Nie and Wager, 2021) by differencing out the propensity score and conditional mean from W_i and Y_i, respectively. These propensity score functions are estimated via auxiliary regression trees with leave-one-out estimates. The purpose of this is to improve finite sample performance of the model, as GRF estimates can be biased in finite samples if the conditioning moderators in X_i are not fully partialled out because of the small sample size. Athey et al. (2019) show that this reduces finite sample bias in these situations. Second, trees are estimated based on the concept of "honesty" developed by Athey and Imbens (2016). The consistency and asymptotic normality of non-parametric estimators relies on cross-training: using a different sample for model estimation than the one used for model training. The reason is that using the same sample for training and estimation introduces dependence between the structure of the model and the estimates themselves. Therefore, an honest causal tree takes the full data sample and extracts a random sub-sample to grow the tree, then further divides this sample into a training sample and an estimation sample. The training sample is used to perform splitting and define the tree structure, the estimation sample is used to populate the leaves and provide $CATE$ estimates. Appendix E summarizes the process for growing trees and obtaining the GRF-weights.

3.3.2 DiDCF Overview

The above description of the GRF algorithm is designed for settings where treatment is credibly conditionally random, as in the case of a randomized experiment. Our study is quasi-experimental and applies difference-in-differences, which relies on the conditional parallel trends assumptions stated above as Assumption 5 (for the $CATT$) and Assumption 6 (for the $CACRT$). We therefore require an operationalization of the GRF algorithm that identifies that $CATT/CACRT$ under Assumptions 5 and 6 rather than conditional independence. One such approach is the Difference-in-Differences Causal Forest (DiDCF) of Gavrilova et al. (2025). In this subsection, we will provide a brief overview of the DiDCF, which is designed for the most straightforward types of event-study: cases with a single treatment group which is affected by a binary treatment (the authors also provide an extension for staggered treatment adoption settings). The next section will discuss our further extension of this approach to settings of staggered treatment adoption by multiple treatment groups under continuous treatment.

The standard DiDCF assumes that all treated units are treated in the same period, and that treatment is both binary and an absorbing state. The target estimand in this case is the canonical

binary $CATT$ within a given period t, which is the $CATT$ corresponding to Definition 1 in the case of binary treatment (and only 2 groups):

$$CATT(t, x \mid w) \equiv \mathbb{E}[Y_{i,t}(1) \mid W_i = 1,\ X_i = x] - \mathbb{E}[Y_{i,t}(0) \mid W_i = 1,\ X_i = x] \tag{16}$$

The DiDCF estimator proceeds in the following sequence:

Standard DiDCF Algorithm:

1. Transform the outcome variable by differencing it relative to a base year, b:

   $$\Delta Y_{i,t} = Y_{i,t} - Y_{i,b}$$

   Conducting analysis on this transformed variable ensures that any forests trained on it are DiD-based estimators. Gavrilova et al. (2025) show that under a similar set of assumptions to our Assumptions 1 – 5 (and some standard regularity assumptions), that a DiD-based estimator identifies the $CATT$ from Equation 16.

2. For each period $t \neq b$, estimate a separate (standard) causal forest (GRF) in the manner of Athey et al. (2019). This produces an estimate of the $CATT$ function in each period, $\widehat{CATT}(t, x \mid w)$. Intuitively, this approach is equivalent to training a single causal forest on the full data, but forcing the first split to be on time period. This is necessary because not doing so would lead to cross-period contamination within leaves (i.e. units from different time periods would fall in the same leaf). If the treatment effect is not uniform across time periods, heterogeneity bias in the same spirit as that highlighted by Sun and Abraham (2021a) in the case of staggered adoption may enter the estimates. It is also important to note that this approach allows estimation of pre-treatment placebo tests of parallel trends by estimating forests in periods $t < b$, where the outcomes of not-yet-treated units are compared to the outcomes of never-treated units.

3. Predict the treatment effect for unit i in period t among only treated units, using trees not trained on unit i.

3.3.3 Extending DiDCF to Staggered and Continuous Treatment

Gavrilova et al. (2025) show that their estimator is consistent and asymptotically normal under Assumptions[24] 1 – 5. Here, we outline an approach to extend the DiDCF to a staggered adoption

[24]Their version of the parallel trends assumption is less strict than our Assumption 5 because they consider only one treated group. Satisfaction of our Assumption 5 implies satisfaction of their parallel trends assumption.

setting with continuous treatment.

Recall from Section 3.1 that $\mathcal{G} \subseteq \{2, ..., T, \infty\}$ is the set of cohorts of initial treatment we consider (after dropping always-treated units), and $\bar{\mathcal{G}} = \mathcal{G} \backslash \{\infty\}$ is the set of ever-treated cohorts. Units in the same cohort have the same base period, which is the period before initial treatment, denoted b_g. The first adjustment to the DiDCF algorithm is that we begin by splitting the data into $\bar{\mathcal{G}}$ separate sub-samples, where each subsample contains only units from cohort g or never-treated units: $\mathcal{S}_g$.

In the case of a binary treatment, one could then estimate a standard DiDCF within each subsample, yielding consistent estimates of the cohort $\times$ time $CATT$ with a corresponding cohort $\times$ time variance. However, in the case of a continuous treatment, this approach would fail to identify the $CATT(g, t, w, x)$ or the $CACRT(g, t, w, x)$ targets of Equations 3 and 4. The reason is because DiDCF calls the standard GRF algorithm within each subsample $\mathcal{S}_g$, which estimates conditional average partial effect parameters for continuous treatments. Conditional average partial effects are akin to the two-way fixed effects estimator which Callaway et al. (2025) decompose into weighted sums of the $CATT$ and $CACRT$ parameters *plus* possible selection bias terms, where the weights generally do not match the true dose distribution.

The selection bias is easiest to see with two dose groups. Take two groups of districts in cohort g and covariate cell x, one with dose h and one with dose l, both positive. A difference-in-differences comparison of the two gives

$$\begin{aligned}
&\mathbb{E}[\Delta Y \mid W_i = h, G_i = g, X_i = x] - \mathbb{E}[\Delta Y \mid W_i = l, G_i = g, X_i = x] \\
&\qquad = \underbrace{CATT(g, t, h, x \mid g, h, x)}_{\text{level effect of dose } h} - \underbrace{CATT(g, t, l, x \mid g, l, x)}_{\text{level effect of dose } l} \\
&\qquad = \underbrace{\mathbb{E}[Y_{i,t}(g, h) - Y_{i,t}(g, l) \mid W_i = h, G_i = g, X_i = x]}_{\text{causal response}} \\
&\qquad\quad + \underbrace{\Big(CATT(g, t, l, x \mid g, h, x) - CATT(g, t, l, x \mid g, l, x)\Big)}_{\text{selection bias}}
\end{aligned} \tag{17}$$

In order to restructure the DiDCF estimand to return $CATT/CACRT$ parameters, notice that within group g, we can construct $\Delta Y_{i,g,t}$ in terms of Equation 13:

$$\begin{aligned}
\Delta Y_{i,g,t} &= Y_{i,g,t} - Y_{i,g,b_g} \\
&= (\lambda_t - \lambda_{b_g}) + \mathbb{1}\{W_i > 0\}\big(\alpha(X_i) + \tau(X_i)W_i\big) + \eta_{i,g,t}
\end{aligned} \tag{18}$$

Recall that the $CATT$ for any positive dose w is given as $CATT(w, g, t, x) = \mathbb{E}[\Delta Y_{i,g,t}|W = w, G = g, X = x] - \mathbb{E}[\Delta Y_{i,g,t}|W = 0, G = g, X = x]$. Estimating the baseline conditional mean function within $\mathcal{S}_g$, denoted

$$m_{g,t}(X_i) = \mathbb{E}[\Delta Y_{i,g,t}|W = 0, G = g, X = x] \tag{19}$$

can be done in a number of ways (in the CS-DiD we do this via an OLS regression). Here, we estimate this function with a standard regression forest on the never-treated districts with 1,000 trees, then define the partially linear model to be estimated only among treated districts:

$$\Delta\widetilde{Y}_{i,g,t} = \alpha(x) + \tau(x)\, w + \varepsilon_{i,g,t} \tag{20}$$

Where $\Delta\widetilde{Y}_{i,g,t} = \Delta Y_{i,g,t} - \hat{m}_{g,t}(X_i)$. Equation 20 can now be estimated with the staggered DiDCF procedure outlined above and will produce consistent estimates of the $CATT(w, g, t, x) = \hat{\alpha}(X_i) + \hat{\tau}(X_i)W_i$ and (under Assumption 6) the $CACRT(w, g, t, x) = \hat{\tau}(X_i)$.

Before formally stating the consistency result, we introduce the regularity assumptions needed to invoke the proof of Proposition 1 in Gavrilova et al. (2025):

Assumption 7 (Regularity)**.** For each cohort $g \in \bar{\mathcal{G}}$ and event time $h \geq 1$, within the cohort-g subsample $\mathcal{S}_g = \{i : G_i = g\} \cup \{i : G_i = \infty\}$, the covariates X_i are continuously distributed in the unit hypercube $(0, 1)^d$, and the differenced outcome $\Delta\widetilde{Y}_{i,g,t} = Y_{i,t} - Y_{i,b_g}$ satisfies the following conditions. (a) *Joint smoothness:* the conditional moments $\mathbb{E}[\Delta\widetilde{Y}^k_{i,g,t} \mid X_i = x, W_i = w]$ are Lipschitz continuous jointly in (x, w) for $k = 1, 2$. (b) *Non-degenerate conditional variance:* $\mathrm{Var}[\Delta\widetilde{Y}_{i,g,t} \mid X_i] > 0$ uniformly over $\mathrm{supp}(X_i)$. (c) *Bounded moments:* higher-order moments of $\Delta\widetilde{Y}_{i,g,t}$ are bounded. (d) *Independence:* observations are independent across districts. Conditions (b)–(d) correspond to Assumption 3 of Gavrilova et al. (2025); condition (a) strengthens their Lipschitz requirement from X_i alone to (X_i, W_i) jointly, as required for consistency of the continuous-treatment causal forest (Nie and Wager, 2021; Athey et al., 2019).

Lemma 3 (Consistency of the Two-Stage DiDCF)**.** Let $\hat{\alpha}(x)$ and $\hat{\tau}(x)$ be obtained from Equations 18–20. Under Assumptions 1 – 5 and 7, for each $g \in \bar{\mathcal{G}}$, $t \geq g$, $w \in \mathcal{W}_+$, and $x \in \mathcal{X}$:

$$\widehat{CATT}(g, t, w, x) = \hat{\alpha}(x) + \hat{\tau}(x)\, w \tag{21}$$

is consistent and asymptotically normal for the building block $CATT(g, t, w, x \mid g, w, x)$. If Assumptions 6 and 4 also hold, then $CATT(g, t, w, x \mid g, w, x) = CATT(g, t, w, x)$, and for

$w \in \mathcal{W}_+^c$ the slope $\hat{\tau}(x)$ is consistent and asymptotically normal for $CACRT(g,t,w,x)$:

$$\frac{\widehat{CACRT}(g,t,w,x) - CACRT(g,t,w,x)}{\sqrt{\text{var}(\widehat{CACRT}(g,t,w,x))}} \xrightarrow{d} \mathcal{N}(0,1) \quad \text{as } N \to \infty. \tag{22}$$

For proof, see Appendix D.

Aggregation. Both stages are fit separately for each cohort g and event time e, so each cohort keeps its own base period b_g and its own never-treated comparison group, and no response function is assumed common across cohorts. Evaluating the fitted functions at each treated district's own dose gives the district-level building block estimates

$$\widehat{CATT}(g,t,W_i,X_i) = \hat{\alpha}_{g,t}(X_i) + \hat{\tau}_{g,t}(X_i)\, W_i, \qquad \widehat{CACRT}(g,t,W_i,X_i) = \hat{\tau}_{g,t}(X_i)$$

To report the event-study parameters of Equations 7 and 8, we average these estimates within cohort and then across cohorts. Because Equation 7 averages over the dose distribution, the dose-specific weights $\phi^{dose,es}$ collapse to the share of treated districts belonging to cohort g among the cohorts observed at event time e, $\phi_g = \mathbb{P}(G_i = g \mid G_i + e \in [2,T],\ G_i \leq T)$, which we estimate by the relative frequency $\hat{\phi}_g = n_g / \sum_{g' \in \bar{\mathcal{G}}} n_{g'}$ following Callaway and Sant'Anna (2021):

$$\widehat{CATT}^{es}(e) = \sum_{g \in \bar{\mathcal{G}}} \hat{\phi}_g \widehat{CATT}_g(e), \qquad \widehat{CATT}_g(e) = \frac{1}{n_g} \sum_{i:G_i=g} \widehat{CATT}(g, b_g + e, W_i, X_i) \tag{23}$$

Treating the cohort-specific averages as independent, the Delta method gives

$$\text{Var}\left(\widehat{CATT}^{es}(e)\right) = \sum_{g \in \bar{\mathcal{G}}} \hat{\phi}_g^2 \, \text{Var}\left(\widehat{CATT}_g(e)\right) \tag{24}$$

We close with some implementation details. In practice, we estimate the following regression with the lm_forest() algorithm from the grf package:

$$\Delta Y_{i,g,t} = m_{g,t}(X_i) + \mathbb{1}\{W_i > 0\}\alpha_{g,t}(X_i) + \tau_{g,t}(X_i)W_i + \varepsilon_{i,g,t}, \qquad i \in \mathcal{S}_g$$

The baseline conditional mean function estimate, $\hat{m}_{g,t}(x)$, comes from an auxiliary regression forest on never-treated units estimated with 1,000 trees. This process provides more statistical power to the causal forests in the splitting process because using the full sample (rather than just treated units) allows us to overcome minimum node size constraints, and therefore allows for deeper trees to be grown. In our setting, this is helpful because each cohort $\times$ event-time cell has a relatively small number of treated units. This approach is equivalent to the 2-stage approach described above when the nuisance estimates of $\hat{m}_{g,t}(x)$ are constant within x groups

– a condition which is true in our setting.

Each causal forest is estimated with 1,500 trees. For each tree, we hold out half of the subsample for honest estimation, and require at least 5 observations in a resulting child node for a split to occur. We use the default rule-of-thumb for determining the number of variables considered at each split point: if k is the number of covariates, select the minimum of $\sqrt{k} + 20$ or k. When training each tree, units are weighted by enrollment share for population representativeness, and we perform clustering at the district level. The variance of $\hat{\tau}(x)$ is constructed using the bootstrap of little bags, which is accomplished by training sets of trees and comparing estimates within and across those sets (Athey et al., 2019).

3.4 Policy Learning via Optimal Decision Trees

A potentially useful exercise which applies our non-parametric estimates of heterogeneous treatment effects is to understand alternative treatment assignment regimes for targeting districts towards charter school expansion. For school districts considering the decision to expand charter schools or not, what would a simple decision rule rooted in our $CACRT$ estimates look like in order to provide guidance for that question? The question we pose is: how should a social planner optimally choose which districts to expand charter schools to maximize the post-treatment outcome defined as either AFGR, Math test scores, or ELA test scores.

Formally, we seek to estimate a decision rule, $\pi \in \Pi$, that maps an individual's observed covariates to a binary decision of treatment assignment. That is, $\pi : X_i \in \mathcal{X} \rightarrow \{0, 1\}$, where $\pi(X_i) = 1$ indicates that the district should marginally expand charter school intensity and $\pi(X_i) = 0$ indicates that the district should not. Kitagawa and Tetenov (2018) specify this optimization problem as one which finds the policy that minimizes empirical regret relative to the best policy, $R(\pi)$, defined as:

$$R(\pi) = \arg\max_{\pi' \in \Pi} \{V(\pi') : \pi' \in \Pi\} - V(\pi) \tag{25}$$

Where $V(\pi) = \mathbb{E}[Y_{i,t}\big(\pi(X_i)\big) - Y_{i,t}(0)]$ is the average impact of implementing policy π relative to treating no one. Kitagawa and Tetenov (2018) solve this problem with inverse propensity score weighting, but Athey and Wager (2021) have proposed writing Equation 25 in terms of the $CACRT$ (in their case the $CATE$) and estimate the policy by solving:

$$\hat{\pi} = \arg\max_{\pi \in \Pi} \left\{ \frac{1}{n} \sum_{i=1}^{n} \big(2\pi(X_i) - 1\big)\, \hat{\Gamma}_{i,t} \right\} \tag{26}$$

Where $\hat{\Gamma}_{i,t}$ is the doubly robust score obtained via the augmented inverse-probability weights

(AIPW) estimator. Chernozhukov et al. (2018) shows that AIPW scores are more efficient when aggregating $CATE$ (or $CACRT$ in our case) estimates to higher-level averages. For instance, if we wanted to estimate the $ACRT$ of charter expansion via the causal forest estimators, doing so by averaging AIPW scores of the $CACRT$s (i.e. via $\widehat{ACRT}(t) = \frac{1}{n}\sum_i^n \hat{\Gamma}_{i,t}$) is more efficient than simply averaging over $CACRT$ estimates. These scores can be constructed as follows (Nie and Wager, 2021):

$$\hat{\Gamma}_{i,t} = \widehat{CACRT}(t, X_i) + \frac{W_{i,t} - \hat{p}(X_i)}{\hat{p}(X_i)(1 - \hat{p}(X_i))}[Y_{i,t} - \hat{\mu}(X_i) - (W_{i,t} - \hat{p}(X_i))\hat{\tau}_t(X_i)] \tag{27}$$

Where $\hat{\mu}(X_i) = \mathbb{E}[Y_{i,t}|X_i = x]$ and $\hat{p}(X_i) = \mathbb{E}[W_{i,t}|X_i = x]$ come from the auxiliary forests estimated during the causal forest training process.

It is important to note that the solution to Equation 26 is the optimal charter expansion policy under an extremely strict set of assumptions. Namely, the solution to Equation 26 assumes that the cost to expand charter schools is equal across all districts and charter intensities. It also assumes that the social welfare function is Utilitarian and perfectly proxied by a single outcome. For this reason, we use the solution policies only to better understand the $CACRT$ function estimates from the causal forests, and do not interpret them as the true optimal treatment allocation policies.

Equation 26 can be solved easily (though inefficiently) by performing an exhaustive search over a limited class of possible policies. Athey and Wager (2021) show that doing this will yield the regret-minimizing policy within that limited class. Here, we search over the space of policies representing depth-2 and depth-3 decision trees. We compute doubly robust scores per cohort $\times$ calendar-time cell from the grf R package. For each treated district, we average scores across all post-treatment event times to obtain a single welfare score $\bar{\Gamma}_i$. We restrict the covariate matrix X_i to the ten moderators with the highest variable importance at $e = 0$. The policytree package (Sverdrup et al., 2020) performs the search to solve Equation 26.

4 Results

We focus this section on the test score outcomes–Math and ELA–because of the measurement issues with graduation mentioned in Section 2 and the fact that graduation standards are more subjective and controlled by schools themselves. But we do include analogous analysis for graduation rates in Appendix Section B.

4.1 Replicating the Average Treatment Effects

Table 1 reports descriptive statistics for the test-score sample, separately for districts that are ever treated by charter school expansion and those that are never treated. Ever-treated districts differ substantially from never-treated districts along several dimensions. They have lower average math and ELA scores, smaller shares of White students, larger shares of Black, Hispanic, and low-income students, and substantially larger enrollments. They are also much more likely to be urban and much less likely to be rural. In contrast, average per-pupil spending and revenue are relatively similar across the two groups. These differences reinforce the importance of accounting for observed district characteristics and fixed unobserved differences across districts in the treatment-effect analysis.

We next estimate average treatment effects of charter-school expansion using the CS-DiD. Table 2 reports both the average treatment effect on the treated (ATT) and the average causal response to treatment ($ACRT$) for math and ELA achievement. Table 2 reports two versions of the CS-DiD estimator. Both are built from the 2×2 comparisons in the continuous-treatment framework of Callaway et al. (2025). This estimates effects separately within cohort-by-period cells, uses never-treated districts as the comparison group, and aggregates the cell-level estimates using cohort-size weights. Columns (1) and (3) exclude district controls, while Columns (2) and (4) include them. All specifications use the one-year-lagged charter enrollment share as the treatment intensity.

Panel A takes the long difference in outcomes and regresses it on a treatment indicator, charter enrollment share, and the covariates. The covariates enter only additively, with one set of coefficients shared by treated and comparison districts. This approach is precise, but it adjusts for covariates in the way that Sant'Anna and Zhao (2020) and Caetano and Callaway (2024) show does not generally recover the ATT under conditional parallel trends. If treatment effects vary with the covariates, or if covariates relate differently to outcome trends in the two groups, the coefficients on the treatment terms become variance-weighted averages of heterogeneous effects. They are then biased for the ATT and ACRT, and the bias grows with the strength of the heterogeneity that motivates our analysis. Panel B instead uses the approach proposed by Callaway et al. (2025), which is related to the imputation estimators of Wooldridge (2021) and Borusyak, Jaravel, and Spiess (2024). Within each cell, we estimate the conditional mean of the untreated outcome change using never-treated districts only, then subtract this prediction from each treated district's outcome change. The resulting difference is then regressed on the covariates, charter share, and their interactions, using treated districts only.

Because comparison districts alone identify the counterfactual trend, Panel B remains consistent under conditional parallel trends when effects vary with the covariates. Because Panel B rests on weaker assumptions about how covariates relate to effects and trends, we treat it as our

preferred estimate. This does come as the cost of precision, but this is less of an issue here with our very large sample.

As in Chen and Harris (2023), our preferred estimates (Panel B) suggest that charter schools have positive market-level effects. The ATT and ACRT are both always positive and, in three of the eight cases, precisely estimated. However, the results are reversed in Panel A. Such a sharp change in results, as noted above, is most likely to arise from bias caused by effect heterogeneity. This further motivates the rest of our paper.

4.2 Distribution of Effects

4.2.1 Overall Distribution Across Geographic Districts

The above estimates of the ATT and ACRT may mask substantial effect heterogeneity and that is the central focus of our study. For the heterogeneity analysis, we use the full-sample CS-DiD framework and report specifications both with and without district controls. For the heterogeneity results, we place emphasis on the DiDCF as our preferred set of estimates and provide results from the linear CS-DiD primarily as a point of reference.

The first set of results regarding market-level effect heterogeneity are the distributions of effects for each school district. We estimate and plot $CATTs$ for each geographic district $\times$ event-time with the DiDCF (Figures 1 and 2 for math and ELA, and B2 for AFGR). These distributions aim to answer the first primary questions of our analysis: is there notable heterogeneity in charter school effects?

For test scores, the effect distributions point to meaningful heterogeneity. The range of math score $CATT$ estimates is from about [-0.3, 0.3], with the distribution appearing Gaussian centered near zero (the $ATT \approx 0.001$ according to DiDCF). These results suggest that more than half of district $\times$ event-time estimates are negative, with the most negatively impacted districts losing about 0.03 s.d. in math performance for every 10pp in additional charter intensity they adopted. However, about 45% of district $\times$ event-time estimates were positive. The largest of which experienced a 0.03 s.d. increase to math scores for the same 10pp increase in charter intensity. The results are nearly identical for ELA scores: the distribution again ranges from about [-0.28, 0.3] with an $ATT = 0.014$.

4.2.2 Effect Heterogeneity by State

Districts are nested within states, which is also the institutional unit where most charter policies are made. Here we estimate equations (1) and (2) where the interaction term H_{is} is just a vector of state indicators. We also restrict to states that have at least five geographic districts contributing to the estimation.

The other study that has estimated state-specific effects is CREDO (2023). These are shown in the last column. Also, at the bottom of each column we report the simple correlation between our estimate and the CREDO estimate, converted to the same scale as ours. We do not expect our estimates to be perfectly correlated for many reasons: (a) our time period does not match theirs; (b) we are identifying effects only from changes in market share, omitting some that were opened at baseline, while CREDO's strategy incorporates the effects of essentially all charter schools; and (c) they are estimating participant effects only and we are capturing the market-level effect, which includes the participant effect but captures a broader array of mechanisms. So, we would expect a positive correlation, but one that might not be close to unity. This is what we see. The correlations range from +0.34 to +0.58 for the first two samples that focus on the overall average treatment effect. The correlations are predictably lower for Sample 3, which focus only on the subset of districts that had zero charter schools at baseline. These estimates, combined with the tests for identification threats in Chen and Harris (2023) reinforce that the effect heterogeneity patterns in our study are valid.

4.3 Analyzing Effect Heterogeneity by District Traits

Most of the rest of the study focuses on explaining the variation across districts and states, using both the CS-DiD and DiDCF methods, with priorty placed on the DiDCF. The factors are broken into four categories: student demographics (e.g., race and district size), market context (e.g., TPS baseline outcomes and teacher supply metrics), school finance (e.g., charter spending, TPS-charter funding gaps, and total average spending), and other charter school policies. We organize the analysis topic by topic, so that we can directly compare the results across methods and summarize the results for each set. We also generally focus our attention on the $CATT$ (except for the policy trees where the $CACRT$ is particularly relevant) because Assumption 5 is more easily satisfied than Assumption 6, and the $CATT$ is therefore more likely to be identified.

4.3.1 Our Framework for Interpreting Results

Interpreting heterogeneous treatment effect estimates in the presence of (possibly) complex nonlinearity is a challenging task. The individualized $CATT$ estimates produced by the DiDCF implicitly incorporate any number of possible higher-order polynomials and/or covariate interactions. We therefore follow a formulaic approach to understanding the $CATT$ functionals: (1) establish variable importance, (2) investigate linear patterns in $CATT$ estimates within important individual covariates, (3) investigate nonlinear patterns within quantiles of important individual covariates and across important covariate interactions. Our interpretation framework focuses on the DiDCF, and the CS-DiD estimates from the linear parametric model (Table **??**) are discussed

in parallel as a point of comparison.

For the first step, establishing variable importance, we rely on the "Variable Importance Factor (VIF)" produced by causal forests within the DiDCF. The VIF is a simple measure of variable importance and is calculated by computing the share of total trees within each forest that split along a given variable, weighted by the depth at which the split occurred so that earlier splits are weighted more heavily. With a large enough sample and forests with a sufficient number of trees, these scores roughly indicate which covariates drive heterogeneity in some fashion, as the forests will asymptotically distinguish between moderators driving the heterogeneity versus those that are simply correlated with the drivers. This is a more reliable indicator of importance than simply considering the statistical significance of CS-DiD coefficients, because CS-DiD coefficients test only for significant *linear* relationships and could therefore miss important nonlinear relationships. In our framework, we focus exclusively on variables that score in the top 10 of VIF. We treat every variable in this list as important in some manner, whether it be linearly or nonlinearly. Because the DiDCF estimates separate forests across event-times, we report these VIF scores in matrix form as Figures 3 and 4 (B3 for AFGR). This allows us to evaluate which variables are important for determining short-run vs. long-run impacts.

The second step of our interpretation framework seeks to establish a "general direction" of the heterogeneity within each important moderator. That is, we look for evidence that a high-VIF moderator exhibits a linear relationship with the $CATT$. We specifically consider two different statistics: (1) Tables 4 and 5 (B5 for AFGR) which report differences in the mean values of each moderator for districts with positive $CATT$ estimates vs. districts with negative $CATT$ estimates, and (2) Tables 6 and 7 (B6 for AFGR) report the Best Linear Predictors (BLP) of the $CATT$ for each outcome among the top 10 highest VIF moderators for DiDCF. The difference-in-means values examine the "profile" of an average positively/negatively impacted district $\times$ event-time. This provides some additional insight into the relative linearity of the relationship between a particular moderator and the $CATT$ estimate. For example, if the mean positively-impacted district for a particular (high VIF) moderator is approximately equal to the mean of the average negatively-impacted district, this is evidence for a non-linear relationship. Conversely, if the two means are notably different, this provides evidence that there is a level difference in moderator values across positively vs. negatively impacted districts. The coefficients in the BLP tables represent the linear relationship between each moderator and the $CATT$ and are obtained by regressing the moderators on of the $CATT$ estimate itself. A significant coefficient provides additional evidence that a linear approximation of the relationship between a particular moderator and the $CATT$ carries useful information about how the $CATT$ changes with that moderator.

The third and final step in our interpretation framework seeks to establish nonlinear relationships between high-VIF moderators and the $CATT$, particularly among the high-VIF moderators

that failed to exhibit any clear linear patterns. To do this, we consider an additional three sets of results:

1. First, we report event studies for each outcome in Figures 5 and 6 for DiDCF, and A1 for CS-DiD (we also report these for $CACRT$ using the DiDCF in Figures C11 and C12). For DiDCF, these event studies segment units by their moderator value (for the top 5 highest VIF moderators at event-time 0) such that districts falling in each quartile are plotted separately. These event-studies allow us to simultaneously assess how each moderator drives heterogeneity across its own distribution in both an overall sense and dynamically. This can provide direct evidence of non-linearities in each moderator's relation to the $CATT$. The event-studies also allow us to provide evidence towards the validity of the conditional parallel trends assumption (particularly Assumption 5) by estimating $CATT$s in pre-treatment periods via a placebo indicator for not-yet-treated districts. The CS-DiD finds no evidence of diverging trends in pre-treatment periods across ever-treated vs. never-treated units for any outcome. The DiDCF also does not generally detect diverging trends in the pre-period for any outcome, but hints that they may be present (if we had more statistical power) for AFGR and possibly ELA scores.[25][26] We provide a deeper discussion of the consequences to our heterogeneity inferences in the event of linear trend divergences in Appendix A. The key insight is that regardless of the presence of district-specific linear time trends, the heterogeneity results remain valid under a plausible set of conditions which appear to be satisfied in our setting.

2. Second, we report shallow-depth decision trees from the policy learning exercise of Athey and Wager (2021) discussed in Section 3.4 in Figures 7 – 10 (B7 and B8 for AFGR). These trees represent the optimal treatment assignment rule among the class of depth-2 (or depth-3) rules to maximize the expected outcome in question. These assignment rules are highly informative of the interactive relationships between moderators within the $CACRT$ function estimate. However, they are not intended to represent the true optimal treatment assignment policy.[27]

3. Finally, in order to further explore non-linearities in treatment effects, we group districts

[25] Because these pre-treatment placebo estimates are computed via a binary indicator for "not-yet-treated," we can only estimate pre-treatment $\widehat{CATT}$s, not $\widehat{CACRT}$s ($CACRT$s cannot be estimated without variation in dose). We include these in the $CACRT$ event-studies, but these are notably the same estimates as in the $CATT$ event studies.

[26] We also report the results of an integrated moments test for the conditional parallel trends assumption proposed by Callaway and Sant'Anna (2021) in Table A1. We find no evidence that parallel trends conditional on covariates fails to hold in our data.

[27] These learned policies are welfare-maximizing under a very strict set of assumptions: the social welfare function must be perfectly proxied by the outcome in question, welfare weights must be Utilitarian, and charter expansion costs must be equivalent across units. We refer the reader to Kitagawa and Tetenov (2018) for a complete discussion of these assumptions.

into 10 subgroups which are defined via k-means matching on X, weighted by VIF. The objective is to identify neighborhoods of districts with similar qualities along the important heterogeneity dimensions, and evaluate the characteristics of districts that perform well vs. those which do not. Figures C10 plots separate event studies for each of these 10 groups, and Tables C8 and C9 (Table B7 for AFGR) report the covariate means within each group, sorted by ATT. In these tables, the best performing group of districts is Group 1 and the lowest performing group is Group 10. We view these tables primarily as adding context and nuance to the quartile event study figures. Therefore, we place them in the appendix.

The following subsections will follow this framework in order to distill the various figures and tables discussed above into broader conclusions of how each moderator group affects charter school impacts.

4.3.2 Trait Set #1: Demographics

Prior research has consistently found that charter schools are more effective for disadvantaged students, broadly defined (Cohodes and Parham, 2021). The theories around this have been based partly on the lower test scores these groups typically have or, relatedly, that these groups are served by schools that are more poorly performing. We consider six measures of demographics including district size (via Log Enrollment) and the share of students who are Black, Hispanic, White, on Free/Reduced Price Lunch, and in Special Education.

Math Test Scores

For Math scores, we find that demographics (or unobserved variables that are heavily correlated with them) are some of the most important drivers of charter school effects. All six demographic variables we tested land in the top 10 VIF scores: Free/Reduced Price Lunch (FRL) % (2), Black % (3), Log Enrollment % (5), Hispanic % (6), White % (8), and Special Ed % (9). In particular FRL % generally grows in importance over time when considering long-run effects.

Every demographic moderator has notable patterns of nonlinearity, but there are a few broad trends in impacts that appear from examining the diff-in-means and BLP tables. Generally, districts which benefited from charter schools had a higher share of FRL and/or Black students, a lower share of White students, and/or were smaller in size.

For FRL %, the diff-in-means suggests that positive $CATT$ districts had more low-income students, and the corresponding event study also suggests that districts in the top 25% of FRL were impacted more. However, the BLP did not detect a linear pattern, and the k-means profiles also show evidence that while districts with the largest $CATT$ estimates tended to have higher FRL shares, this moderator appears to be interactive. This interactivity is corroborated in the

policy trees, as FRL % appears as a critical decision variable in both the depth-2 and depth-3 trees. The policy trees seem to elaborate on the finding that better performing districts tend to, but don't always, have more FRL students. Specifically, districts with low FRL % should still expand charter schools if Total Spending is low *and* Black % is also low, or if Special Ed % is low. Meanwhile, if FRL share is high, it is sufficient to consider Baseline Math performance (expand if it is sufficiently low). This provides evidence that charter schools tend to be more effective in more demographically homogeneous areas: those with higher incomes + fewer minority students *or* those with lower incomes + more minority students or those with lower baseline performance.

For the ethnic variables (Black %, Hispanic %, and White %), the findings are similar to those of FRL %. The difference-in-means and k-means profiles agree that more Black students and/or fewer White students tend to be associated with positively-impacted districts. The event study for Black % also hints that the bottom 25% of Black share districts tended to have lower impacts than the rest of the population, but does not suggest an obvious linear pattern. The depth-3 policy tree elaborates on this: districts with a high Black share are positively impacted if they *also* have low spending and high enrollment. Districts with a low black share can be positively impacted by charter schools if they are simultaneously higher income (i.e. low FRL %) and low spending. While White % does not appear in the policy trees nor does it have an event study (not top 5 VIF), the BLP suggests that the $CATT$ declines with more White students.

Finally, regarding district size (Log Enrollment), the patterns suggest nonlinearity. There is some evidence that smaller districts may be more positively impacted by charter schools than larger districts. The diff-in-means, BLP, and event study all suggest this. However, the k-means profiles show that while the best performing districts were of smaller size more frequently, so were the worst performing districts. This suggests Log Enrollment is likely interactive in some way. This is corroborated by the depth-3 policy tree: low enrollment districts should expand only if they are also high spending districts with low student-teacher ratios. If the district is high enrollment, they may also benefit from expansion if spending is low and Black share is high.

To summarize, each method suggests both sides of the same coin: that positively impacted districts are more often those with more non-white, low-income students. However, these demographic indicators seem to be highly nonlinear in their relationships to the $CATT$. In particular, they are informative when interacted with measures of funding or baseline Math scores. High minority and/or low-income districts, when paired with low baseline test scores or low spending, are the most likely to benefit from charter school expansion. Simultaneously, high income and/or low minority districts can be positively impacted by charter schools as well, indicating that demographic homogeneity may be the environment most favorable to charter schools.

ELA Test Scores

For ELA scores, demographics remain among the most important drivers of heterogeneity. Each of the six demographic measures we considered again scored in the top 10 VIF: FRL % (2), Log Enrollment % (4), Black % (5), Hispanic % (6), White % (7), and Special Ed % (10).

In general, most of the conclusions we derived from the Math score outcome are also true for ELA scores. There is a broad trend that districts with more FRL students, more non-White students, and smaller total enrollment are positively impacted more frequently. However, like with Math scores, these trends are highly nonlinear and there remains evidence that both tails of the income distribution are served well by charter schools while the districts caught in the middle are often negatively impacted.

For FRL %, the diff-in-means and event study agree that districts with more FRL students often have more positive $CATT$ estimates. However, the k-means profiles suggest that while it is true that the most positively impacted groups of districts did have the highest FRL shares, some positively impacted groups also had low FRL shares. This is corroborated by the BLP which does not find a significant relationship between FRL and the $CATT$. Therefore, some degree of interactivity is to be expected regarding student income. Despite there being some interactivity evidence, neither policy tree found that the FRL interactions were informative enough to split on.

For the ethnic measures (Black %, Hispanic %, White %), the evidence is fairly clear that more non-White students is often associated with more positive $CATT$ estimates. Both the diff-in-means and BLP agree with this, and the k-means profiles suggest that the most positively impacted groups typically had the lowest shares of White students while the most negatively impacted groups had the highest share of White students. Like with Math scores, however, there is evidence that both tails of the White % distribution can be served well by charter schools. Several of the positively impacted k-means groups had a high White share, indicating there is (likely similar to Math) interactivity in the share of minority students in a district. Despite this interactivity likely being present, the policy trees again did not find it informative enough to split on. One may surmise that the nature of the interactivity is likely similar to that of Math scores, but in the case of ELA it reveals less information about the $CATT$ than measures of spending and/or Enrollment.

Lastly, for Log Enrollment, the same patterns we discussed for Math scores are largely present in ELA scores. There is a linear trend that districts smaller in size are more often positively impacted by charter schools, but further examination into the non-linear measures reveals this is only part of the story. The event study suggests that the bottom quartile of districts was usually more positively impacted, but that the difference wasn't particularly large or consistent across time periods. Meanwhile, the k-means profiles highlight significant interactivity despite the fact that the most positively impacted group of districts was the smallest in size. The depth-3 policy tree elaborates on this evidence: small districts should expand charter schools if they are

also high spending with low student-teacher ratios. Meanwhile large districts can still expand if spending is sufficiently low.

To summarize, positively impacted districts tend to be larger, lower income districts with higher levels of non-White students. However, as with Math scores, the story seems to be that both tails of the income distribution are served well by additional charter schools. Fortunately, these demographic findings are similar across the two test score outcomes, indicating there may not be a trade-off across outcomes.

4.3.3 Trait Set #2: Market Context

We included three types of market context variables: region type (urban, suburban, town, and rural), attractiveness to young teachers (proxied by if a university is located in the city), and baseline academic performance in traditional public schools.

Math Test Scores

Baseline Math ranks #1 in VIF (by a wide margin) as the most important predictor of short-run charter school impacts. Despite remaining at #1 for most periods, it does diminish somewhat when considering long-run impacts in favor of FRL % and Teacher Salary which both gain on it. No other market context variable carries notable information according to VIF.

The relationship between a district's baseline Math scores and its $CATT$ is fairly clear: charter schools are more effective in districts that are initially low performing. Nearly every measure we report points to this. The diff-in-means suggest that positive $CATT$ districts have notably lower baseline Math scores than negative $CATT$ districts, the event study suggests that the bottom quartile of performers at baseline are impacted more in nearly every period, the k-means profiles display a clear pattern where the most positively impacted groups are all lower performing at baseline, and the depth-2 policy tree recommends charter expansion to low performing districts with high FRL share.

ELA Test Scores

Baseline ELA ranks #1 again in VIF as the primary predictor of short-run charter school impacts. It generally remains important over time as well, though like with Math scores, it diminishes over time in importance. No other market context variable carries notable information according to VIF.

Like with Math scores, there is ample evidence that charter schools tend to be more effective in initially lower-performing districts. The diff-in-means, event study, and k-means profiles each suggest that the relationship between baseline ELA scores and the $CATT$ is linear in that lower initial performance is associated with larger $CATT$s. The policy trees do not split on baseline

ELA, indicating any interactive effects it has are either weak or uninformative.

Our takeaway from these findings is that much of the prior literature has improperly attributed the relationship between baseline performance and the $CATT$ to demographics which are highly correlated with baseline performance. As we have demonstrated, demographics appear to be extremely nonlinear and interactive with one another and with other moderators in their own relationships with the $CATT$. Meanwhile, baseline performance has cut through this correlation and captured the consistently highest VIF score with the clearest-cut effect heterogeneity. Charter schools appear to be uniquely successful in bringing low performing districts closer to the mean.

4.3.4 Trait Set #3: School Funding

We include several different measures of school finance including Total Revenue (per-pupil), Total Spending (per-pupil), Charter Spending (per-pupil), Teacher Salaries, TPS-Charter Spending Gap, and average Student-Teacher Ratio within the district.

Math Test Scores

School finance variables appear to drive heterogeneity in Math score effects less than demographics or market context, though they do appear to have some significance. Student-Teacher Ratio (4), Total Per-Pupil Spending (7), and the TPS-Charter Spending Gap (10) each score in the top 10 for VIF (and notably, Total Per-Pupil Revenue (11) and Teacher Salary (12) just miss the top 10 and have VIF scores of at least 0.04).

The linear measures we report suggest that districts more positively impacted by charter schools tend to spend more per-student and have lower Student-Teacher Ratios. For both moderators, the diff-in-means and k-means profiles suggest that the districts with the highest $CATT$ estimates are higher spenders. Though it is worth noting that the BLP does not find a significant linear relationship for either moderator and the k-means profile for Student-Teacher Ratio indicates some nonlinearity is likely present as the cluster with the highest Student-Teacher Ratio has a positive ATT. In fact, both the Student-Teacher Ratio and Total Spending Per-Pupil appear as key splitting variables in the depth-3 policy tree, highlighting their interactions are informative. Total Spending appears especially informative as an interaction. Districts with high spending should expand charter schools when combined with a low Student-Teacher Ratio and low Enrollment, or when combined with a high Student-Teacher Ratio and a low Special Ed share. Meanwhile districts with low Total Spending may still see charter school benefits if those districts also have higher incomes or have high Enrollment and a large share of Black students.

ELA Test Scores

School finance variables are about equally (or perhaps slightly more) important towards determining ELA score impacts as they are for Math. Three of our six school funding variables appear in the top 10 of VIF: Student-Teacher Ratio (3), Total Spending Per-Pupil (8), and Teacher Salary (9), where the TPS-Charter Spending Gap (11) and Total Revenue Per-Pupil (12) just miss the top 10.

While the overall relationships between the finance variables and the $CATT$ are similar for ELA as they are for Math, the variable interactions appear to be much more prominent for ELA than for Math. For Student-Teacher Ratio, the k-means profiles suggest that the most positively impacted district cluster had the lowest Student-Teacher Ratios, but otherwise there aren't clear linear patterns in the effects. The event study supports this, as the smallest quartile of Student-Teacher Ratios has the largest $CATT$ estimates, but otherwise does not suggest a trend. This is particularly evident in the policy trees, where Student-Teacher Ratio appears as a splitting variable in both. The takeaways are broadly similar: expand charter schools in districts with low Student-Teacher ratios with high salaries and/or high spending.

Like with Math scores, districts that spend more tend to have larger $CATT$ estimates. The diff-in-means and k-means profiles both suggest a pattern where positive-$CATT$ districts spend more than negative-$CATT$ districts. The depth-3 policy tree elaborates on this finding and suggests several interactions that spending may undertake with Student-Teacher Ratio (discussed above) and Log Enrollment. Generally, charter schools should be expanded in districts with high spending and low Student-Teacher Ratios, or in districts with low spending but with low enrollments. It is also worth explicitly mentioning that the most positively impacted districts have the highest salaries for teachers.

4.3.5 Trait Set #4: State Charter Policy

Math Test Scores

We considered six policy dimensions from the charter industry groups: (1) whether the state has an Equitable Funding agreement, (2, 3) if there are clear charter school startup/renewal policies, (4) if charter schools are automatically exempt from state/district regulations, (5) if there are caps on charter school growth within a district, and (6) if the state has performance-based contracts. These variables all have extremely low VIF scores, with only Expemtions from State/Local Regulations (VIF rank 13) scoring > 0.

Perhaps not coincidentally, these variables also reveal almost no information about $CATT$ estimates. The DiDCF diff-in-means and k-means tables suggest that positive $CATT$ districts are slightly more likely to be exempt from state/local regulations and are slightly less likely to have clear renewal policies, but otherwise there is not much else to glean from these figures. Because these variables have low VIFs, they do not appear in the BLP or quartile event studies,

and (unsurprisingly) do not show up in either of the learned policy trees.

ELA Test Scores

Like with Math scores, these variables score very low on VIF and are generally uninformative to the $CATT$. Exempt from State/District Regulations and Clear Renewal policies never score higher than 0.01 on VIF. They therefore do not appear in most of our tests and are not present in the policy trees. Like with Math scores, positive $CATT$ districts are slightly more likely to be located in districts exempt from regulations and/or unclear renewal policies.

4.3.6 Trait Set #5: Diminishing Returns

Another important heterogeneity question is estimating the dose-response functions themselves: $ATT(w)$ and $ACRT(w)$. In other words, one may be interested in understanding how the marginal impact of charter schools changes as we vary the charter intensity, holding constant all other covariates. For this type of heterogeneity, the CS-DiD and DiDCF are the wrong estimators. Recall from Lemma 3 that our DiDCF procedure specifies the $CATT(\cdot)$ as a partially linear model: $\hat{\alpha}(x)+\hat{\tau}(x)w$. This functional form is meant to flexibly model heterogeneity in treatment effects across covariates x but *within* dose levels, w. It is therefore locally linear within doses rather than a smooth estimator of dose-responses. However, one can still use data-driven nonparametric estimators to obtain estimates of $ATT(w)$ and $ACRT(w)$ by following the approach of Chen et al. (2025), which is the preferred approach of Callaway et al. (2025) as well. Their method involves regressing $\Delta\widetilde{Y}_{i,g,t}$ on flexible K-dimensional transformations of W, which take the form of cubic B-splines, $\psi^K(W)$, among treated units:

$$\Delta\widetilde{Y}_{i,g,t} = \psi^K(W)'\beta_K + \varepsilon_{i,g,t}$$

The nonparametric estimators of $ATT(w)$ and $ACRT(w)$ are then formed as:

$$\widehat{ATT}(w) = \Big(\psi^K(w)\Big)'\hat{\beta}_K, \quad \widehat{ACRT}(w) = \Big(\partial\psi^K(w)\Big)'\hat{\beta}_K$$

Where $\partial\psi^K(s) = [d\psi_{K1}(s)/ds, ..., d\psi_{KK}(s)/ds]$ and $\hat{\beta}_K$ is the K-dimensional vector of OLS estimates. We refer the reader to Chen et al. (2025) and Callaway et al. (2025) for further details on construction of the optimal sieve dimension, K, and construction of confidence bands.[28] We focus on the $\widehat{ACRT}(w)$ function which emphasizes how the marginal effect of additional charter schools changes as the dose increases. For test scores, these are displayed in Figure 11. We find

[28]It is worth mentioning that identification of the dose-response functions in this manner requires the assumptions undertaken by Callaway et al. (2025), one of which is an unconditional parallel trends assumption that is stronger than the conditional parallel trends assumption we invoke in our main analysis.

evidence for diminishing marginal returns in both outcomes. While the estimates are expectedly noisy at high dose levels due to lack of power, the marginal impacts of additional charter schools diminish to near-zero at around 50-60% charter share.

4.3.7 Comparison with CS-DiD

* Needs updated Table 3 first *

5 Conclusion

A common theme of the economics of education is that different educational units (teachers, schools, and so on) vary widely in their quality. A second common theme is that measurable traits in those units (teacher certification, school size, and so on) do not explain much if any of this variation. With more than one thousand school districts having at least one charter school, it is now possible to test whether that pattern holds for charter schools.

In some respects, we find the same pattern. First, do charter effects vary? Do some "bloom" and others "wilt"? Yes, as with teachers and schools, some of the effects, especially with respect to high school graduation are very large. We also see some, albeit less, variation in effects across states.

Second, can we explain that variation? Here, we see several noteworthy patterns. Districts with more low-income students benefit more from having charter schools. These results are not driven by the fact that we are controlling for so many factors at the same time in the main results (Table 4). They also hold when we estimate equation (1) with smaller blocks of variables.

Somewhat surprisingly, charter schools seem to improve geographic district outcomes when those districts were higher performing to begin with. This may be less surprising, however, when we note that some of the states with larger effects, especially Massachusetts (see Appendix Table C10), also have high baseline achievement levels; conversely, some of the states with smaller effects (Arkansas and South Carolina) have very low performance levels.

State policies may explain some of the variation in results across states. Charter policies that require less transparency in charter applications and renewal decisions are associated with more positive charter effects. One theory that could explain this is that authorizers, in states with stronger transparency, may be pressed to make their decisions more on the basis of public information, as opposed to private information that might have that is also indicative of future charter performance. Less surprising is that charter schools seem more effective, as least with respect to test scores, when state policies provide more funding to charter schools. However, as noted in section II, the role of charter funding is more complicated than the state-level policy ratings account for. We certainly expect the charter participant effects to be more positive when

charter schools have more funding but, for purposes of understanding the market-level effects, we also have to be concerned with what charter entry means for TPS in the same districts, which may influence the competitive responses. We are in the process of collecting data, at the district level, which will allow us to better understand these issues.

We see only limited evidence of diminishing marginal returns. The effects are larger for test scores when we focus only on the districts that have a first charter entry during the panel, which is suggestive of DMR. But this pattern no longer holds when we consider the full distribution of baseline charter entry. We also see little clear evidence that charter effects vary based on changes in total geographic district enrollment.

Another pattern throughout the paper is that the results differ between the high school outcomes and test outcomes. This is unsurprising for three reasons: (a) our test scores results pertain to elementary/middle schools while high school graduation pertains to a different set of schools—high schools; (b) the time span of the test score data differs from that of high school graduation, so the schools are also of different vintages; and (c) test scores and high school graduation likely have different education production functions—we do not expect schools to be equally effective in producing all outcomes. When we limit only to test scores, we see suggestive evidence that districts with declining total enrollment see more negative charter effects. We are also conducting analysis of the possibility that charter effects might display diminishing marginal returns, i.e., that the effect of charter schools is larger for initial entrants.

This work is still preliminary. In ongoing work, we are also: (a) applying shrinkage estimation techniques to understand how much of the variation is driven by measurement error; (b) collecting additional data about school types (e.g., CMO status) so we can test whether the effects are larger when certain types of schools enter; (c) collecting better data, at the district level, on charter school spending and teacher supply, so we can estimate the effects of these additional traits; and (d) carrying out additional analysis of the roles played by diminishing.

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Figures

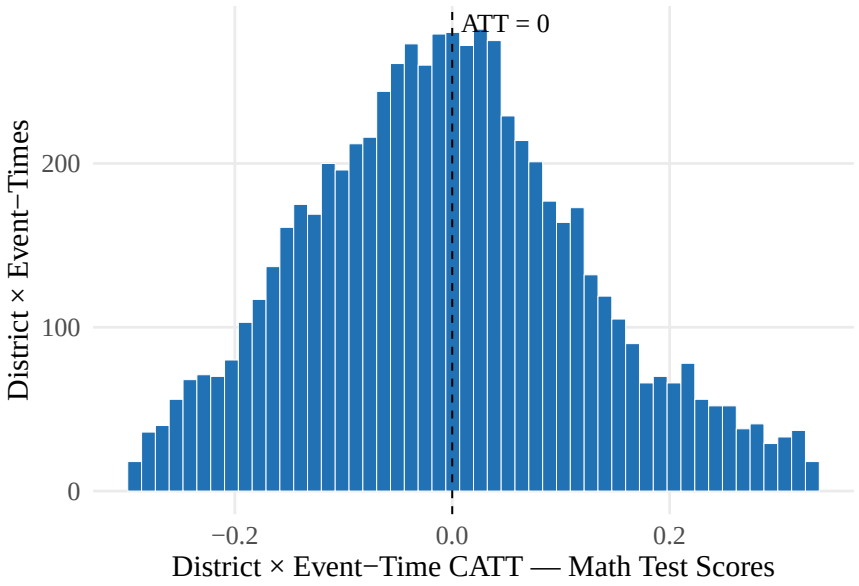


Figure 1. DiDCF CATT Distribution: Math Scores

Notes: Distribution of cohort × event-time Level-CATT estimates from the DiDCF applied to math test scores (Math). The vertical dashed line marks the ATT after fully aggregating across doses, event-times, and covariates. We drop the top and bottom 5% of CATT estimates from this figure.

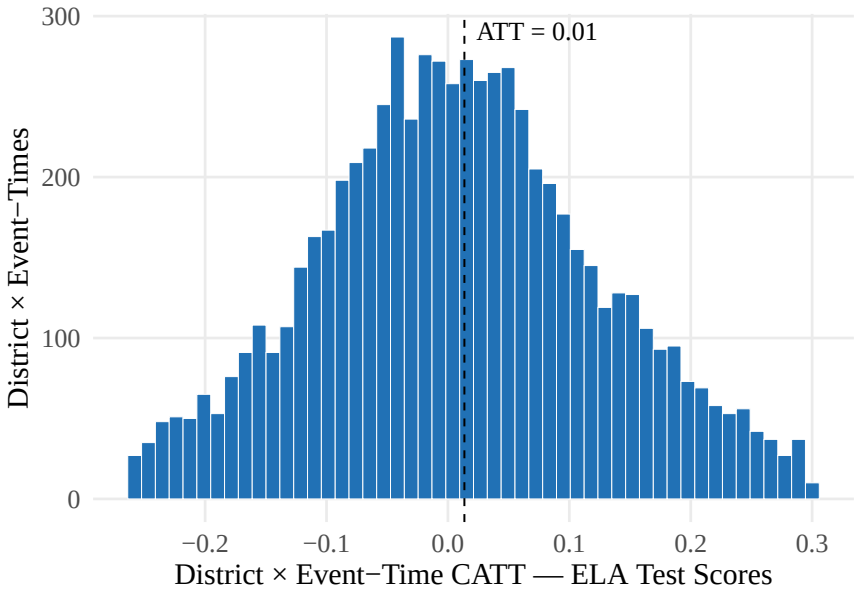


Figure 2. DiDCF CATT Distribution: ELA Scores

Notes: Distribution of cohort × event-time Level-CATT estimates from the DiDCF applied to ELA test scores (ELA). The vertical dashed line marks the ATT after fully aggregating across doses, event-times, and covariates. We drop the top and bottom 5% of CATT estimates from this figure.



Figure 3. DiDCF Variable Importance Factors: Math Scores

. *Notes*: Heatmap of variable importance factors (VIF) across all covariates and post-treatment event-times ($h = 1, \ldots, 7$) from the DiDCF applied to math test scores. Cell values show the (depth-weighted) share of splits on each covariate at each event-time. Grey cells indicate covariates absent from the data.

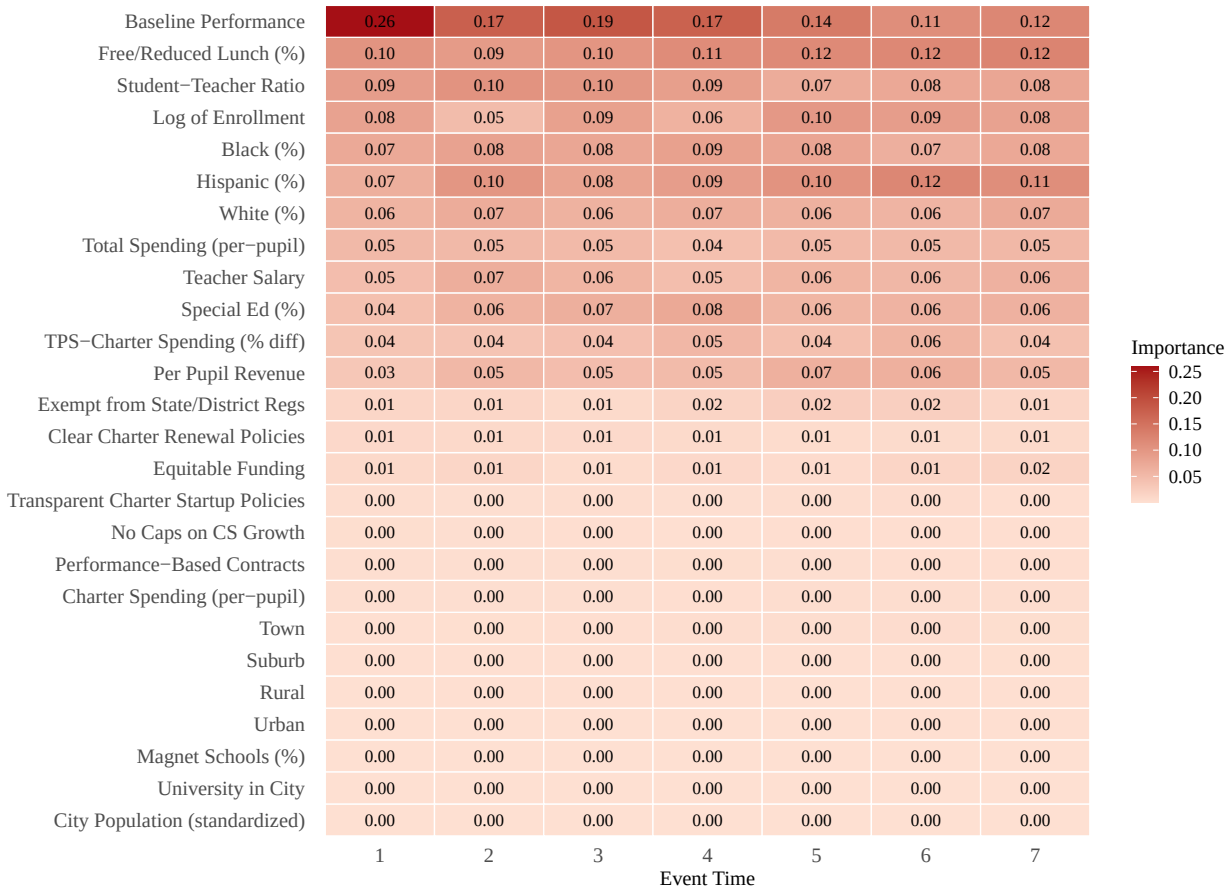
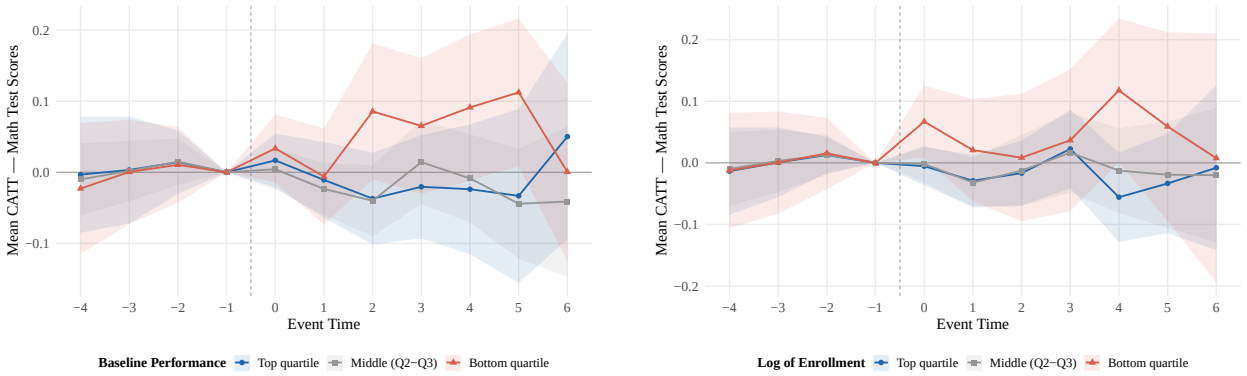


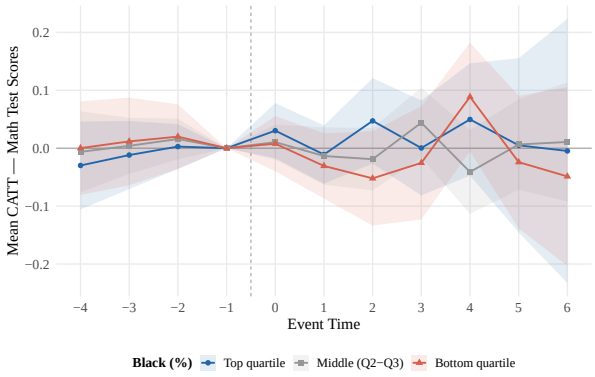
Figure 4. DiDCF Variable Importance Factors: ELA Scores

Notes: Heatmap of variable importance factors (VIF) across all covariates and post-treatment event-times ($h = 1, \ldots, 7$) from the DiDCF applied to ELA test scores. Cell values show the (depth-weighted) share of splits on each covariate at each event-time. Grey cells indicate covariates absent from the data.



(a) Baseline Math Scores

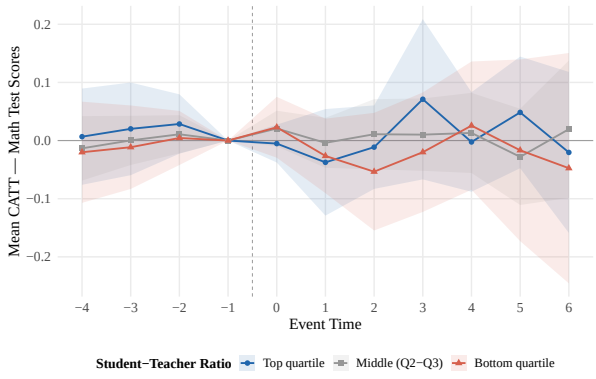
(b) Log Enrollment



(c) Black (%)



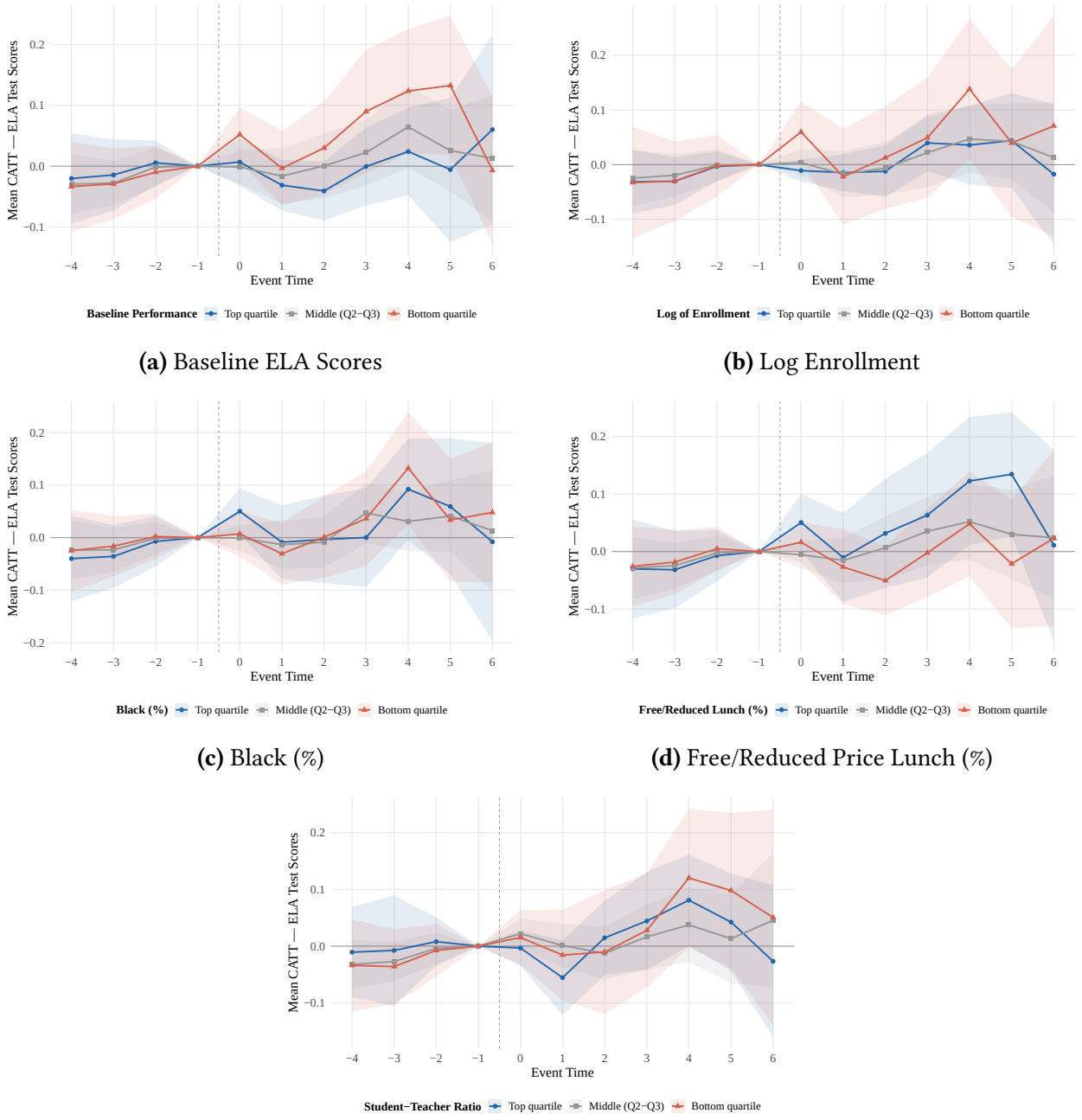
(d) Free/Reduced Price Lunch (%)



(e) Student-Teacher Ratio

Figure 5. DiDCF Quartile Event Studies (CATT): Math Scores

Notes: Each panel shows cohort-weighted ATT estimates by event-time, separately for districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF at $h = 1$ for math scores.



(a) Baseline ELA Scores

(b) Log Enrollment

(c) Black (%)

(d) Free/Reduced Price Lunch (%)

(e) Student-Teacher Ratio

Figure 6. DiDCF Quartile Event Studies (CATT): ELA Scores

Notes: Each panel shows cohort-weighted ATT estimates by event-time, separately for districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF at $h = 1$ for ELA scores.

Figures 7–10 display optimal policy trees estimated via the Athey and Wager (2021) framework applied to DiDCF doubly robust scores. Each outcome is shown at depth 2 (four terminal leaves) and depth 3 (eight terminal leaves). Blue leaves recommend expanding charter market share; red

leaves recommend against expansion. Internal nodes display the splitting variable and threshold; "Yes" (condition satisfied, left branch) and "No" (condition not satisfied, right branch) labels accompany each edge. Policy trees for the graduation rate outcome are in Appendix B as Figures B7 and B8.

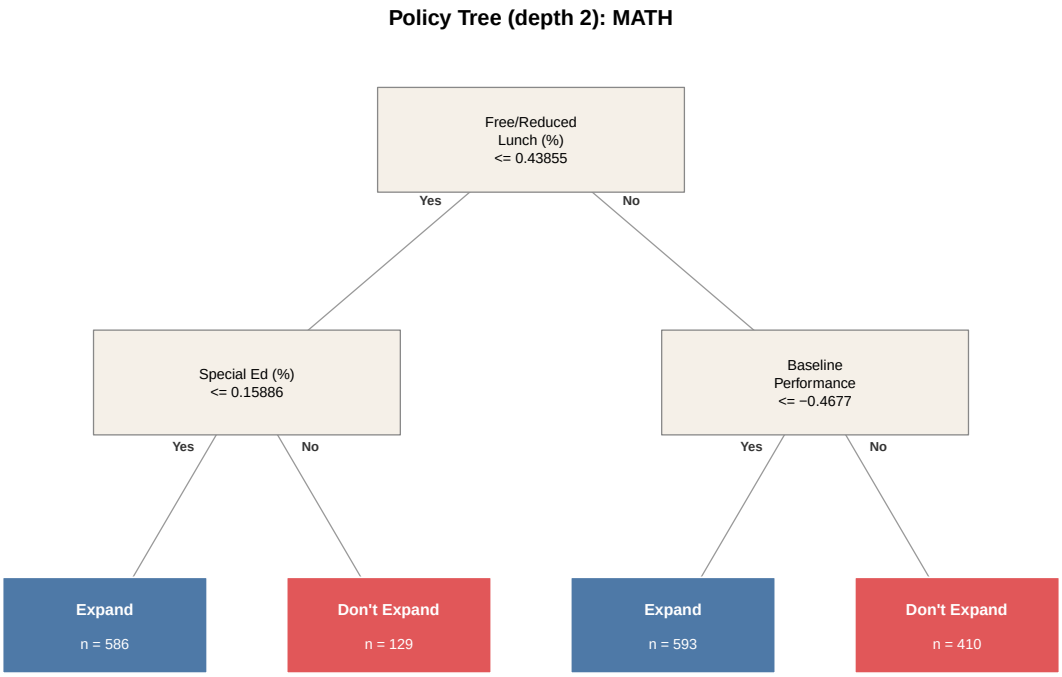


Figure 7. Optimal Policy Tree (Depth 2): Math Scores

Notes: Depth-2 optimal policy tree estimated via the Athey and Wager (2021) framework applied to DiDCF doubly robust scores for the math test scores outcome (Math). Each internal node shows the split variable and threshold; leaf colors indicate the recommended policy action (blue = expand charter market share; red = do not expand). The sample consists of treated districts with non-missing DR scores averaged across post-treatment event times, restricted to the ten highest-VIF moderators at $h = 1$. The doubly-robust scores in question refer to those associated with CACRT such that the policy prescription refers to the binary decision to "marginally increase" charter intensity or not.

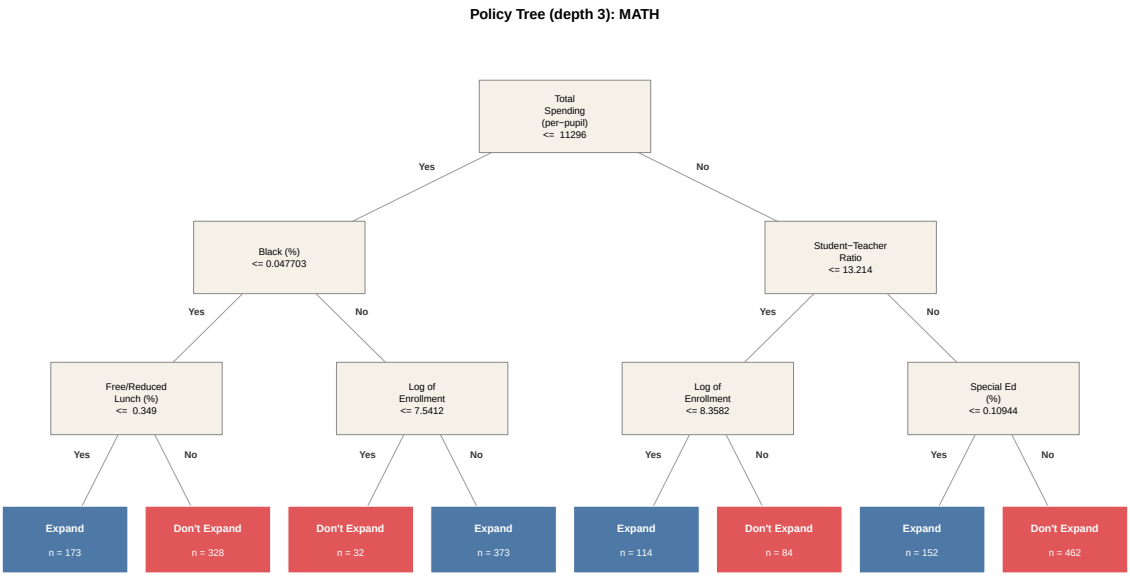


Figure 8. Optimal Policy Tree (Depth 3): Math Scores

Notes: Depth-3 optimal policy tree estimated via the Athey and Wager (2021) framework applied to DiDCF doubly robust scores for the math test scores outcome (Math). Each internal node shows the split variable and threshold; leaf colors indicate the recommended policy action (blue = expand charter market share; red = do not expand). The sample consists of treated districts with non-missing DR scores averaged across post-treatment event times, restricted to the ten highest-VIF moderators at $h = 1$. The doubly-robust scores in question refer to those associated with CACRT such that the policy prescription refers to the binary decision to “marginally increase" charter intensity or not.

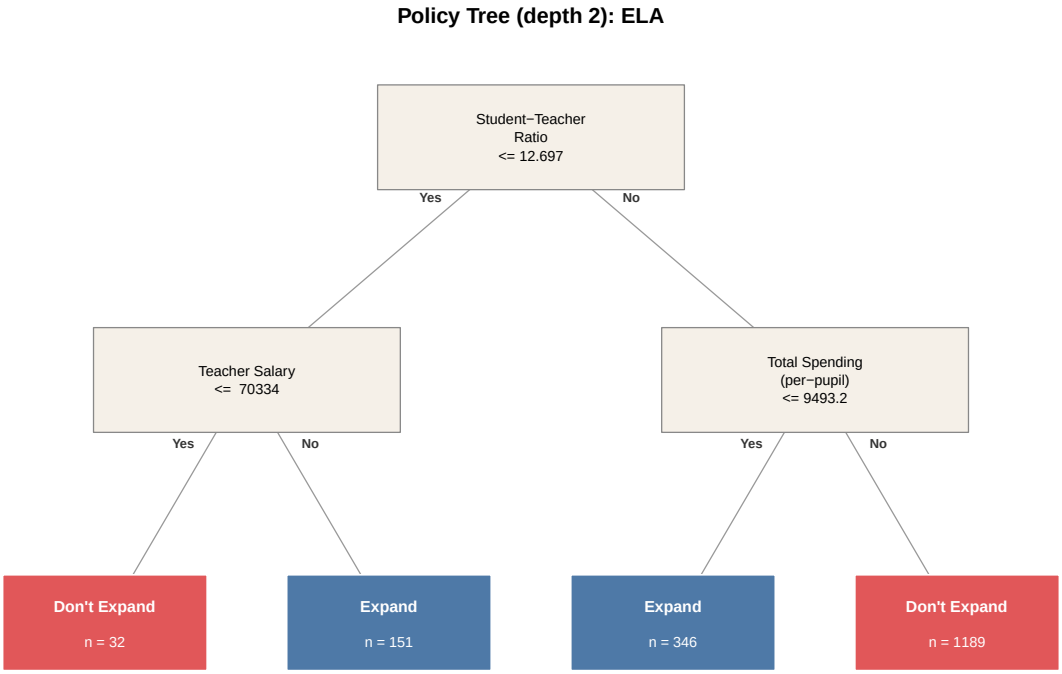


Figure 9. Optimal Policy Tree (Depth 2): ELA Scores

Notes: Depth-2 optimal policy tree estimated via the Athey and Wager (2021) framework applied to DiDCF doubly robust scores for the ELA test scores outcome (ELA). Each internal node shows the split variable and threshold; leaf colors indicate the recommended policy action (blue = expand charter market share; red = do not expand). The sample consists of treated districts with non-missing DR scores averaged across post-treatment event times, restricted to the ten highest-VIF moderators at $h = 1$. The doubly-robust scores in question refer to those associated with CACRT such that the policy prescription refers to the binary decision to “marginally increase" charter intensity or not.

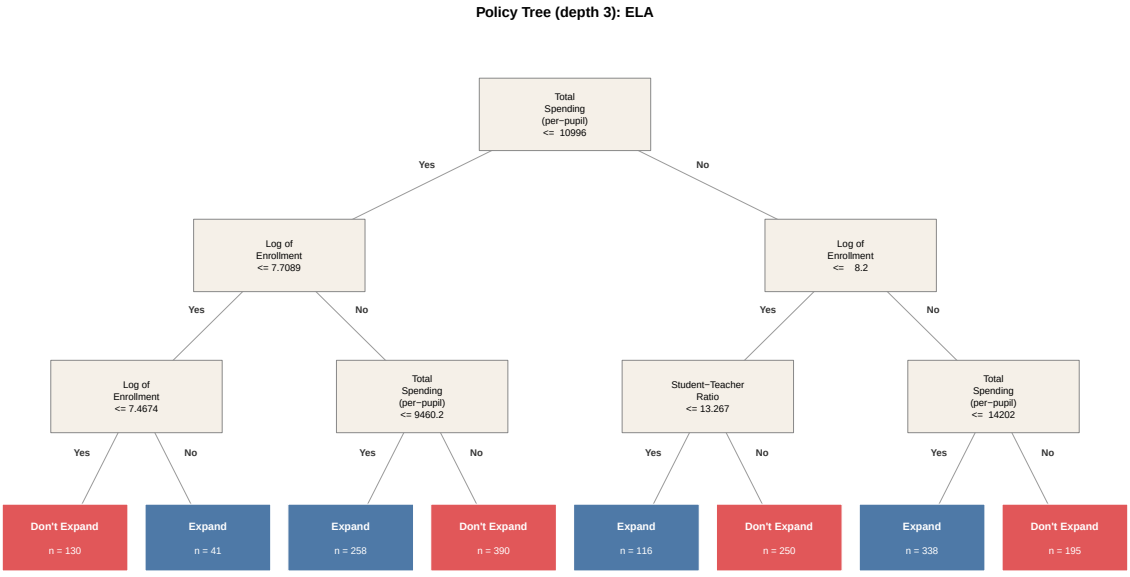
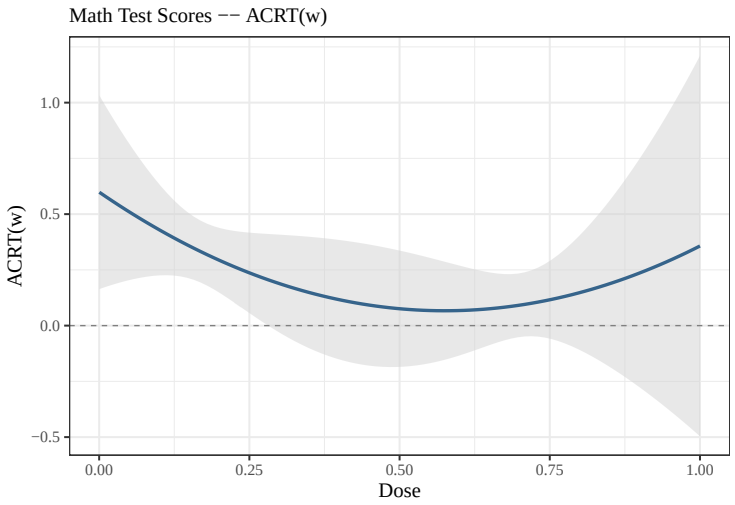
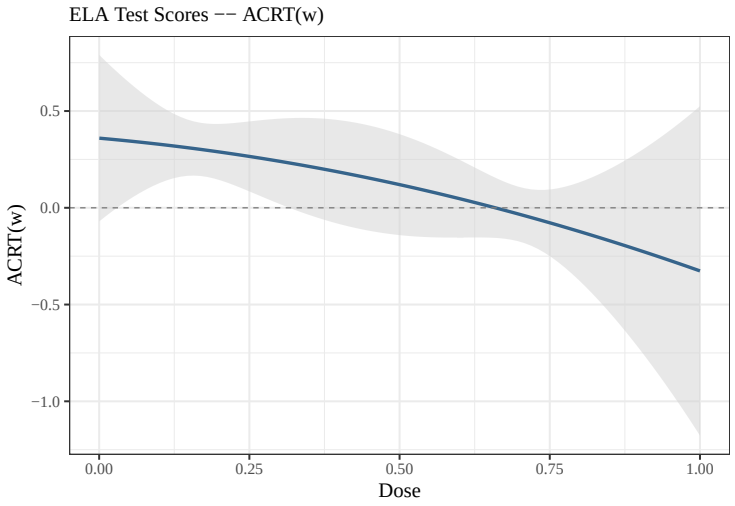


Figure 10. Optimal Policy Tree (Depth 3): ELA Scores

Notes: Depth-3 optimal policy tree estimated via the Athey and Wager (2021) framework applied to DiDCF doubly robust scores for the ELA test scores outcome (ELA). Each internal node shows the split variable and threshold; leaf colors indicate the recommended policy action (blue = expand charter market share; red = do not expand). The sample consists of treated districts with non-missing DR scores averaged across post-treatment event times, restricted to the ten highest-VIF moderators at $h = 1$. The doubly-robust scores in question refer to those associated with CACRT such that the policy prescription refers to the binary decision to “marginally increase" charter intensity or not.



(a) Math Achievement



(b) ELA Achievement

Figure 11. Diminishing Returns to additional Charter Intensity

Notes: This figure reports $ACRT(w)$ parameters across dose levels for both test score outcomes. These figures are constructed using the Chen et al. (2025) regression approach, where $\Delta\widetilde{Y}_{i,g,t}$ is regressed on cubic B-splines among treated districts. 95% confidence intervals are displayed. These figures are interpreted as the average marginal effect of additional charter schools evaluated at dose w.

Tables

Table 1. Descriptive Statistics for the Test-Score Sample

Variable	Ever-Treated	Never-Treated	Difference	Range
Panel A: Outcomes				
Math test score	-0.253	0.278	-0.531^{***}	(-10.328, 7.533)
ELA test score	-0.277	0.285	-0.563^{***}	(-7.424, 6.256)
Panel B: Student Demographics				
White student share	0.408	0.690	-0.281^{***}	(0.000, 1.000)
Black student share	0.238	0.112	0.126^{***}	(0.000, 1.000)
Hispanic student share	0.285	0.136	0.149^{***}	(0.000, 1.000)
Free or reduced-price lunch share	0.557	0.405	0.152^{***}	(0.000, 1.000)
Special education share	0.125	0.129	-0.005^{***}	(0.000, 0.568)
Panel C: Market Setting				
Urban district	0.483	0.110	0.373^{***}	(0.000, 1.000)
Suburban district	0.400	0.432	-0.032^{***}	(0.000, 1.000)
Town district	0.050	0.173	-0.124^{***}	(0.000, 1.000)
Rural district	0.068	0.285	-0.217^{***}	(0.000, 1.000)
City population (standardized)	-0.019	-0.009	-0.009^{***}	(-1.873, 3.129)
District enrollment size (log)	10.165	8.390	1.775^{***}	(2.639, 13.441)
Number of magnet schools	16.620	0.466	16.154^{***}	(0.000, 266.000)
University presence (standardized)	-0.072	-0.002	-0.070^{***}	(-0.572, 6.346)
Panel D: School Revenue and Spending				
Per-pupil spending ($1,000s)	12.3	12.2	0.1^{***}	(8.3, 23.2)
Per-pupil revenue ($1,000s)	12.0	12.1	-0.1^{***}	(8.5, 22.8)
TPS–charter spending gap	0.124	0.003	0.121^{***}	(-0.657, 1.927)
Equitable operational funding	0.538	0.463	0.076^{***}	(0.000, 1.000)
Per-pupil charter-school spending ($1,000s)	8.3	0.3	8.0^{***}	(0.0, 29.8)
Teacher salary ($1,000s)	99.7	94.7	5.1^{***}	(63.8, 139.6)
Student–teacher ratio	16.504	15.529	0.974^{***}	(10.162, 21.815)
Panel E: Charter Characteristics				
CMO share	0.080	0.001	0.078^{***}	(0.000, 1.000)
Panel F: Other State Charter-School Policies				
Regulatory exemptions	0.635	0.560	0.075^{***}	(0.000, 1.000)
No caps on charter-school growth	0.830	0.677	0.153^{***}	(0.000, 1.000)
Clear (non)renewal procedures	0.673	0.608	0.065^{***}	(0.000, 1.000)
Transparency	0.567	0.497	0.070^{***}	(0.000, 1.000)

Notes: Ever-treated districts have positive charter enrollment at some point in the sample; never-treated districts do not. The Difference column reports ever-treated minus never-treated means. Estimates are weighted by grade-level enrollment. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 2. Average Effects of Charter-School Expansion: CS-DiD

	Math		ELA	
	(1)	(2)	(3)	(4)
Panel A: CS-DiD, Full Sample				
ATT	0.000	0.001	0.008	0.009
	(0.006)	(0.006)	(0.006)	(0.006)
ACRT	0.044	−0.006	0.121**	0.052
	(0.041)	(0.035)	(0.054)	(0.042)
R^2	0.866	0.867	0.890	0.891
Observations	476,721	475,572	476,721	475,572
Panel B: CS-DiD, Last Three Years				
ATT	−0.000	−0.000	0.007	0.006
	(0.008)	(0.008)	(0.008)	(0.008)
ACRT	−0.029	−0.027	0.116**	0.118**
	(0.025)	(0.023)	(0.056)	(0.059)
R^2	0.912	0.912	0.916	0.916
Observations	112,469	112,296	112,469	112,296
Panel C: Staggered CS-DiD				
ATT	−0.067***	−0.062***	−0.047***	−0.044***
	(0.009)	(0.009)	(0.009)	(0.009)
ACRT	0.328***	0.291***	0.239***	0.199***
	(0.059)	(0.058)	(0.051)	(0.049)
Treated observations	11,240	11,240	11,240	11,240
District fixed effects	Yes	Yes	Yes	Yes
State-by-grade-by-year fixed effects	Yes	Yes	Yes	Yes
District controls	No	Yes	No	Yes

Notes: All three panels report the ATT and ACRT using charter enrollment share W. Panels A and B use the pooled parametric CS-DiD specification with $D = 1\{W > 0\}$, district fixed effects, and state-by-grade-by-year fixed effects. For positive treatment, the treatment component is $\beta_0 + \beta_1 W$, so the ATT is $\beta_0 + \beta_1 E[W \mid W > 0]$, while the ACRT is β_1. Panel A uses the full sample, while Panel B applies the final-three-year sample restriction. Panel C implements the staggered-adoption DiD separately by treatment cohort and post-treatment calendar year, using the period immediately preceding treatment as the cohort-specific base period and never-treated districts as the comparison group. Cohort–time estimates in Panel C are aggregated using treated-observation shares; the Panel C sample-count row reports treated cohort–time observations. All regressions are weighted by grade-level enrollment. Standard errors in Panels A and B are clustered at the district level. Panel C uses district-clustered standard errors within each cohort–time regression; the reported aggregate standard errors combine cell-specific variances but do not incorporate cross-cell covariance arising from reuse of never-treated districts. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 3. Heterogeneity in the Effects of Charter-School Expansion: CS-GDD

		Math	ELA
Moderator	Component	(1)	(2)
Baseline Treatment Effects			
Positive charter exposure	Level	−0.045	−0.219***
		(0.097)	(0.084)
Charter enrollment share	CACRT	−0.283	1.419
		(1.879)	(1.727)
Student Demographics			
White student share	Level	0.020	0.624**
		(0.300)	(0.284)
	CACRT	−0.767	−7.454**
		(3.265)	(3.304)
Black student share	Level	−0.101	0.291
		(0.314)	(0.301)
	CACRT	−2.332	−6.157*
		(3.426)	(3.339)
Hispanic student share	Level	0.043	0.327
		(0.297)	(0.297)
	CACRT	−1.145	−4.380
		(3.215)	(3.342)
Free or reduced-price lunch share	Level	0.071	0.173
		(0.118)	(0.134)
	CACRT	1.018	−0.788
		(1.255)	(1.247)
Special education share	Level	−0.529	−0.744
		(0.601)	(0.559)
	CACRT	2.294	6.733
		(5.212)	(4.754)
Market Setting			
Baseline outcome	Level	−0.046**	−0.049*
		(0.021)	(0.026)
	CACRT	0.048	0.050
		(0.134)	(0.160)
Urban district	Level	−0.084	0.005
		(0.057)	(0.055)
	CACRT	1.032*	−0.801
		(0.623)	(0.614)
Suburban district	Level	−0.202***	−0.050
		(0.057)	(0.053)
	CACRT	2.472***	0.060
		(0.534)	(0.493)
Town district	Level	−0.275***	−0.109**
		(0.057)	(0.053)
	CACRT	3.176***	1.140**
		(0.472)	(0.467)
Log enrollment	Level	0.051**	0.067***
		(0.022)	(0.022)
	CACRT	−1.055***	−1.026***
		(0.206)	(0.201)
Magnet schools	Level	0.013	−0.013
		(0.010)	(0.009)

Continued on next page

Table 3 continued

Moderator	Component	Math (1)	ELA (2)
	CACRT	−0.369 (0.292)	0.272 (0.275)
School Revenue and Spending			
Per-pupil spending ($1,000s)	Level	0.003 (0.012)	0.002 (0.011)
	CACRT	−0.022 (0.103)	−0.071 (0.093)
Per-pupil revenue ($1,000s)	Level	0.026* (0.014)	0.006 (0.014)
	CACRT	−0.043 (0.138)	−0.027 (0.140)
TPS–charter spending gap	Level	0.032 (0.123)	0.182* (0.110)
	CACRT	−3.336*** (0.872)	−2.170** (0.957)
Equitable operational funding	Level	−0.089 (0.129)	−0.063 (0.121)
	CACRT	1.308 (1.357)	−1.455 (1.272)
Charter-school spending ($1,000s)	Level	0.021** (0.010)	0.019** (0.010)
	CACRT	−0.240** (0.096)	−0.027 (0.082)
Teacher salary ($1,000s)	Level	−0.004** (0.002)	−0.000 (0.002)
	CACRT	0.051*** (0.016)	0.031* (0.018)
Student–teacher ratio	Level	0.021** (0.010)	−0.004 (0.009)
	CACRT	−0.055 (0.090)	0.143* (0.085)
Other State Charter-School Policies			
Regulatory exemptions	Level	−0.019 (0.102)	−0.083 (0.088)
	CACRT	0.191 (0.890)	0.320 (0.778)
No caps on charter-school growth	Level	−0.163 (0.132)	0.109 (0.116)
	CACRT	3.615** (1.466)	1.245 (1.292)
Clear renewal and nonrenewal procedures	Level	−0.047 (0.138)	−0.074 (0.126)
	CACRT	2.205 (1.477)	2.237* (1.359)
Transparency	Level	0.252** (0.128)	0.081 (0.115)
	CACRT	−2.242* (1.277)	−0.484 (1.189)
Performance-based contracts	Level	0.251 (0.178)	0.165 (0.148)

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Table 3 continued

Moderator	Component	Math (1)	ELA (2)
	CACRT	-4.237^{**} (1.984)	1.139 (1.882)
Missing-Value Indicators			
TPS–charter spending gap missing	Level	-1.450^{***} (0.426)	-1.746^{***} (0.388)
	CACRT	-1.232^{**} (0.611)	-0.929 (0.600)
Charter-school spending missing	Level	1.637^{***} (0.417)	1.905^{***} (0.385)
	CACRT	-1.632^{**} (0.745)	0.343 (0.710)
Treated observations		7,547	7,547

Notes: The table reports heterogeneity estimates from the staggered, continuous-dose CS-GDD, estimated separately in each cohort g by calendar-year t cell using charter enrollment share W and $D = 1\{W > 0\}$. Within a cell, the estimator is equivalent to the first-differenced regression $Y_t - Y_{g-1} = \gamma_0 + X'\gamma + D(\alpha_0 + X'\alpha) + W(\tau_0 + X'\tau) + \varepsilon$ on cohort-g and never-treated district-grades. It is estimated in two stages: the outcome change of never-treated district-grades is regressed on the moderators X, which removes the counterfactual trend $\gamma_0 + X'\gamma$ (not reported), and the residualized change of treated district-grades is then regressed on $(1, X, W, WX)$. Hence $CATT(w, X) = \alpha_0 + X'\alpha + w(\tau_0 + X'\tau)$ and $CACRT(X) = \tau_0 + X'\tau$. Reported coefficients average the cell estimates over post-treatment cells, weighted by each cell's number of treated observations. Moderators are centered at their grade-level-enrollment-weighted sample means, so the baseline rows report the level component α_0 and CACRT τ_0 at the mean moderator vector. For each moderator k, the Level row reports α_k, the coefficient on $D \times X_k$, which is the treatment-by-moderator interaction reported in a binary-treatment difference-in-differences; the CACRT row reports τ_k, the coefficient on $W \times X_k$. Because the dose is continuous, the gradient of the CATT with respect to X_k at dose w is $\alpha_k + \tau_k w$, so α_k is that gradient as $w \to 0^+$. An alternative summary evaluates the gradient at the mean treated dose, $\alpha_k + \tau_k \bar{w}$; we do not report it. Urban, suburban, and town coefficients are interpreted relative to rural districts. Moderators are measured in the first observed year for each district-grade, except the baseline outcome, which is Y_{g-2} centered within each cell. State policy measures are rescaled to $[0, 1]$ before centering. Missing moderator values are set to the sample mean and flagged by the missing-value indicators. Per-pupil spending, per-pupil revenue, charter-school spending, and teacher salary are expressed in thousands of dollars. Both stages are weighted by grade-level enrollment. Standard errors are clustered at the district level and account for estimation of the first stage. Standard errors are shown in parentheses below point estimates. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 4. DiDCF Differences-in-Means: Math Scores

Variable	Mean (pos. CATT)	Mean (neg. CATT)	Difference	p-value
Baseline Performance	-0.31	-0.05	-0.26	0.00
Free/Reduced Lunch (%)	0.53	0.46	0.06	0.00
Black (%)	0.20	0.16	0.04	0.00
Student-Teacher Ratio	15.97	16.36	-0.40	0.01
Log of Enrollment	8.43	8.65	-0.22	0.00
Hispanic (%)	0.21	0.18	0.03	0.02
Total Spending (per-pupil)	12182.03	11781.46	400.58	0.02
White (%)	0.55	0.61	-0.06	0.00
Special Ed (%)	0.13	0.13	0.00	0.39
TPS-Charter Spending (% diff)	0.16	0.22	-0.06	0.01
Per Pupil Revenue	11932.57	11781.86	150.71	0.36
Teacher Salary	93007.23	95199.46	-2192.23	0.02
Exempt from State/District Regs	0.64	0.62	0.02	0.03
Clear Charter Renewal Policies	0.61	0.63	-0.02	0.05
Equitable Funding	0.49	0.51	-0.02	0.23
No Caps on CS Growth	0.82	0.81	0.01	0.16
Town	0.23	0.20	0.04	0.08
Charter Spending (per-pupil)	10973.88	10281.14	692.74	0.00
Transparent Charter Startup Policies	0.54	0.54	0.00	0.89
Performance-Based Contracts	0.55	0.56	-0.01	0.28
Suburb	0.35	0.33	0.02	0.32
Rural	0.27	0.25	0.02	0.34
Urban	0.15	0.22	-0.08	0.00
Magnet Schools (%)	0.22	0.36	-0.14	0.06
University in City	0.01	0.02	-0.01	0.35
City Population (standardized)	0.01	0.00	0.01	0.27

Notes: For each moderator, the table reports the mean value among districts with positive CATT estimates (Mean^{+}), the mean among districts with negative CATT estimates (Mean^{-}), their difference, and the associated p-value of the difference. Rows sorted by variable importance factor.

Table 5. DiDCF Differences-in-Means: ELA Scores

Variable	Mean (pos. CATT)	Mean (neg. CATT)	Difference	p-value
Baseline Performance	-0.33	-0.14	-0.19	0.00
Free/Reduced Lunch (%)	0.51	0.47	0.04	0.00
Student-Teacher Ratio	16.13	16.24	-0.11	0.46
Log of Enrollment	8.46	8.65	-0.19	0.00
Black (%)	0.18	0.17	0.01	0.40
Hispanic (%)	0.21	0.18	0.03	0.01
White (%)	0.56	0.61	-0.05	0.00
Total Spending (per-pupil)	12305.87	11585.57	720.30	0.00
Teacher Salary	94969.76	93290.98	1678.78	0.07
Special Ed (%)	0.13	0.13	0.00	0.21
TPS-Charter Spending (% diff)	0.21	0.17	0.03	0.12
Per Pupil Revenue	12197.91	11461.22	736.69	0.00
Exempt from State/District Regs	0.62	0.65	-0.03	0.01
Clear Charter Renewal Policies	0.62	0.63	-0.02	0.06
Equitable Funding	0.49	0.51	-0.02	0.19
Transparent Charter Startup Policies	0.53	0.55	-0.02	0.01
No Caps on CS Growth	0.81	0.82	-0.01	0.30
Performance-Based Contracts	0.55	0.57	-0.03	0.00
Charter Spending (per-pupil)	10738.63	10445.32	293.31	0.15
Town	0.23	0.19	0.04	0.04
Suburb	0.36	0.32	0.04	0.10
Rural	0.24	0.27	-0.03	0.21
Urban	0.16	0.22	-0.05	0.01
Magnet Schools (%)	0.27	0.33	-0.06	0.46
University in City	0.01	0.01	0.00	0.91
City Population (standardized)	0.01	-0.01	0.02	0.03

Notes: For each moderator, the table reports the mean value among districts with positive CATT estimates (Mean^{+}), the mean among districts with negative CATT estimates (Mean^{-}), their difference, and the associated p-value of the difference. Rows sorted by variable importance factor.

Table 6. DiDCF Best Linear Predictor: Math Scores

	Estimate	Std. Error	p-value
(Intercept)	0.42	0.09	0.00
Baseline Performance	-0.01	0.01	0.31
Black (%)	-0.01	0.05	0.80
Hispanic (%)	-0.09	0.05	0.07
Free/Reduced Lunch (%)	0.01	0.04	0.87
White (%)	-0.11	0.05	0.03
Special Ed (%)	-0.16	0.14	0.25
Log of Enrollment	-0.03	0.01	0.00
Total Spending (per-pupil) ($1,000s)	0.00	0.00	0.15
Student-Teacher Ratio	0.00	0.00	0.90
TPS-Charter Spending (% diff)	-0.01	0.02	0.77
TPS-Charter Spending (% diff) (missing)	0.06	0.02	0.01

Notes: Best linear predictor (BLP) regression of cohort-weighted CATT estimates on district moderators, applied to math test scores. Each row reports a separate OLS coefficient from regressing the mean post-treatment CATT on the listed covariate. The "missing" indicator on the second instance of Charter Spending refers to a binary indicator for if the district was missing data for Charter Spending.

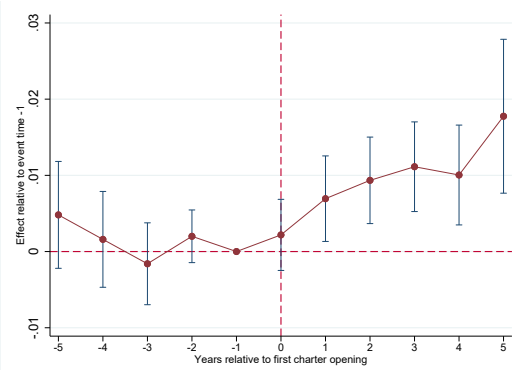
Table 7. DiDCF Best Linear Predictor: ELA Scores

	Estimate	Std. Error	p-value
(Intercept)	0.39	0.08	0.00
Baseline Performance	0.00	0.01	0.43
Black (%)	-0.03	0.04	0.45
Hispanic (%)	0.01	0.04	0.88
Free/Reduced Lunch (%)	0.04	0.03	0.21
White (%)	-0.07	0.04	0.07
Special Ed (%)	0.06	0.12	0.60
Log of Enrollment	-0.03	0.00	0.00
Total Spending (per-pupil) ($1,000s)	0.00	0.00	0.70
Student-Teacher Ratio	0.00	0.00	0.03
Teacher Salary ($1,000s)	0.00	0.00	0.79

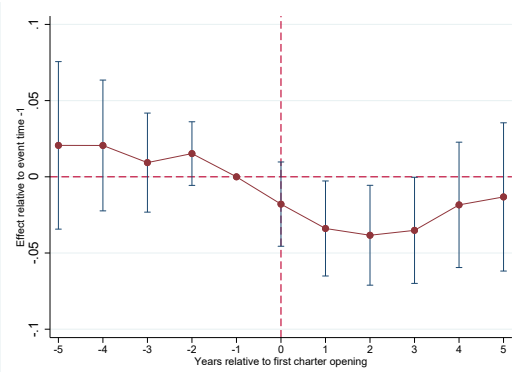
Notes: Best linear predictor (BLP) regression of cohort-weighted CATT estimates on district moderators, applied to ELA test scores. Each row reports a separate OLS coefficient from regressing the mean post-treatment CATT on the listed covariate. The "missing" indicator on the second instance of Charter Spending refers to a binary indicator for if the district was missing data for Charter Spending.

Appendix

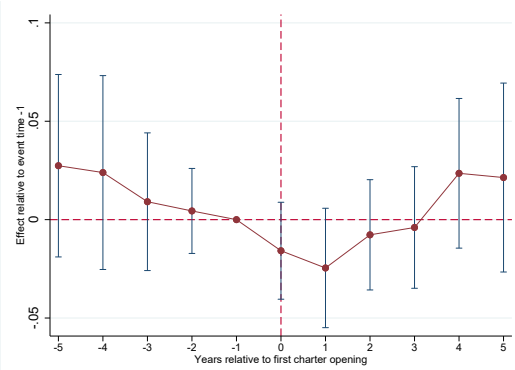
A Tests of Identifying Assumptions



(a) Graduation Rate



(b) Math Achievement



(c) ELA Achievement

Figure A1. Pre-Trends and Event-Study Estimates of Charter-School Effects

Notes: This figure reports event-study estimates from five years before to five years after the first observed year with a positive charter enrollment share. Event time -1 is the omitted reference period. Panel (a) reports estimates for graduation rate, and Panels (b) and (c) report estimates for math and ELA achievement, respectively. Vertical bars indicate 95% confidence intervals. All specifications include district controls and fixed effects, are weighted by enrollment, and cluster standard errors at the district level.

Table A1. Conditional Parallel Trends Pre-Test

	p-value, by random sample							
Outcome	(1)	(2)	(3)	(4)	(5)	Median	Reject at 5%	Treated
Graduation (AFGR)	0.33	0.67	0.28	0.50	0.56	0.50	0 of 5	171
Math	0.38	0.78	0.58	0.79	0.49	0.58	0 of 5	267
ELA	0.31	0.74	0.48	0.60	0.46	0.48	0 of 5	267

Notes: The table reports the integrated moments test of Callaway and Sant'Anna (2021) for the conditional parallel trends assumption holding in all pre-treatment time periods for all groups. For tractability, we conduct this exercise 5 times with 5 random subsamples of the data. In no draw for any outcome do we reject the null hypothesis that parallel trends holds conditional on covariates in our data.

A.1 District-Specific Time Trends

One important possibility is the existence of district-specific time trends differentially affecting districts that adopt charter schools vs. districts that do not. While our pre-trend placebo tests for the CS-DiD model do not show evidence for these, the pre-trend placebo tests we estimate for the DiDCF suggest they may be present, particularly with ELA scores.[29] However, trend violations which correlate with binary treatment status, but not with overall dose, will not bias our estimates of heterogeneity. Specifically, they will not bias our "ranking" of districts most affected vs. least affected by charter entry nor will they bias our estimates of how the $CATT/CACRT$ varies with x.

To illustrate this, recall the DiDCF's partially linear specification of the $CATT$ from Equation 20:

$$\mathbb{E}[\Delta\widetilde{Y} \mid X] = \hat{\alpha}(x) + \hat{\tau}(x)\mathbb{E}[W \mid X] \tag{A1}$$

Now, imagine we introduce an unobserved random variable within the set of treated districts, δ_i, into the data-generating process which creates differential trends that are uncorrelated with dose:

$$\Delta\widetilde{Y}_i = \alpha(x_i) + \tau(x_i)W_i + \delta_i + \varepsilon_i; \quad \text{Cov}(\delta, W \mid X) = 0 \tag{A2}$$

Lemma 4. *Under Equation A2, the $CACRT$ bias equals:*

$$\hat{\tau}(x) - \tau(x) = \frac{Cov(\delta, W \mid X = x)}{Var(W \mid X = x)} = 0 \tag{A3}$$

[29]It is worth noting that the mechanics of the DiDCF estimator do not allow us to explicitly control for district-specific time trends (linear or otherwise) as is possible in the parametric model.

The CATT bias equals:

$$\begin{aligned}\hat{\alpha}(x) - \alpha(x) &= \mathbb{E}[\delta \mid X = x] - \big(\hat{\tau}(x) - \tau(x)\big)\,\mathbb{E}[W \mid X = x] \\ &= \mathbb{E}[\delta \mid X = x]\end{aligned} \tag{A4}$$

For proof, see Appendix D

Lemma 4 states that under the existence of an unobservable covariate causing differential trends across treated districts relative to untreated districts, so long as the unobservable is otherwise uncorrelated with dose (and Conditional Strong PTA holds), the estimates of the $CACRT$ will be unaffected and all heterogeneity conclusions one may draw from them will remain correct. The unobservable *will*, however, bias the $CATT$ estimate by a scalar equal to the conditional mean of the unobservable.

Given that the $CATT$ will suffer bias under this type of contamination, under which conditions can we be confident that heterogeneity in the $CATT$ will remain valid?

Lemma 5. *Define $\hat{\theta}_i = \theta_i + \eta_i$ as the estimated $CATT$ for district i, which is the sum of the true $CATT$ (θ_i) and the prediction error (η_i). The correlation between the true $CATT$ and the $CATT$ prediction can be stated as:*

$$\rho(\hat{\theta}, \theta) = \frac{1 + rs}{\sqrt{1 + 2rs + s^2}} \tag{A5}$$

Where $r = Corr(\theta, \eta)$ and $s = \frac{sd(\eta)}{sd(\theta)}$.

Proof. Let $\sigma_\theta^2 = Var(\theta)$. Then $\text{Cov}(\theta, \hat{\theta}) = \text{Var}(\theta) + \text{Cov}(\theta, \eta) = \sigma_\theta^2(1 + rs)$ and $\text{Var}(\hat{\theta}) = \text{Var}(\theta) + 2\text{Cov}(\theta, \eta) + \text{Var}(\eta) = \sigma_\theta^2(1 + 2rs + s^2)$. Divide: $\rho = \sigma_\theta^2(1 + rs)/[\sigma_\theta^2\sqrt{1 + 2rs + s^2}]$, which is Equation A5.

Lemma 5 says that the $CATT$ estimates, despite being biased, preserve their ranking of units under two special cases. First, if $s = 0$ (i.e. the contamination is equal across treated units and is unrelated to X), then $\rho = 1$ and all $CATT$ heterogeneity results are valid. Second, if $r = 1$ (i.e. if the contamination is proportional to the $CATT$), then $\rho = 1$ again for any s, because $(1 + s)/\sqrt{(1 + s)^2} = 1$. The case where $CATT$ rankings are not perfectly preserved is if both $s > 0$ and $r < 1$ (i.e. the contamination is not equal across units nor is it proportional to the true $CATT$). For example, in the case where the contamination is completely uncorrelated to the true $CATT$ (i.e. where $r = 0$), $\rho = 1/\sqrt{1 + s^2}$ and the quality of the heterogeneity estimates degrades as the variance in η grows relative to the variance in θ. It also degrades rather slowly: e.g. if $s = 0.5$ (the variance in the error is half as large as the variance in the $CATT$), the correlation between true $CATT$s and estimated $CATT$s is still $\rho \approx 0.9$.

In the case where the contamination takes on a district-specific linear time trend, then $\delta_i = \gamma_i h$, where γ_i is the slope specific to district i and h is the event-time. In this setting, $\text{Cov}(\delta, W \mid X) = h \cdot \text{Cov}(\gamma, W \mid X)$. In order for Lemma 4 to hold, it must be that each district's linear trend is uncorrelated with it's eventual dose in every period h: $\text{Cov}(\gamma, W \mid X) = 0$. We can check this by estimating each treated district's own pre-treatment trend, $\hat{\gamma}_i$ and regressing $\hat{\gamma}_i$ on $(1, W_i, X_i)$ to check if these slopes are conditionally correlated with the district's realized dose. We report these results in Table A2. We find no evidence that linear time trends in the pre-treatment years are correlated with dose, conditional on X. This provides support that under Strong Conditional Parallel Trends (Assumption 6), our estimates of $CACRT$ (i.e. $\tau(x)$) parameters remain unbiased even in the presence of linear trend divergence.

For the $CATT$ heterogeneity to remain valid under district-specific linear time trends, we require the linear trends to be either homogeneous across X neighborhoods or directly proportional to the true $CATT$. Because true $CATT$s are unknown, it is difficult to provide evidence for the latter condition. However, one can seek visual evidence for the former condition by revisiting the neighborhood event-study plots of Figures C10 and B5. Recall, these figures defined neighborhoods of districts based on covariate similarity using k-means clustering and estimated event-studies within each group. Evidence for homogeneous pre-trend divergence would appear as low variance in $CATT$ estimates during the pre-treatment periods. This is clearly evident in AFGR and ELA scores – the two outcomes where pre-trends seem to be diverging. We view this as strong evidence supporting the validity of the $CATT$ heterogeneity results for both potentially problematic outcomes.

Table A2. Correlation of District-Specific Linear Time Trends and Charter Intensity

Outcome	Event time	N	Dose coef.	SE	t	p
afgr	pooled	5732	0.0092	0.0070	1.32	0.187
math	pooled	7880	0.0085	0.0445	0.19	0.848
ela	pooled	7880	-0.0061	0.0410	-0.15	0.881

Notes: This table reports the results of a regression on district-specific linear trends on that district's charter intensity, conditional on X.

B AFGR Results

B.1 Average Treatment Effect

In this appendix section, we study the main results concerning the average freshman graduation rate (AFGR) outcome. We begin with the ATT as estimated by the CS-DiD approach. Table B3 displays these results for each of the three samples we consider, though we focus on the full sample (Panel A). Unlike math and ELA test scores, charter schools have a positive and statistically significant ATT of 0.019, indicating that a 10pp increase in the charter market share yields a 0.19pp increase in AFGR. The DiDCF estimates an $ATT = 0.004$ as seen in Figure B2.

B.2 Distribution of Treatment Effects

The distribution of $CATT$ estimates per geographic district $\times$ event-time for AFGR can be found in Table B2 for the DiDCF. Our estimates suggest there exists notable heterogeneity in $CATT$s, with the 5th and 95th percentiles of DiDCF effects ranging from approximately [-0.06, 0.07]. This indicates there exist districts where a 10pp increase in charter intensity leads to as much as a 6pp *decrease* in AFGR, while there are simultaneously districts where the same 10pp increase in charter share leads to a 7pp *increase* in AFGR.

B.2.1 Effect Heterogeneity by State

* Feng TBD *

B.3 Effect Heterogeneity by District Traits

B.3.1 Trait Set # 1: Demographics

Like with test scores, we find that demographics are important drivers of charter school effects when considering graduation rates. Each of the six demographic variables score in the top 10 of VIF: Log Enrollment (1), Free-Reduced Lunch (FRL) % (2), Hispanic % (4), Special Ed % (7), White % (8), and Black % (10). Notably, while Log Enrollment is the most important variable for *short run* impacts according to DiDCF, it diminishes quickly in importance when considering long run effects in favor of FRL % and Hispanic %.

The demographic measures generally do not show evidence of linear relationships with the $CATT$ when considering AFGR. Districts with more FRL students seem to be positively impacted more often than not. Specifically, the event study suggests that districts in the top quartile of FRL share are impacted more. The k-means table reinforces this by showing that the two groups most positively impacted by charter schools are also the two groups with the largest FRL share.

However, despite these signs that more low income students is associated with larger $CATT$ estimates, the effects seem to be nonlinear beyond the top quartile, and the diff-in-means and BLP tables do not find notable linear relationships between FRL share and the $CATT$.

The ethnic variables (Black %, Hispanic %, and White %) appear to be highly nonlinear in their relationships with the $CATT$. The diff-in-means and BLP for Hispanic % suggest that more Hispanic students is associated with more positive effects, but the event study and k-means profiles suggest the relationship is much more interactive than the linear measures indicate. Specifically, the k-means table suggests that while the 2nd most positively impacted group of districts had a 78% Hispanic share, several other Hispanic-dominated district groupings fared much worse; the event study similarly shows that both the top *and* bottom quartiles of Hispanic % are impacted more positively than the middle 50%. The depth-3 policy tree sheds some light on Hispanic %'s interactivity. While it does not split on Hispanic % directly, it is clear that in order for a district to be both low Black % and low White %, the district is likely high Hispanic %. For these high Hispanic districts, charter school expansion is likely to be beneficial if the district is sufficiently low performing at baseline.

Black % seems particularly interactive when considering AFGR. While the BLP suggests that more Black students is associated with larger $CATT$s (as with Math and ELA), the diff-in-means does not find a notable difference between positive vs. negative $CATT$ districts, and the k-means profiles suggest that while the best performing district group had a very high Black share, other districts with a high Black share did not fare as strongly. The depth-3 policy tree sheds some light on the interactivity for Black share. Specifically, high Black share districts should expand charter school presence if they are simultaneously high FRL (i.e. low income) or high Student-Teacher Ratio (i.e. low spending). Conversely, low Black share districts can also expand charter schools if they are simultaneously large and predominantly White %.

District size (in terms of Log Enrollment) seems to also be related to the $CATT$ in an interactive (and important) manner. The various tables and figures suggest different directions. The diff-in-means weakly suggests that smaller districts are more often better off. The BLP finds no relationship. The k-means table seems to suggest that bigger districts are more often positively impacted. The event studies show some evidence that smaller districts are impacted more. These conflicting results are only reconcilable by a highly interactive relationship, which both policy trees corroborate. Enrollment appears to interact with both spending variables (depth-2 tree) and ethnic demographics (depth-3 tree). In particular, larger districts are more often better off but only when the TPS-Charter spending gap is large or if the district is predominantly White.

To summarize, these findings are similar to those of the two test score outcomes, though the relationships are more interactive and nonlinear. While it is more often the case that districts with more low-income and/or ethnic minority students (in larger districts) are more often pos-

itively impacted by charters, it is by no means guaranteed that a district with low-incomes or high non-White share will be positively impacted. Like with math scores, the "demographically homogeneous" districts tend to be the ones more positively impacted.

B.3.2 Trait Set # 2: Market Context

Baseline AFGR ranks #6 in VIF score and remains important over time, indicating it is a consistent driver of heterogeneity. No other market context variable carries notable information according to VIF. Perhaps because indicators like Urban vs. Suburban vs. Rural are better proxied by Log Enrollment according to DiDCF.

The relationship between a district's baseline AFGR and its $CATT$ appears to be more non-linear and much weaker than with test scores. Our linear measures do not provide much evidence one way or another, but the BLP does weakly suggest that lower Baseline AFGR districts tend to have larger $CATT$ estimates, and the k-means table is fairly clear (though non linear in some groups) that the most positively impacted groups of districts are those with lower Baseline AFGR. Because Baseline AFGR was #6 in VIF, it does not get an event study, and the only instance of Baseline AFGR appearing in the policy trees was the final split of the depth-3 tree where districts with high White and/or Hispanic shares and low baseline AFGRs should expand charter schools.

To summarize: like with test scores, districts with lower Baseline AFGR are more often positively impacted. But this trend is not as clear as it was with test scores.

B.3.3 Trait Set #3: School Funding

School finance variables appear to drive heterogeneity in AFGR effects less than for test scores, but with some important interactions. The TPS-Charter Spending Difference (3), Student-Teacher Ratio (5), and Per-Pupil Revenue (9) each score in the top 10 for VIF.

These variables appear to have largely nonlinear relationships with the $CATT$. As was discussed in the **Demographics** subsection, the TPS-Charter Spending Gap appears to be interactive with both district Enrollment and Per-Pupil Revenue. Specifically, bigger spending gaps are associated with more positive $CATT$ estimates in larger districts, and smaller gaps are associated with positive effects in districts with low revenue per-student.

The other two district finance variables, Student-Teacher Ratio and Teacher Salary, do not seem to have linear or particularly meaningful relationships with the $CATT$. The diff-in-means, BLP, and k-means profiles do not exhibit clear trends. Student-Teacher Ratio does appear in the depth-3 policy tree, interacting with Black student share and FRL. Specifically, a high Student-Teacher ratio can be associated with positive $CATT$s in low-income, high-Black share districts.

B.3.4 Trait Set #4: State Charter Policy

Like with Math and ELA test scores, the policy setting variables generally score low on VIF. Districts where charter schools are automatically exempt from TPS regulations score #11, and districts with Equitable Funding agreements score #14. No other policy variable generates a VIF score greater than 0.01. Because these variables have low VIF scores, they do not appear in the BLP or quartile event studies, and (unsurprisingly) do not show up in either of the learned policy trees.

Our metrics have little to say about how $CATT$s vary with these moderators. Neither the CS-DiD nor the DiDCF diff-in-means suggest a linear relationship. The k-means profiles suggest that these variables are likely interactive as (for example) the best and worst performing groups have a similar share of districts which are exempt from state/local regulations. Regardless, these interactions are likely weak drivers of $CATT$ heterogeneity given they do not appear in the policy trees and have low VIFs.

B.3.5 Trait Set #5: Diminishing Returns

We refer the reader to Section 4.3.6 for a review of the methodology for estimating the dose-response and marginal-response functions. Unlike with test scores where diminishing marginal return patterns were fairly clear, we do not find clear evidence of this for AFGR. We generally lack power to estimate precise $ACRT(w)$ parameters at any dose level for AFGR, and Figure B9 suggests there could be a U-shaped pattern in marginal effects. At low charter share (between 0-10%) there are diminishing-but-positive returns to additional charter schools. Marginal returns from 10-75% charter share are negative, before turning positive again at very high doses.

Table B3. Average Effects of Charter-School Expansion: Graduation Rate

	Graduation Rate	
	(1)	(2)
Panel A: DiD, Full Sample		
ATT	0.019^{***}	0.017^{***}
	(0.007)	(0.004)
ACRT	0.034	0.060
	(0.061)	(0.055)
R^2	0.889	0.893
Observations	132,876	131,435
Panel B: DiD, Last Three Years		
ATT	-0.000	0.001
	(0.013)	(0.014)
ACRT	0.174^{*}	0.171^{*}
	(0.105)	(0.103)
R^2	0.909	0.909
Observations	22,482	22,265
Panel C: Staggered DiD		
ATT	0.011^{***}	0.011^{***}
	(0.001)	(0.001)
ACRT	0.016	0.033
	(0.026)	(0.025)
Treated observations	5,669	5,663
District fixed effects	Yes	Yes
State-by-year fixed effects	Yes	Yes
District controls	No	Yes

Notes: All three panels report the ATT and ACRT using charter enrollment share W. Panels A and B use the pooled parametric DiD specification with $D = 1\{W > 0\}$, district fixed effects, and state-by-year fixed effects. For positive treatment, the treatment component is $\beta_0 + \beta_1 W$, so the ATT is $\beta_0 + \beta_1 E[W \mid W > 0]$, while the ACRT is β_1. Panel A uses the full sample, while Panel B applies the final-three-year sample restriction. Panel C implements the staggered-adoption CS-DiD separately by treatment cohort and post-treatment calendar year, using the period immediately preceding treatment as the cohort-specific base period and never-treated districts as the comparison group. Cohort–time estimates in Panel C are aggregated using treated-observation shares; the Panel C sample-count row reports treated cohort–time observations. Regressions are weighted by first-period high school enrollment. Standard errors in Panels A and B are clustered at the district level. Panel C uses district-clustered standard errors within each cohort–time regression; the reported aggregate standard errors combine cell-specific variances but do not incorporate cross-cell covariance arising from reuse of never-treated districts. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table B4. Heterogeneity in the Effects of Charter-School Expansion on Graduation: Stacked CS-GDD

		Graduation
Moderator	Component	(1)
Baseline Treatment Effects		
Positive charter exposure	Level	−0.013
		(0.017)
Charter enrollment share	CACRT	0.160
		(0.212)
Student Demographics		
White student share	Level	0.005
		(0.045)
	CACRT	−0.874
		(0.599)
Black student share	Level	0.040
		(0.054)
	CACRT	−0.841
		(0.608)
Hispanic student share	Level	0.032
		(0.048)
	CACRT	−0.729
		(0.588)
Free or reduced-price lunch share	Level	−0.064*
		(0.037)
	CACRT	−0.045
		(0.392)
Special education share	Level	−0.003
		(0.067)
	CACRT	0.322
		(0.758)
Market Setting		
Baseline outcome	Level	0.016
		(0.034)
	CACRT	−0.178
		(0.264)
Urban district	Level	0.024*
		(0.013)
	CACRT	−0.260*
		(0.142)
Suburban district	Level	0.013
		(0.012)
	CACRT	−0.012
		(0.140)
Town district	Level	0.010
		(0.011)
	CACRT	−0.061
		(0.135)
Log enrollment	Level	−0.010**
		(0.004)
	CACRT	0.119**
		(0.047)
Magnet schools	Level	

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Table B4 continued

Moderator	Component	Graduation (1)
	CACRT	
School Revenue and Spending		
Per-pupil spending ($1,000s)	Level	−0.011* (0.006)
	CACRT	0.060 (0.043)
Per-pupil revenue ($1,000s)	Level	0.012 (0.007)
	CACRT	−0.082* (0.044)
TPS–charter spending gap	Level	−0.010 (0.026)
	CACRT	−0.419 (0.283)
Equitable operational funding	Level	−0.006 (0.013)
	CACRT	0.043 (0.169)
Charter-school spending ($1,000s)	Level	0.002 (0.002)
	CACRT	−0.004 (0.020)
Teacher salary ($1,000s)	Level	0.000 (0.001)
	CACRT	0.002 (0.005)
Student–teacher ratio	Level	0.000 (0.002)
	CACRT	−0.035 (0.026)
Other State Charter-School Policies		
Regulatory exemptions	Level	0.032 (0.028)
	CACRT	−0.192 (0.287)
No caps on charter-school growth	Level	0.005 (0.024)
	CACRT	0.085 (0.280)
Clear renewal and nonrenewal procedures	Level	0.006 (0.030)
	CACRT	0.637 (0.441)
Transparency	Level	−0.004 (0.043)
	CACRT	−0.088 (0.467)
Performance-based contracts	Level	0.002 (0.025)

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Table B4 continued

Moderator	Component	Graduation (1)
	CACRT	0.454 (0.350)
Missing-Value Indicators		
TPS–charter spending gap missing	Level	0.000 (0.015)
	CACRT	−0.437 (0.289)
Charter-school spending missing	Level	0.004 (0.021)
	CACRT	0.377 (0.274)
Treated observations		5,732

Notes: The table reports heterogeneity estimates from the stacked, staggered, continuous-dose CS-GDD, using charter enrollment share W and $D = 1\{W > 0\}$. For each cohort g and post-treatment calendar year t, the outcome change $Y_t - Y_{g-1}$ of never-treated units is first regressed on the moderators X, which removes the counterfactual trend $\gamma_{0,gt} + X'\gamma_{gt}$ (not reported). The residualized changes of treated units in all post-treatment cells are then stacked and regressed on cell-specific intercepts $\alpha_{0,gt}$ and dose slopes $\tau_{0,gt}$ and on (X, WX) with coefficients common to all cells, so $CATT_{gt}(w, X) = \alpha_{0,gt} + X'\alpha + w(\tau_{0,gt} + X'\tau)$ and $CACRT_{gt}(X) = \tau_{0,gt} + X'\tau$. Each cell compares cohort g with never-treated units from base year $g - 1$; only the moderator coefficients are pooled across cells. Moderators are centered at their enrollment-weighted sample means, and the baseline rows report the averages of $\alpha_{0,gt}$ and $\tau_{0,gt}$ over post-treatment cells, weighted by each cell's number of treated observations. For each moderator k, the Level row reports α_k, the coefficient on $D \times X_k$, which is the treatment-by-moderator interaction reported in a binary-treatment difference-in-differences; the CACRT row reports τ_k, the coefficient on $W \times X_k$. Because the dose is continuous, the gradient of the CATT with respect to X_k at dose w is $\alpha_k + \tau_k w$, so α_k is that gradient as $w \to 0^+$. An alternative summary evaluates the gradient at the mean treated dose, $\alpha_k + \tau_k \bar{w}$; we do not report it. Units are districts. Urban, suburban, and town coefficients are interpreted relative to rural districts. Moderators are measured in the first observed year for each unit, except the baseline outcome, which is Y_{g-2} centered within each cell. State policy measures are rescaled to $[0, 1]$ before centering. Missing moderator values are set to the sample mean and flagged by the missing-value indicators. Per-pupil spending, per-pupil revenue, charter-school spending, and teacher salary are expressed in thousands of dollars. Both stages are weighted by the graduation-rate denominator. Standard errors are clustered at the district level and account for estimation of the first stage. Standard errors are shown in parentheses below point estimates. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

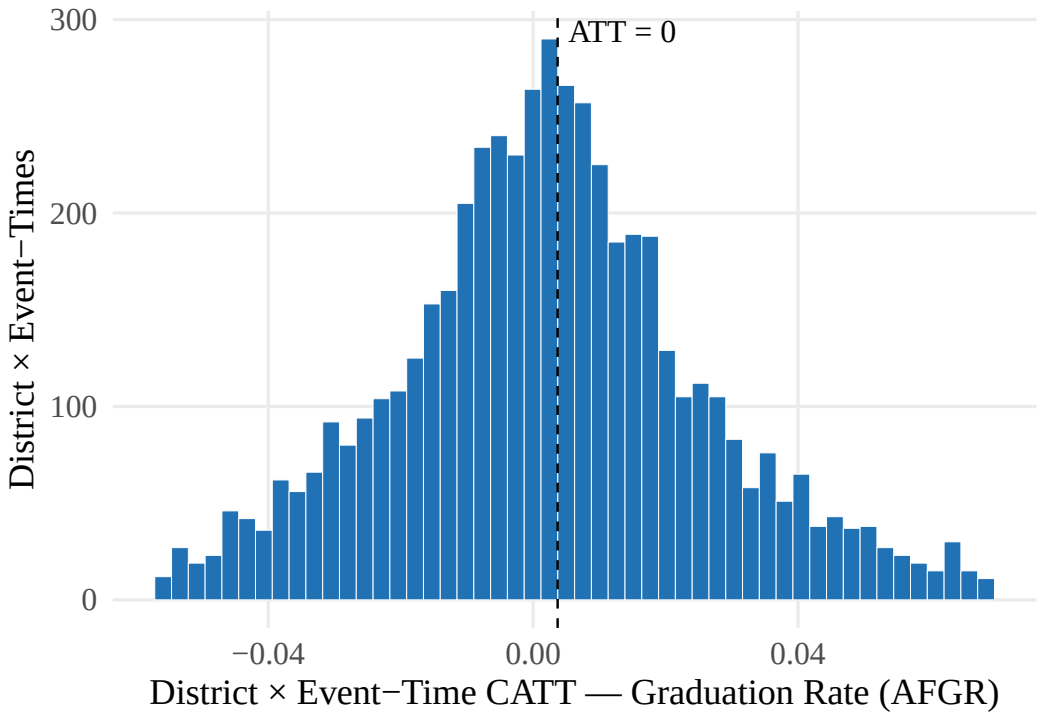


Figure B2. DiDCF CATT Distribution: Graduation Rate (AFGR)

Notes: Distribution of cohort × event-time Level-CATT estimates from the DiDCF applied to graduation rates (AFGR). The vertical dashed line marks the ATT after fully aggregating across doses, event-times, and covariates. We drop the top and bottom 5% of CATT estimates from this figure.

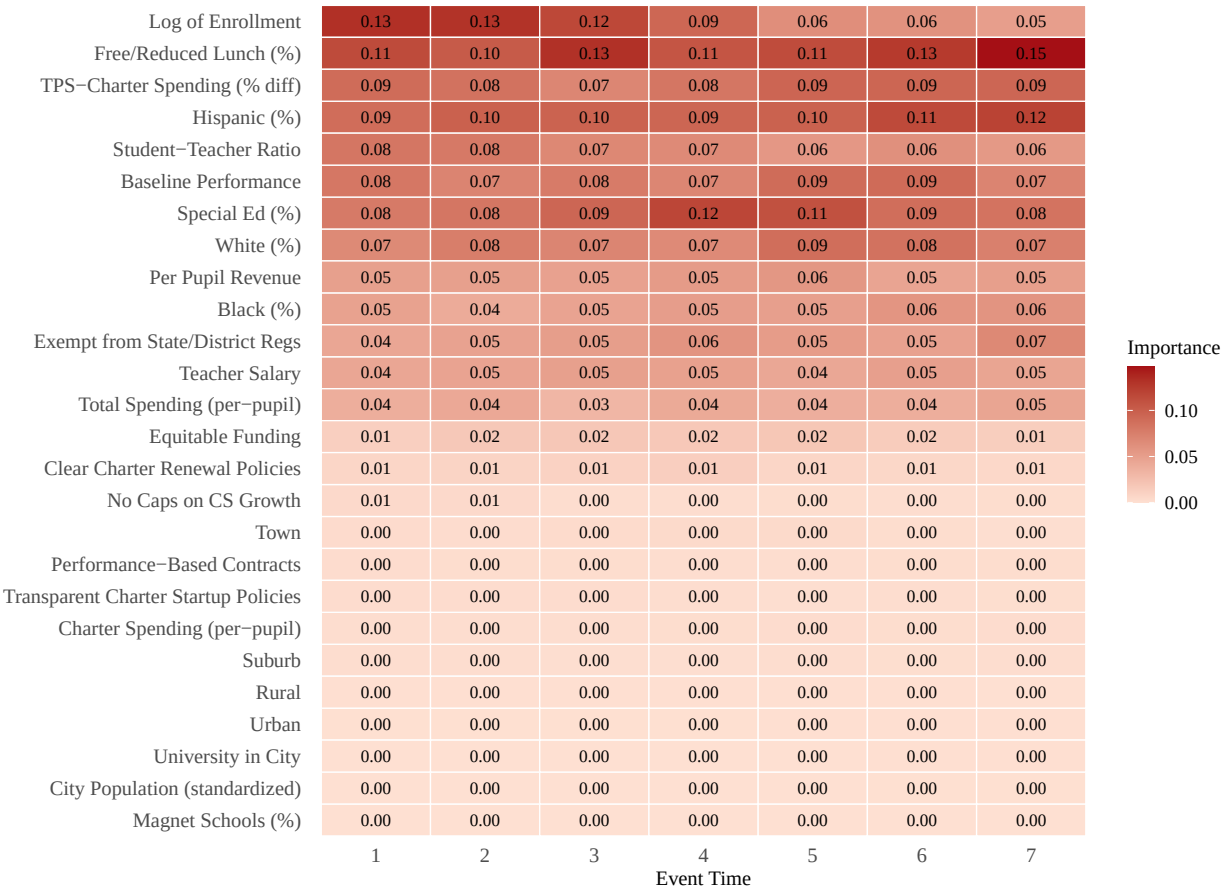
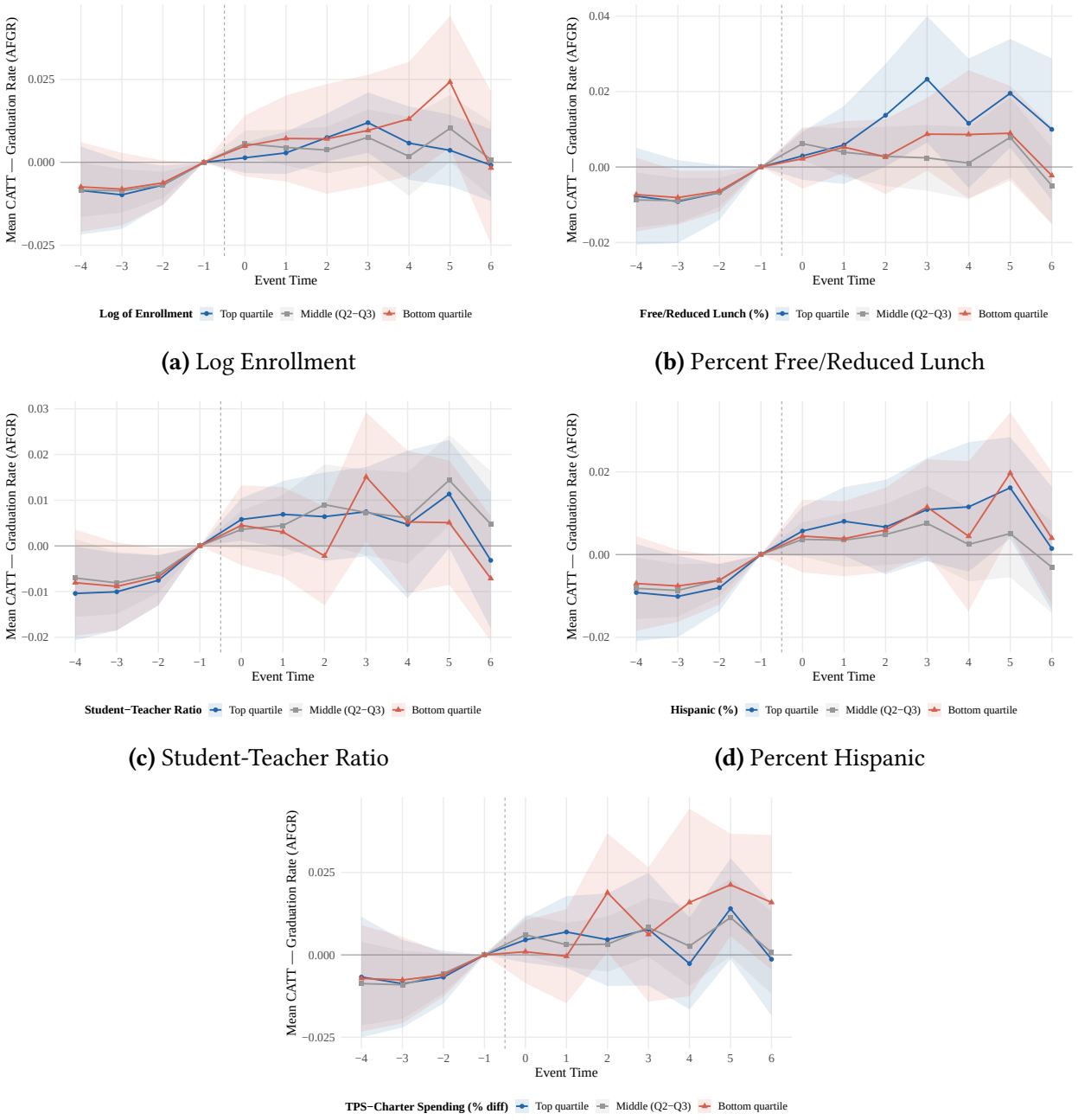


Figure B3. DiDCF Variable Importance Factors: Graduation Rate (AFGR)

Notes: Heatmap of variable importance factors (VIF) across all covariates and post-treatment event-times ($h = 1, \ldots, 7$) from the DiDCF applied to graduation rates. Cell values show the (depth-weighted) share of splits on each covariate at each event-time. Grey cells indicate covariates absent from the data.



(a) Log Enrollment

(b) Percent Free/Reduced Lunch

(c) Student-Teacher Ratio

(d) Percent Hispanic

(e) % Difference TPS vs. Charter Spending

Figure B4. DiDCF Quartile Event Studies (CATT): Graduation Rate (AFGR)

Notes: Each panel shows cohort-weighted ATT estimates by event-time, separately for districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF at $h = 1$ for graduation rates.

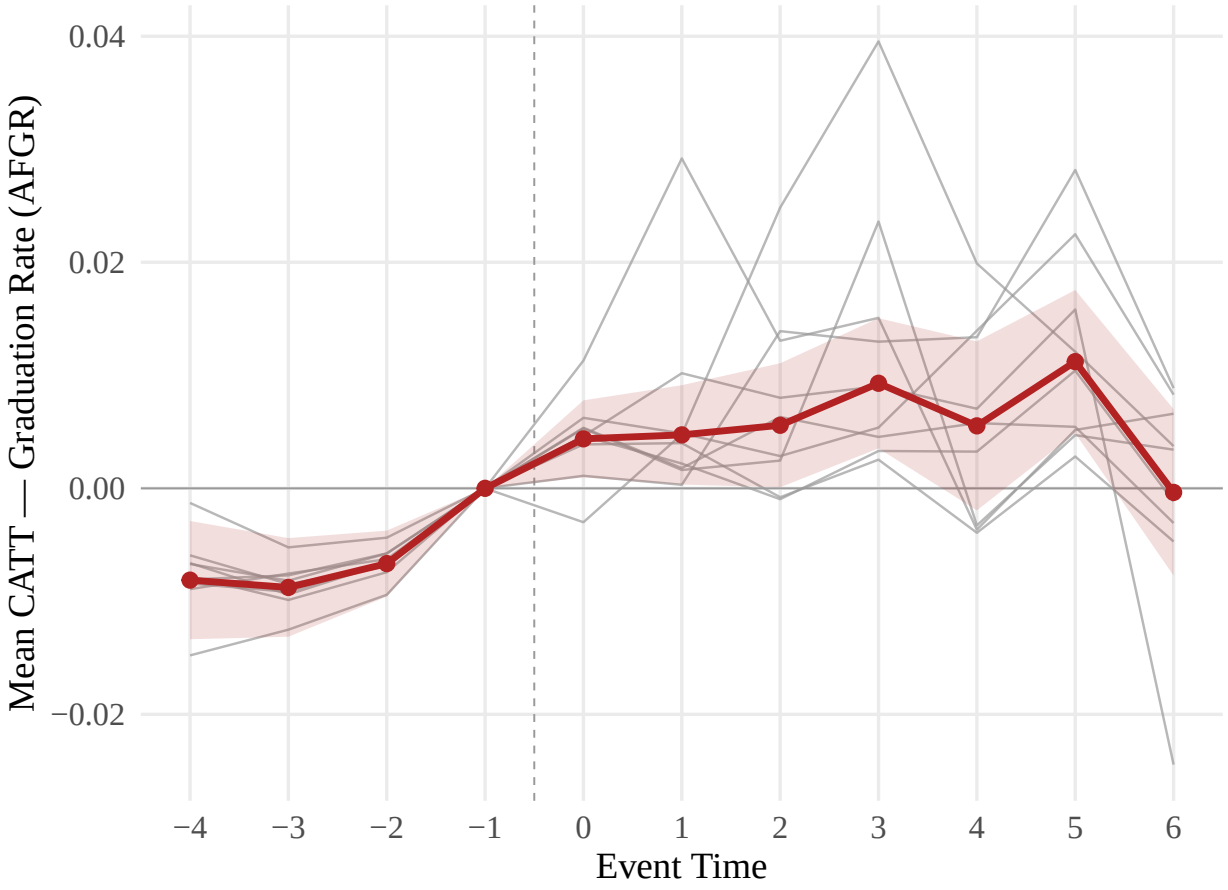
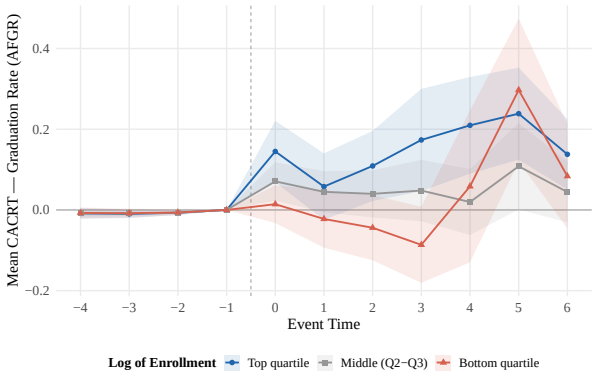
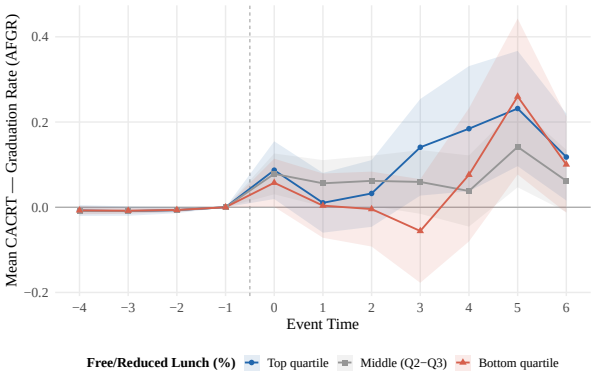


Figure B5. DiDCF K-Means Matched Event Studies (CATT): AFGR

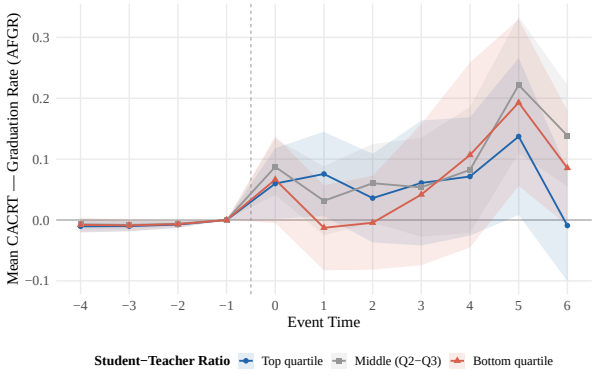
Notes: This figure reports cohort-weighted ATT estimates by event-time, separately for districts in each of 10 similarity groups determined by VIF-weighted k-means clustering. The overall ATT is presented as the red line.



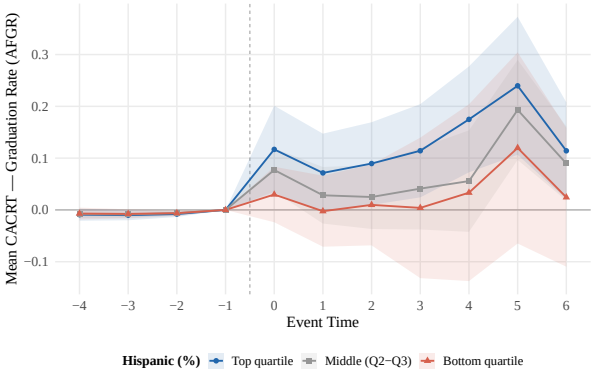
(a) Log Enrollment



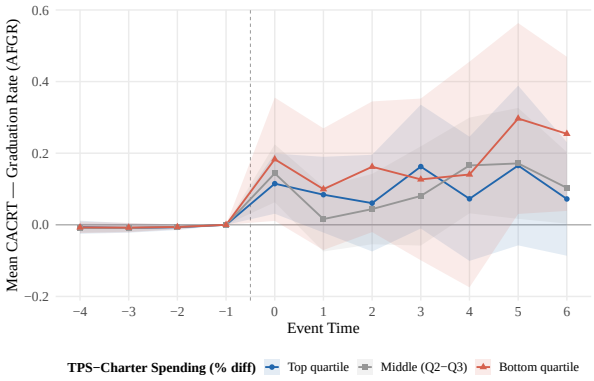
(b) Percent Free/Reduced Lunch



(c) Student-Teacher Ratio



(d) Percent Hispanic



(e) % Difference TPS vs. Charter Spending

Figure B6. DiDCF Quartile Event Studies (CACRT): Graduation Rate (AFGR)

Notes: Each panel shows cohort-weighted ACRT estimates by event-time, separately for districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF at $h = 1$ for graduation rates.

Table B5. DiDCF Differences-in-Means: Graduation Rate (AFGR)

Variable	Mean (pos. CATT)	Mean (neg. CATT)	Difference	p-value
Log of Enrollment	8.69	8.75	-0.05	0.57
Free/Reduced Lunch (%)	0.24	0.22	0.02	0.18
TPS-Charter Spending (% diff)	0.23	0.19	0.05	0.13
Hispanic (%)	0.13	0.10	0.03	0.01
Student-Teacher Ratio	18.01	17.85	0.16	0.44
Baseline Performance	0.75	0.76	-0.01	0.56
Special Ed (%)	0.10	0.11	-0.01	0.00
White (%)	0.70	0.73	-0.03	0.13
Per Pupil Revenue	6294.47	6311.38	-16.91	0.90
Black (%)	0.12	0.12	0.00	0.86
Exempt from State/District Regs	0.63	0.61	0.02	0.29
Teacher Salary	62925.65	61217.41	1708.24	0.08
Total Spending (per-pupil)	6319.05	6250.99	68.06	0.60
Equitable Funding	0.49	0.46	0.03	0.25
Clear Charter Renewal Policies	0.63	0.61	0.02	0.17
No Caps on CS Growth	0.78	0.80	-0.01	0.36
Town	0.20	0.21	-0.01	0.64
Performance-Based Contracts	0.51	0.53	-0.02	0.16
Transparent Charter Startup Policies	0.50	0.52	-0.02	0.12
Charter Spending (per-pupil)	10595.83	10081.06	514.77	0.08
Suburb	0.30	0.31	-0.01	0.68
Rural	0.24	0.22	0.02	0.56
Urban	0.26	0.25	0.01	0.76
University in City	-0.01	0.00	-0.01	0.31
City Population (standardized)	0.00	0.01	0.00	0.85
Magnet Schools (%)	0.00	0.00	0.00	NaN

Notes: For each moderator, the table reports the mean value among districts with positive CATT estimates (Mean^{+}), the mean among districts with negative CATT estimates (Mean^{-}), their difference, and the associated p-value of the difference. Rows sorted by variable importance factor.

Table B6. DiDCF Best Linear Predictor: Graduation Rate (AFGR)

	Estimate	Std. Error	p-value
(Intercept)	-0.03	0.02	0.21
Baseline Performance	-0.01	0.01	0.13
Black (%)	0.02	0.01	0.02
Hispanic (%)	0.05	0.01	0.00
Free/Reduced Lunch (%)	0.01	0.01	0.12
White (%)	0.03	0.01	0.00
Special Ed (%)	-0.03	0.02	0.08
Log of Enrollment	0.00	0.00	0.63
Per Pupil Revenue ($1,000s)	0.00	0.00	0.59
Student-Teacher Ratio	0.00	0.00	0.81
TPS-Charter Spending (% diff)	0.01	0.01	0.38
TPS-Charter Spending (% diff) (missing)	-0.01	0.00	0.21

Notes: Best linear predictor (BLP) regression of cohort-weighted CATT estimates on district moderators, applied to graduation rates. Each row reports a separate OLS coefficient from regressing the mean post-treatment CATT on the listed covariate. Estimates above zero indicate that districts with higher values of the covariate experience larger positive effects of charter entry.

Table B7. DiDCF Differences-in-Means (k-means matched groups): AFGR

	Covariate Grouping								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log of Enrollment	8.92	9.40	10.07	8.80	10.57	9.51	7.57	7.05	8.14
Free/Reduced Lunch (%)	0.46	0.58	0.18	0.13	0.42	0.27	0.14	0.38	0.11
TPS-Charter Spending (% diff)	0.27	0.06	0.12	0.40	0.32	0.19	0.19	0.05	0.14
Hispanic (%)	0.05	0.78	0.10	0.04	0.13	0.31	0.02	0.13	0.03
Student-Teacher Ratio	16.08	18.50	17.18	20.11	17.36	22.18	15.71	15.84	18.44
Baseline Performance	0.59	0.63	0.70	0.82	0.52	0.75	0.87	0.74	0.80
Special Ed (%)	0.10	0.10	0.13	0.11	0.09	0.10	0.12	0.15	0.04
White (%)	0.40	0.15	0.69	0.90	0.22	0.49	0.93	0.72	0.91
Per Pupil Revenue	5969.98	5700.17	5733.96	6672.18	7864.86	5222.69	6596.92	8085.67	5422.08
Black (%)	0.46	0.04	0.18	0.03	0.62	0.08	0.02	0.04	0.04
Exempt from State/District Regs	0.72	0.71	0.64	0.49	0.54	0.76	0.50	0.70	0.71
Teacher Salary	54171.63	61972.37	56402.06	73471.05	77147.06	69418.35	55852.93	61532.26	56791.41
Total Spending (per-pupil)	5938.67	5945.22	5754.55	6730.30	7853.69	5361.01	6555.30	7671.86	5419.73
Equitable Funding	0.45	0.75	0.43	0.51	0.50	0.86	0.18	0.58	0.46
Clear Charter Renewal Policies	0.71	0.65	0.64	0.57	0.62	0.73	0.44	0.63	0.75
No Caps on CS Growth	0.78	0.74	0.89	0.80	0.75	0.78	0.80	0.79	0.69
Town	0.19	0.25	0.02	0.23	0.00	0.09	0.33	0.25	0.33
Performance-Based Contracts	0.59	0.58	0.62	0.56	0.56	0.32	0.49	0.51	0.50
Transparent Charter Startup Policies	0.57	0.51	0.57	0.59	0.51	0.51	0.39	0.56	0.43
Charter Spending (per-pupil)	10857.94	9553.39	9526.70	9098.00	12508.40	10032.24	11629.57	13351.44	9586.67
Suburb	0.16	0.22	0.40	0.46	0.16	0.51	0.14	0.14	0.35
Rural	0.29	0.03	0.13	0.09	0.00	0.07	0.45	0.57	0.22
Urban	0.36	0.50	0.44	0.22	0.84	0.33	0.07	0.04	0.10
University in City	0.01	-0.03	-0.03	0.02	-0.11	-0.01	0.00	0.02	0.00
City Population (standardized)	-0.00	-0.05	-0.06	0.00	0.15	0.02	0.00	0.04	0.00
Magnet Schools (%)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ATT	0.01	0.01	-0.00	-0.00	0.00	0.00	-0.00	-0.01	-0.01
ACRT	-0.04	-0.05	0.04	0.01	-0.01	-0.02	-0.10	-0.02	-0.03

Notes: This table reports the within-group means for each of 10 covariate neighborhoods assigned through VIF-weighted k-means matching. Rows are sorted by VIF in descending order, columns are sorted by within-group ATT, then by within-group $ACRT$ so that group (1) represents the best performing districts and group (10) are the worst performing districts.



Figure B7. Optimal Policy Tree (Depth 2): Graduation Rate (AFGR)

Notes: Depth-2 optimal policy tree estimated via the Athey and Wager (2021) framework applied to DiDCF doubly robust scores for the graduation rate outcome (AFGR). Each internal node shows the split variable and threshold; leaf colors indicate the recommended policy action (blue = expand charter market share; red = do not expand). The sample consists of treated districts with non-missing DR scores averaged across post-treatment event times, restricted to the ten highest-VIF moderators at $e = 1$. The doubly-robust scores in question refer to those associated with CACRT such that the policy prescription refers to the binary decision to “marginally increase" charter intensity or not.

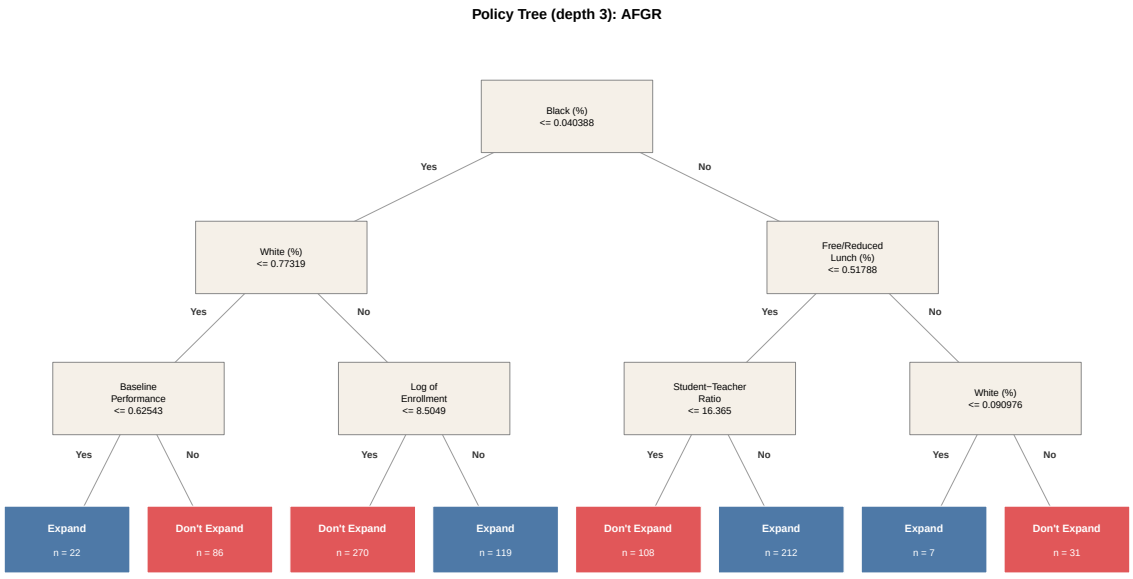


Figure B8. Optimal Policy Tree (Depth 3): Graduation Rate (AFGR)

Notes: Depth-3 optimal policy tree estimated via the Athey and Wager (2021) framework applied to DiDCF doubly robust scores for the graduation rate outcome (AFGR). Each internal node shows the split variable and threshold; leaf colors indicate the recommended policy action (blue = expand charter market share; red = do not expand). The sample consists of treated districts with non-missing DR scores averaged across post-treatment event times, restricted to the ten highest-VIF moderators at $e = 1$. The doubly-robust scores in question refer to those associated with CACRT such that the policy prescription refers to the binary decision to "marginally increase" charter intensity or not.

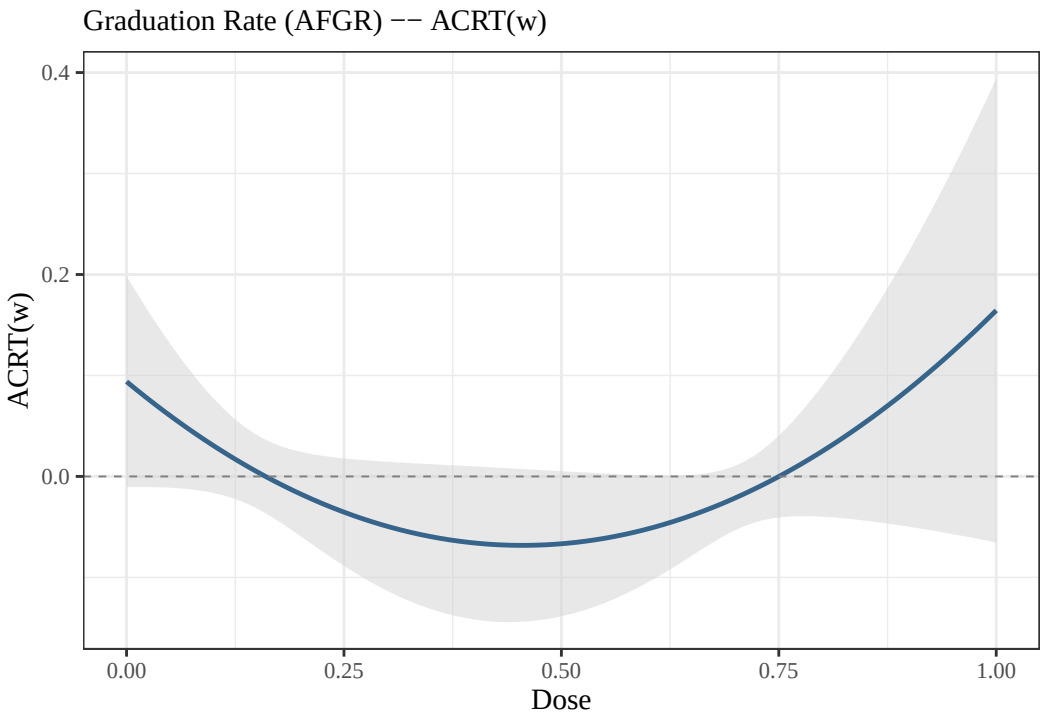
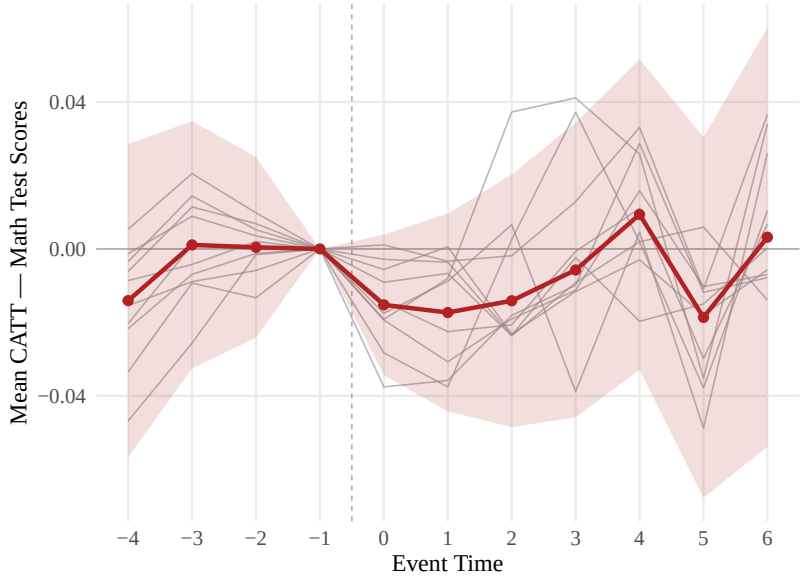


Figure B9. Diminishing Returns to additional Charter Intensity (DiDCF) – AFGR

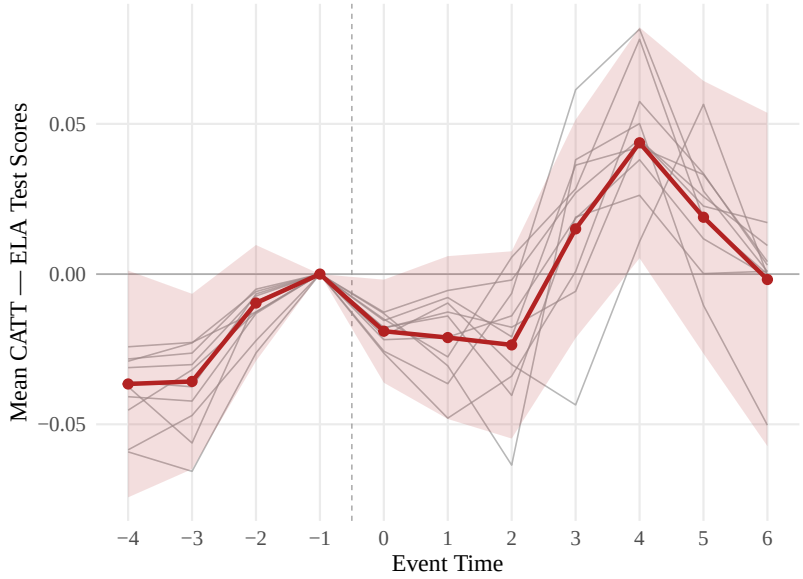
Notes: This figure reports $ACRT(w)$ parameters (Callaway et al., 2025) across dose levels for the graduation rate outcome. 95% confidence intervals are displayed. This figure is interpreted as the average marginal effect of additional charter schools evaluated at dose w.

C Additional Effect Hetero

C.1 k-Means Matched Neighborhoods



(a) Math Scores



(b) ELA Scores

Figure C10. DiDCF K-Means Matched Event Studies (CATT): Math and ELA

Notes: These figures report cohort-weighted ATT estimates by event-time, separately for districts in each of 10 similarity groups determined by VIF-weighted k-means clustering. The overall ATT is presented as the red line.

Table C8. DiDCF Differences-in-Means (k-means matched groups): Math Scores

	Covariate Grouping									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Baseline Performance	-0.68	-1.86	-0.83	-1.19	-0.72	1.41	-0.36	0.38	0.19	0.61
Free/Reduced Lunch (%)	0.64	0.81	0.49	0.70	0.65	0.19	0.65	0.42	0.40	0.31
Black (%)	0.32	0.74	0.03	0.07	0.47	0.05	0.06	0.17	0.03	0.04
Student-Teacher Ratio	13.17	14.46	18.42	21.92	14.45	15.24	15.64	14.92	14.19	19.93
Log of Enrollment	8.65	8.21	7.50	9.24	9.34	8.63	9.26	9.47	7.48	8.79
Hispanic (%)	0.34	0.05	0.16	0.67	0.10	0.06	0.71	0.16	0.05	0.12
Total Spending (per-pupil)	23236.39	12763.07	10336.27	10976.54	11543.54	13710.96	10271.01	10375.80	12806.54	10516.98
White (%)	0.27	0.17	0.75	0.19	0.39	0.82	0.19	0.63	0.89	0.76
Special Ed (%)	0.16	0.14	0.12	0.10	0.14	0.13	0.10	0.11	0.15	0.12
TPS-Charter Spending (% diff)	0.30	0.11	0.59	0.10	0.14	0.20	0.04	0.11	0.16	0.40
Per Pupil Revenue	23000.65	12292.82	10404.56	10653.91	11596.50	13385.77	9628.86	10091.90	13142.41	10437.44
Teacher Salary	130718.60	94376.71	94000.02	120905.76	88405.78	98787.16	83677.90	84147.72	84027.14	102056.40
Exempt from State/District Regs	0.42	0.70	0.66	0.78	0.69	0.56	0.71	0.67	0.55	0.57
Clear Charter Renewal Policies	0.57	0.68	0.65	0.75	0.71	0.52	0.58	0.65	0.49	0.70
Equitable Funding	0.49	0.45	0.52	0.91	0.52	0.38	0.60	0.45	0.27	0.69
No Caps on CS Growth	0.87	0.92	0.86	0.78	0.91	0.78	0.75	0.85	0.72	0.81
Town	0.00	0.37	0.31	0.17	0.21	0.11	0.17	0.09	0.30	0.21
Charter Spending (per-pupil)	13994.86	12204.52	8151.90	11998.39	10853.28	11420.24	9802.15	9328.18	11814.67	8440.17
Transparent Charter Startup Policies	0.53	0.70	0.55	0.50	0.66	0.51	0.52	0.52	0.44	0.53
Performance-Based Contracts	0.53	0.64	0.50	0.28	0.64	0.57	0.67	0.60	0.57	0.51
Suburb	0.63	0.24	0.16	0.44	0.24	0.62	0.30	0.43	0.21	0.47
Rural	0.13	0.30	0.41	0.12	0.20	0.16	0.12	0.26	0.47	0.07
Urban	0.24	0.10	0.12	0.27	0.35	0.11	0.41	0.21	0.02	0.25
Magnet Schools (%)	0.65	0.46	0.02	0.24	1.11	0.02	0.00	0.35	0.01	0.33
University in City	0.31	-0.02	0.00	-0.02	0.02	0.00	0.00	-0.01	0.00	0.03
City Population (standardized)	0.10	-0.03	0.00	-0.04	0.03	0.00	0.00	-0.02	0.00	0.01
ATT	0.08	0.06	0.04	0.02	0.01	-0.00	-0.02	-0.02	-0.03	-0.05
ACRT	0.19	0.32	0.38	0.39	-0.03	0.37	0.23	0.03	0.21	0.31

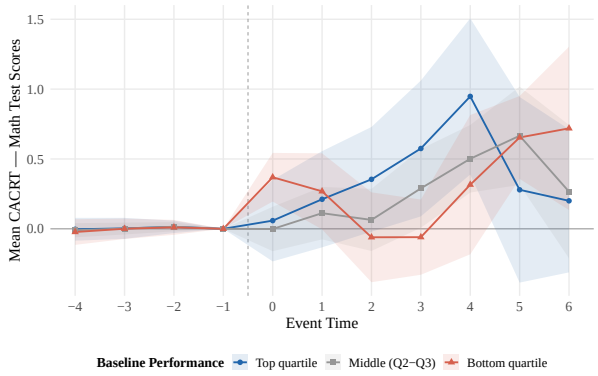
Notes: This table reports the within-group means for each of 10 covariate neighborhoods assigned through VIF-weighted k-means matching. Rows are sorted by VIF in descending order, columns are sorted by within-group ATT, then by within-group $ACRT$ so that group (1) represents the best performing districts and group (10) are the worst performing districts.

Table C9. DiDCF Differences-in-Means (k-means matched groups): ELA Scores

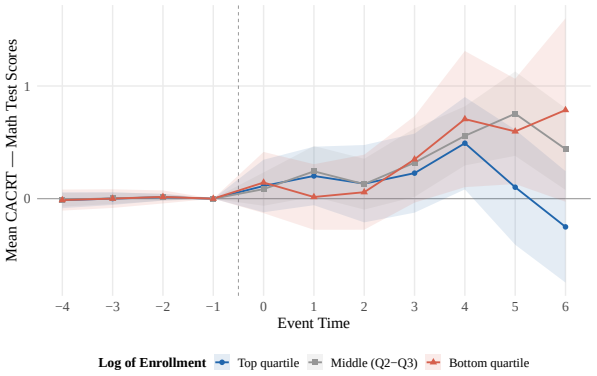
	Covariate Grouping									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Baseline Performance	-0.80	-1.64	1.20	-1.44	-0.90	-0.69	0.09	0.59	0.55	-0.40
Free/Reduced Lunch (%)	0.64	0.79	0.19	0.74	0.64	0.50	0.42	0.35	0.32	0.58
Student-Teacher Ratio	12.84	14.33	14.67	21.96	15.88	18.57	13.89	21.32	17.53	14.65
Log of Enrollment	8.60	8.38	8.79	9.32	9.20	7.66	7.29	9.28	8.49	9.62
Black (%)	0.31	0.69	0.07	0.06	0.06	0.04	0.03	0.06	0.04	0.38
Hispanic (%)	0.35	0.06	0.07	0.70	0.68	0.16	0.06	0.27	0.07	0.12
White (%)	0.28	0.20	0.79	0.17	0.21	0.75	0.87	0.52	0.86	0.47
Total Spending (per-pupil)	23652.06	12513.45	14253.22	10888.53	10279.50	10580.29	12320.37	11524.07	10371.49	11082.58
Teacher Salary	129084.33	92837.61	99519.33	123620.81	83535.83	95544.28	79037.68	117136.45	91415.81	87332.28
Special Ed (%)	0.16	0.14	0.13	0.10	0.10	0.13	0.14	0.11	0.13	0.13
TPS-Charter Spending (% diff)	0.31	0.13	0.13	0.11	0.04	0.58	0.17	0.71	0.21	0.12
Per Pupil Revenue	23432.27	12128.30	13704.25	10836.85	9700.59	10562.54	12611.07	10635.49	10616.23	11184.93
Exempt from State/District Regs	0.41	0.70	0.58	0.78	0.73	0.64	0.57	0.69	0.55	0.69
Clear Charter Renewal Policies	0.56	0.70	0.53	0.75	0.60	0.65	0.48	0.69	0.67	0.68
Equitable Funding	0.48	0.44	0.37	0.92	0.59	0.54	0.25	0.77	0.55	0.51
Transparent Charter Startup Policies	0.53	0.70	0.55	0.50	0.51	0.55	0.42	0.55	0.50	0.60
No Caps on CS Growth	0.88	0.92	0.75	0.78	0.76	0.85	0.73	0.78	0.82	0.90
Performance-Based Contracts	0.55	0.63	0.56	0.28	0.66	0.51	0.57	0.31	0.61	0.62
Charter Spending (per-pupil)	13758.42	11750.26	12369.59	12105.31	9759.85	8152.96	11411.80	8505.22	9038.63	10528.67
Town	0.00	0.36	0.06	0.17	0.19	0.33	0.33	0.11	0.22	0.13
Suburb	0.66	0.20	0.69	0.43	0.29	0.18	0.08	0.42	0.49	0.33
Rural	0.16	0.33	0.11	0.12	0.14	0.36	0.58	0.04	0.15	0.20
Urban	0.19	0.12	0.13	0.27	0.37	0.13	0.00	0.43	0.14	0.34
Magnet Schools (%)	0.69	0.37	0.02	0.23	0.00	0.03	0.01	0.69	0.04	1.19
University in City	0.00	0.00	0.00	-0.02	0.00	0.00	0.00	0.27	0.00	-0.01
City Population (standardized)	0.00	-0.01	0.00	-0.05	0.00	0.00	0.00	0.09	0.00	0.00
ATT	0.08	0.07	0.05	0.03	0.03	0.03	0.01	-0.02	-0.02	-0.02
ACRT	0.36	0.25	-0.02	0.55	0.36	0.21	0.12	0.06	0.03	-0.18

Notes: This table reports the within-group means for each of 10 covariate neighborhoods assigned through VIF-weighted k-means matching. Rows are sorted by VIF in descending order, columns are sorted by within-group ATT, then by within-group $ACRT$ so that group (1) represents the best performing districts and group (10) are the worst performing districts.

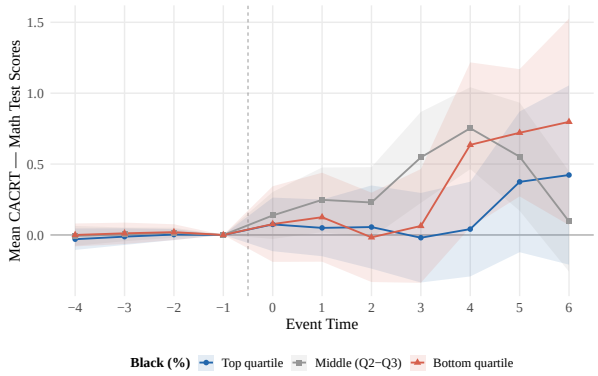
C.2 CACRT Event Studies



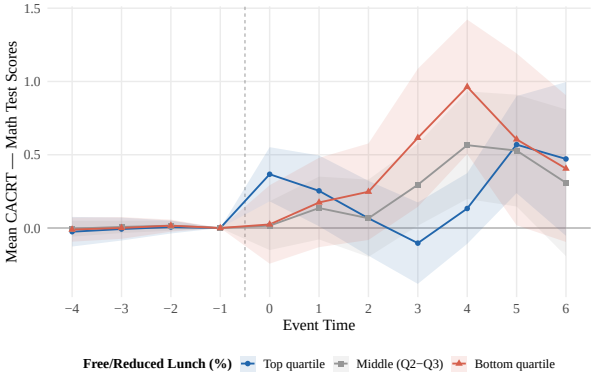
(a) Baseline Math Scores



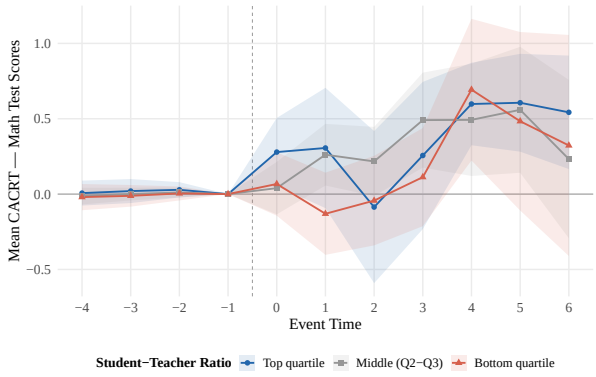
(b) Log Enrollment



(c) Black (%)



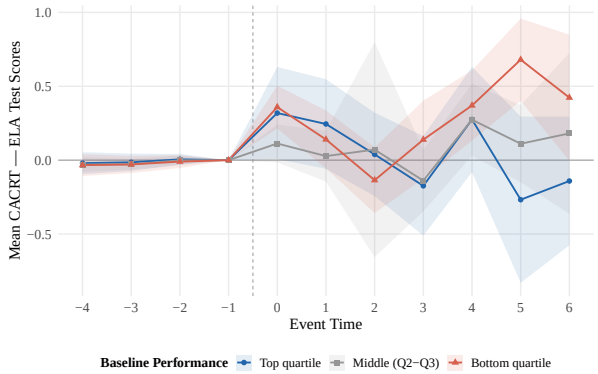
(d) Free/Reduced Price Lunch (%)



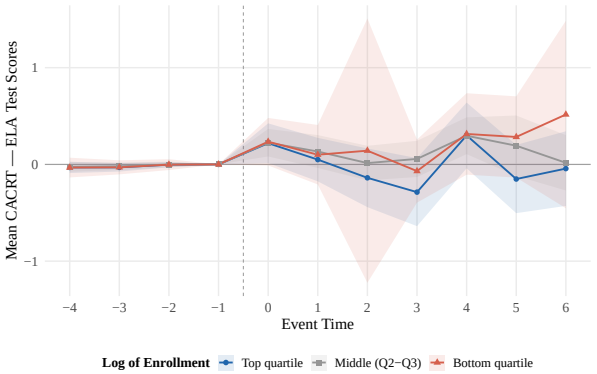
(e) Student-Teacher Ratio

Figure C11. DiDCF Quartile Event Studies (CACRT): Math Scores

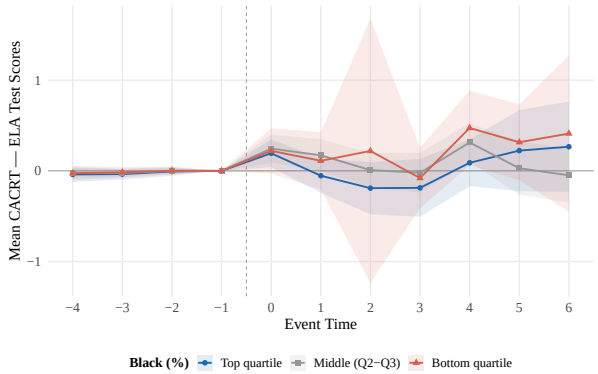
Notes: Each panel shows cohort-weighted ACRT estimates by event-time, separately for districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF at $h = 1$ for math scores.



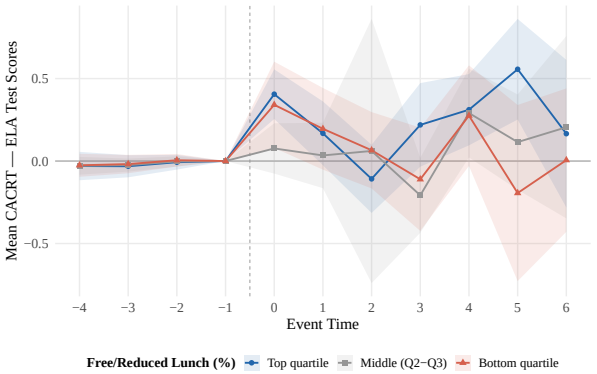
(a) Baseline ELA Scores



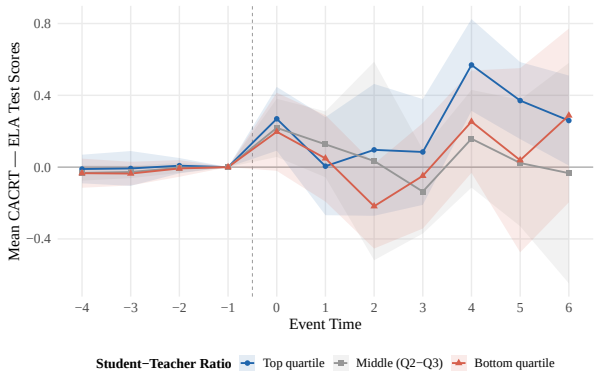
(b) Log Enrollment



(c) Black (%)



(d) Free/Reduced Price Lunch (%)



(e) Student-Teacher Ratio

Figure C12. DiDCF Quartile Event Studies (CACRT): ELA Scores

Notes: Each panel shows cohort-weighted ACRT estimates by event-time, separately for districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF at $h = 1$ for ELA scores.

Table C10. State-by-State Treatment Effects: Comparison with CREDO (2023)

	DiD Estimates		CREDO (2023)	
State	Math	ELA	Math	ELA
AL	—	—	—	—
AK	−0.20	0.05	—	—
AZ	0.04	0.08*	0.18**	3.51**
AR	0.35***	0.11	−0.18	0.53
CA	0.03	0.05	0.70**	1.93**
CO	0.07	0.05	2.28*	2.63*
CT	−0.25	0.09	—	—
DE	−0.33*	−0.61**	—	—
DC	—	—	5.61**	0.88**
FL	−0.09	0.02	−0.18**	1.40**
GA	−0.22	−0.23	—	—
HI	—	—	—	—
ID	−0.32	−0.28	1.40*	2.98*
IL	−0.03	0.22***	8.42**	7.02**
IN	−0.45	−0.31	−4.74	0.53
IA	—	—	—	—
KS	0.10	0.18	—	—
KY	0.00	0.00	—	—
LA	0.11	0.19**	1.05	0.53
ME	—	—	—	—
MD	−0.47	−0.60**	6.49**	6.49**
MA	−0.01	0.19**	7.19**	7.19**
MI	−0.10	−0.22	4.21**	6.32**
MN	−0.24	−0.22	1.40**	3.68**
MS	0.00	0.00	9.82*	6.84*
MO	—	—	—	—
MT	0.00	0.00	—	—
NE	0.00	0.00	0.53	1.40
NV	—	—	—	—
NH	−0.50	−0.76	—	—
NJ	0.58***	0.59***	5.61*	5.79*

Continued on next page

Table C10 continued

	DiD Estimates		CREDO (2023)	
State	Math	ELA	Math	ELA
NM	−0.36	−0.28	1.23	1.93
NY	−0.13	0.51	12.81**	13.16**
NC	0.05	0.15*	−2.98**	2.28**
ND	0.00	0.00	—	—
OH	−0.08	0.02	−6.67**	−0.70**
OK	−0.77***	−0.43	—	—
OR	−0.07	−0.06	−5.61*	−3.16*
PA	−0.09	−0.10	−3.68	−0.18
RI	0.75*	0.60	15.44**	15.79**
SC	−0.04	0.17	−8.25**	−1.40**
SD	0.00	0.00	—	—
TN	—	—	6.84**	5.96**
TX	−0.23***	0.01	−0.88**	4.21**
UT	0.30*	0.20	−2.46	−0.35
VT	—	—	—	—
VA	—	—	—	—
WA	—	—	6.84	4.56
WV	0.00	0.00	—	—
WI	0.14	0.01	2.81	2.63
WY	—	—	—	—
Correlation	0.30	0.41	—	—

Notes: This table compares the state-level DiD estimates with the corresponding estimates reported in Figures 1.10 and 1.11 of CREDO (2023). The final row reports the correlation between the DiD estimates and the corresponding CREDO estimates. The CREDO estimates are divided by 5.7 to convert learning days into effect-size units. Dashes indicate that an estimate is unavailable. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

D Proofs

Proof of Lemma 1

Proof. Recall Definition 1:

$$CATT(g,t,w,x \mid g,w,x) \equiv \mathbb{E}[Y_{i,t}(g,w) - Y_{i,t}(0) \mid G_i = g,\ W_i = w,\ X_i = x]$$

Add and subtract the base period untreated outcome inside Definition 1:

$$\begin{aligned} CATT(g,t,w,x \mid g,w,x) = \mathbb{E}\big[Y_{i,t}(g,w) - Y_{i,b_g}(0) \mid G_i = g,\ W_i = w,\ X_i = x\big] \\ - \mathbb{E}\big[Y_{i,t}(0) - Y_{i,b_g}(0) \mid G_i = g,\ W_i = w,\ X_i = x\big] \end{aligned}$$

The second term is the counterfactual untreated trend of cohort g. Under Assumption 5, this is equal to the observed trend of never-treated districts with covariates x:

$$\begin{aligned} &\mathbb{E}\big[Y_{i,t}(0) - Y_{i,b_g}(0) \mid G_i = g,\ W_i = w,\ X_i = x\big] \\ &\quad = \mathbb{E}\big[Y_{i,t}(0) - Y_{i,b_g}(0) \mid G_i = \infty,\ W_i = 0,\ X_i = x\big] \end{aligned}$$

From here, we can replace potential outcomes with observed outcomes. For cohort-g districts, consistency gives $Y_{i,t} = Y_{i,t}(g,w)$ in the post-period, and because $b_g < g$, Assumption 3 ensures $Y_{i,b_g} = Y_{i,b_g}(g,w) = Y_{i,b_g}(0)$. For never-treated districts, $Y_{i,t} = Y_{i,t}(\infty,0) \equiv Y_{i,t}(0)$ in every period. Substituting:

$$\begin{aligned} CATT(g,t,w,x \mid g,w,x) = \mathbb{E}\big[Y_{i,t} - Y_{i,b_g} \mid G_i = g,\ W_i = w,\ X_i = x\big] \\ - \mathbb{E}\big[Y_{i,t} - Y_{i,b_g} \mid G_i = \infty,\ W_i = 0,\ X_i = x\big] \end{aligned}$$

which is the statement of Lemma 1.

Proof of Lemma 2

Proof. Recall Definition 2:

$$CACRT(g,t,w,x \mid g,w,x) \equiv \frac{\partial}{\partial l}\,\mathbb{E}[Y_{i,t}(g,l) \mid G_i = g,\ W_i = w,\ X_i = x]\Big|_{l=w}$$

Add and subtract the base period outcome under dose l inside the expectation. Because

$b_g < g$, Assumption 3 gives $Y_{i,b_g}(g,l) = Y_{i,b_g}(0)$, so the base period term does not depend on l:

$$\begin{aligned}\mathbb{E}[Y_{i,t}(g,l) \mid G_i = g,\ W_i = w,\ X_i = x] = \mathbb{E}\big[Y_{i,t}(g,l) - Y_{i,b_g}(g,l) \mid G_i = g,\ W_i = w,\ X_i = x\big] \\ + \mathbb{E}\big[Y_{i,b_g}(0) \mid G_i = g,\ W_i = w,\ X_i = x\big]\end{aligned}$$

The first term is the trend that dose group w would have followed under dose l. Under Assumption 6(a), this is equal to the trend of the districts that actually received dose l:

$$\begin{aligned}&\mathbb{E}\big[Y_{i,t}(g,l) - Y_{i,b_g}(g,l) \mid G_i = g,\ W_i = w,\ X_i = x\big] \\ &\quad = \mathbb{E}\big[Y_{i,t}(g,l) - Y_{i,b_g}(g,l) \mid G_i = g,\ W_i = l,\ X_i = x\big]\end{aligned}$$

From here, we can replace potential outcomes with observed outcomes. For cohort-g districts with dose l, consistency gives $Y_{i,t} = Y_{i,t}(g,l)$ in the post-period, and Assumption 3 ensures $Y_{i,b_g} = Y_{i,b_g}(g,l) = Y_{i,b_g}(0)$. Substituting:

$$\begin{aligned}\mathbb{E}[Y_{i,t}(g,l) \mid G_i = g,\ W_i = w,\ X_i = x] = \mathbb{E}\big[Y_{i,t} - Y_{i,b_g} \mid G_i = g,\ W_i = l,\ X_i = x\big] \\ + \mathbb{E}\big[Y_{i,b_g}(0) \mid G_i = g,\ W_i = w,\ X_i = x\big]\end{aligned}$$

Differentiating with respect to l at $l = w$, the second term drops out and

$$CACRT(g,t,w,x \mid g,w,x) = \frac{\partial}{\partial l}\,\mathbb{E}\big[Y_{i,t} - Y_{i,b_g} \mid G_i = g,\ W_i = l,\ X_i = x\big]\Big|_{l=w}$$

which is the statement of Lemma 2. This is the conditional analogue of the second part of Theorem C.1 in Callaway et al. (2025).

Proof of Lemma 3

Proof. Within $\mathcal{S}_g$, the Stage 1 estimand in Equation 19 is

$$m_{g,t}(x) = \mathbb{E}[\Delta Y_{i,g,t} \mid G_i = \infty,\ W_i = 0,\ X_i = x]$$

Subtract $m_{g,t}(x)$ from the long difference of a treated district with dose w and apply Lemma 1:

$$\begin{aligned}\mathbb{E}[\Delta\widetilde{Y}_{i,g,t} \mid G_i = g,\ W_i = w,\ X_i = x] &= \mathbb{E}[\Delta Y_{i,g,t} \mid G_i = g,\ W_i = w,\ X_i = x] - m_{g,t}(x) \\ &= CATT(g,t,w,x \mid g,w,x)\end{aligned}$$

The recentering must be conditional on X_i: subtracting the scalar $\mathbb{E}[\Delta Y_{i,g,t} \mid W_i = 0]$ instead

would leave the x-dependence of $m_{g,t}(x)$ inside $\alpha(x)$. This is the conditional analogue of the recentering in Equation (4.2) of Callaway et al. (2025).

Stage 2 applies the GRF moment condition of Equation 14 to $\Delta\widetilde{Y}_{i,g,t}$ with continuous treatment W_i among treated districts. This solves for the coefficients of the linear projection of $\Delta\widetilde{Y}_{i,g,t}$ on W_i within the covariate cell:

$$\tau(x) = \frac{\mathrm{Cov}(\Delta\widetilde{Y}_{i,g,t},\ W_i \mid G_i = g,\ X_i = x)}{\mathrm{Var}(W_i \mid G_i = g,\ X_i = x)}$$

$$\alpha(x) = \mathbb{E}[\Delta\widetilde{Y}_{i,g,t} \mid G_i = g,\ X_i = x] - \tau(x)\,\mathbb{E}[W_i \mid G_i = g,\ X_i = x]$$

Consistency and asymptotic normality of $\hat{\tau}(x)$ for $\tau(x)$ follow from Theorem 6 of Athey et al. (2019), with the convergence rate given by Theorem 1 of Nie and Wager (2021). These results require conditional moments that are Lipschitz jointly in (x, w), bounded conditional moments, and a conditional density of W_i given X_i bounded away from zero. Assumption 7(a)–(c) delivers the first two and Assumption 2 the third. By the continuous mapping theorem, $\hat{\alpha}(x)$ and $\widehat{CATT}(g,t,w,x) = \hat{\alpha}(x) + \hat{\tau}(x)\,w$ are consistent and asymptotically normal for $\alpha(x)$ and $CATT(g,t,w,x \mid g,w,x)$.

Now add Assumption 6. Under Assumption 5 alone, the building block is local to the districts that chose dose w, so comparing fitted values across doses mixes the causal response with the selection bias term of Equation 17. Assumption 6 removes that term. For any $l \in \mathcal{W}_+$, add and subtract the base period outcome:

$$\begin{aligned} CATT(g,t,w,x \mid g,l,x) = {} & \mathbb{E}[Y_{i,t}(g,w) - Y_{i,b_g}(g,w) \mid G_i = g,\ W_i = l,\ X_i = x] \\ & - \mathbb{E}[Y_{i,t}(0) - Y_{i,b_g}(0) \mid G_i = g,\ W_i = l,\ X_i = x] \end{aligned}$$

Under Assumption 6(a) the first term does not depend on l, and under Assumption 5 the second equals the never-treated trend, which does not depend on l either. Hence $CATT(g,t,w,x \mid g,l,x) = CATT(g,t,w,x \mid g,w,x) = CATT(g,t,w,x)$. Lemma 2 identifies the derivative of this function at w as $CACRT(g,t,w,x)$, and under Equation 20 that derivative is $\tau(x)$. The asymptotic normality of $\hat{\tau}(x) = \widehat{CACRT}(g,t,w,x)$ established above is then Equation 22.

Proof of Lemma 4

Proof. From Equation A1, we know:

$$\hat{\tau}(x) = \frac{\mathrm{Cov}(\Delta\widetilde{Y}, W \mid X)}{\mathrm{Var}(W \mid X)}$$

Plug Equation A2 into $\text{Cov}(\Delta\widetilde{Y}, W \mid X)$. Noting that in order to move from the second line to the third, we notice that within x, $\alpha(x)$ and $\tau(x)$ are scalars. Therefore $\text{Cov}\Big(\alpha(x), W \mid X\Big) = 0$ and $\text{Cov}\Big(\tau(x)W, W \mid X\Big) = \tau(x)\text{Cov}\Big(W, W \mid X\Big) = \tau(x)\text{Var}(W \mid X)$. Also, $\text{Cov}(\varepsilon \mid X) = 0$ by assumption:

$$\begin{aligned}
\text{Cov}(\Delta\widetilde{Y}, W \mid X) &= \text{Cov}\Big(\alpha(x) + \tau(x)W + \delta + \varepsilon, W \mid X\Big) \\
&= \text{Cov}\Big(\alpha(x), W \mid X\Big) + \text{Cov}\Big(\tau(x)W, W \mid X\Big) + \text{Cov}\Big(\delta, W \mid X\Big) + \text{Cov}\Big(\varepsilon, W \mid X\Big) \\
&= \tau(x)\text{Var}\Big(W \mid X\Big) + \text{Cov}\Big(\delta, W \mid X\Big)
\end{aligned}$$

Divide this result by $\text{Var}(W \mid X)$:

$$\begin{aligned}
\hat{\tau}(x) &= \frac{\tau(x)\text{Var}(W \mid X) + \text{Cov}(\delta, W \mid X)}{\text{Var}(W \mid X)} \\
&= \tau(x) + \underbrace{\frac{\text{Cov}(\delta, W \mid X)}{\text{Var}(W \mid X)}}_{=0} \\
&= \tau(x)
\end{aligned}$$

For the second half of Lemma 4, note that the correct model would be on the true DGP (i.e. the expectation of Equation A2). This value is $\mathbb{E}[\Delta\widetilde{Y} \mid X] = \mathbb{E}[\alpha(x) \mid X] + \mathbb{E}[\tau(x)W \mid X] + \mathbb{E}[\delta \mid X] + \mathbb{E}[\varepsilon \mid X] = \alpha(x) + \mathbb{E}[\tau(x)W \mid X] + \mathbb{E}[\delta \mid X]$. The bias between our DiDCF CATT and the true CATT is therefore:

$$\hat{\alpha}(x) - \alpha(x) = \mathbb{E}[\delta \mid X] - \big(\hat{\tau}(x) - \tau(x)\big)\mathbb{E}[W \mid X]$$

The first part of this proof showed that $\hat{\tau}(x) - \tau(x) = 0$, so $\hat{\alpha}(x) - \alpha(x) = \mathbb{E}[\delta \mid X]$, which is the second claim in Lemma 4.

E Technical Summary of GRF Estimation Procedure

A generalized random forest grows B causal trees indexed by $b = 1, ..., B$. Each tree is grown by the following steps:

1. From the full data sample, draw a random subsample without replacement.
2. Further divide the subsample into a training sample, $\mathcal{J}_1$, and an estimation sample, $\mathcal{J}_2$, which are non-overlapping partitions.

3. With training sample $\mathcal{J}_1$, perform greedy and recursive splitting by grid searching over the covariate space and selecting the single covariate, $x_k \in X$, and value, $x_k = c$, that maximizes the squared difference in treatment effects across the resulting child nodes. The objective function is given as:

$$\max_{x_k, c} \quad N_{C_1} N_{C_2} (\hat{\tau}_{C_1} - \hat{\tau}_{C_2})^2$$

Where C_1 and C_2 denote the resulting child nodes such that N_{C_1} and N_{C_2} are the number of units in each resulting child node and $\hat{\tau}_{C_1}$ and $\hat{\tau}_{C_2}$ are the resulting ATEs within each child node. For computational tractability (i.e. in order to avoid estimating linear regressions for each potential split), Athey et al. (2019) show that this process reduces to splitting on a pseudo-outcome corresponding to a first-order gradient-based approximation of the subsample ATEs.

(a) Denote a parent node, P, as any node that is currently being assessed for an optimal split. Compute $\hat{\alpha}_P$ and $\hat{\tau}_P$ by solving Equation 15, which can be alternatively expressed as:

$$\sum_{i=1}^{N_P} \omega_i(x)(Y_i - \alpha(x) - \tau(x)W_i) = 0$$

(b) Also compute an estimate of the gradient of the expectation (once, with all of the data) of $\psi_{\alpha(x),\tau(x)} = Y_i - \alpha(x) - \tau(x)W_i$, denoted $A_p = \nabla \mathbb{E}[\psi_{\hat{\alpha}(x)_P, \hat{\tau}(x)_P} | X_i \in P]$:

$$A_P = \frac{1}{|\{i : X_i \in P\}|} \sum_{\{i: X_i \in P\}} \nabla \psi_{\hat{\alpha}(x)_P, \hat{\tau}(x)_P}$$

(c) Compute the pseudo-outcomes as:

$$\rho_i = -\xi' A_P^{-1} \psi_{\hat{\alpha}(x)_P, \hat{\tau}(x)_P}$$

Where ξ is simply a vector that picks out the $\tau(x)$ coordinate from the $\big(\alpha(x), \tau(x)\big)$ vector.

(d) Finally, use the peudo-outcome in a standard classification and decision tree (CART), which just selects the split that maximizes the sum of the average influence functions between the resulting child nodes:

$$\widetilde{\Delta}(C_1, C_2) = \sum_{j=1}^{2} \frac{1}{|\{i : X_i \in C_j\}|} \Big(\sum_{|\{i: X_i \in C_j\}|} \rho_i \Big)^2$$

Intuitively, the influence function captures how much the treatment effect in the child node is likely to differ from the treatment effect estimate in the parent node. In the simplest case of a randomized experiment where the above process refers to OLS-based identification (without a constant, for simplicity), this first-order approximation looks like:

$$\hat{\tau}_{C_j} \approx \hat{\tau}_P - \frac{1}{\sum_{i \in C_j} W_i^2} \sum_{i \in C_j} W_i (Y_i - W_i \hat{\tau}_P)$$

(e) With the two child nodes, repeat steps (a) – (d) on each, treating each node as a new parent node until a full tree, $\mathcal{T}$, is trained.

4. Within tree $\mathcal{T}$, collect all observations in the estimation subsample $\mathcal{J}_2$ that fall in the same leaf as x, $L_b(x)$. Compute the GRF-weights as the share of total trees where unit i falls into the same leaf as x:

$$\omega_i = \frac{1}{B} \sum_{b=1}^{B} \frac{1}{|L_b(x)|} \mathbb{1}\{X_i \in L_b(x)\}$$